SYSTEMS ENGINEERING HANDBOOK

SYSTEMS ENGINEERING HANDBOOK

A GUIDE FOR SYSTEM LIFE CYCLE PROCESSES AND ACTIVITIES

FIFTH EDITION

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HISTORY OF CHANGES

Revision	Revision date	Change description and rationale
Original	Jun 1994	Draft *Systems Engineering Handbook* (SEH) created by INCOSE members from several defense/aerospace companies—including Lockheed, TRW, Northrop Grumman, Ford Aerospace, and the Center for Systems Management—for INCOSE review.
1.0	Jan 1998	Initial SEH release approved to update and broaden coverage of SE process. Included broad participation of INCOSE members as authors. Based on Interim Standards EIA 632 and IEEE 1220.
2.0	Jul 2000	Expanded coverage on several topics, such as functional analysis. This version was the basis for the development of the Certified Systems Engineering Professional (CSEP) exam.
2.0A	Jun 2004	Reduced page count of SEH v2 by 25% and reduced the US DoD-centric material wherever possible. This version was the basis for the first publicly offered CSEP exam.
3.0	Jun 2006	Significant revision based on ISO/IEC 15288:2002. The intent was to create a country- and domain-neutral handbook. Significantly reduced the page count, with elaboration to be provided in appendices posted online in the INCOSE Product Asset Library (IPAL).
3.1	Aug 2007	Added detail that was not included in SEH v3, mainly in new appendices. This version was the basis for the updated CSEP exam.
3.2	Jan 2010	Updated version based on ISO/IEC/IEEE 15288:2008. Significant restructuring of the handbook to consolidate related topics.
3.2.1	Jan 2011	Clarified definition material, architectural frameworks, concept of operations references, risk references, and editorial corrections based on ISO/IEC review.
3.2.2	Oct 2011	Correction of errata introduced by revision 3.2.1.
4.0	Jul 2015	Significant revision based on ISO/IEC/IEEE 15288:2015, inputs from the relevant INCOSE working groups (WGs), and to be consistent with the Guide to the Systems Engineering Body of Knowledge (SEBoK).
5.0	Jul 2023	Significant revision based on ISO/IEC/IEEE 15288:2023 and inputs from the relevant INCOSE working groups (WGs). Significant restructuring of the handbook based inputs from INCOSE stakeholders.

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PREFACE

The objective of the International Council on Systems Engineering (INCOSE) *Systems Engineering Handbook* (SEH) is to describe key Systems Engineering (SE) process activities. The intended audience is the SE practitioner. When the term "SE practitioner" is used in this handbook, it includes the new SE practitioner, a product engineer, an engineer in another discipline who needs to perform SE, or an experienced SE practitioner who needs a convenient reference.

The descriptions in this handbook show what each SE process activity entails, in the context of designing for required performance and life cycle considerations. On some projects, a given activity may be performed very informally; on other projects, it may be performed very formally, with interim products under formal configuration control. This document is not intended to advocate any level of formality as necessary or appropriate in all situations. The appropriate degree of formality in the execution of any SE process activity is determined by the following:

The need for communication of what is being done (across members of a project team, across organizations, or over time to support future activities)

The level of uncertainty

The degree of complexity

The consequences to human welfare

On smaller projects, where the span of required communications is small (few people and short project life cycle) and the cost of rework is low, SE activities can be conducted very informally and thus at low cost. On larger projects, where the span of required communications is large (many teams that may span multiple geographic locations and organizations and long project life cycle) and the cost of failure or rework is high, increased formality can significantly help in achieving project opportunities and in mitigating project risk.

In a project environment, work necessary to accomplish project objectives is considered "in scope"; all other work is considered "out of scope." On every project, "thinking" is always "in scope." Thoughtful tailoring and intelligent application of the SE processes described in this handbook are essential to achieve the proper balance between the risk of missing project technical and business objectives on the one hand and process paralysis on the other hand. Part IV provides tailoring and application guidance to help achieve that balance.

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HOW TO USE THIS HANDBOOK

PURPOSE

This handbook defines the "state-of-the-good-practice" for the discipline of Systems Engineering (SE) and provides an authoritative reference to understand the SE discipline in terms of content and practice.

APPLICATION

This handbook is consistent with ISO/IEC/IEEE 15288 (2023), *Systems and software engineering—System life cycle processes*, hereafter referred to as ISO/IEC/IEEE 15288, to ensure its usefulness across a wide range of application domains for engineered systems and products, as well as services. ISO/IEC/IEEE 15288 is an international standard that provides system life cycle process outcomes, activities, and tasks, whereas this handbook further elaborates on the activities and practices necessary to execute the processes.

This handbook is also consistent with the *Guide to the Systems Engineering Body of Knowledge*, hereafter referred to as the SEBoK (2023), to the extent practicable. In many places, this handbook points readers to the SEBoK for more detailed coverage of the related topics, including a current and vetted set of references. The SEBoK also includes coverage of "state-of-the-art" in SE.

For organizations that do not follow the principles of ISO/IEC/IEEE 15288 or the SEBoK to specify their life cycle processes, this handbook can serve as a reference to practices and methods that have proven beneficial to the SE community at large and that can add significant value in new domains, if appropriately selected, tailored, and applied. Part IV provides top-level guidance on the application of SE in selected product sectors and domains.

Before applying this handbook in a given organization or on a given project, it is recommended that the tailoring guidelines in Part IV be used to remove conflicts with existing policies, procedures, and standards already in use within an organization. Not every process will apply universally. Careful selection from the material is recommended. Reliance on process over progress will not deliver a system. Processes and activities in this handbook do not supersede any international, national, or local laws or regulations.

USAGE

This handbook was developed to support the users and use cases shown in Table 0.1. Primary users are those who will use the handbook directly. Secondary users are those who will typically use the handbook with assistance from SE practitioners. Other users and use cases are possible.

TABLE 0.1 Handbook users and use cases

User	Type	Use cases
Seasoned SE Practitioner. Those who need to reinforce, refresh, and renew their SE knowledge	Primary	• Adapt or refer to handbook to suit individual applicability • Explore good practices • Identify blind spots or gaps by providing a good checklist to ensure necessary coverage • References to other sources for more in-depth understanding
Novice SE Practitioner: Those who need to start using SE	Primary	• Support structured, coherent, and comprehensive learning • Understand the scope (breadth and depth) of systems thinking and SE practices
INCOSE Certification: Systems Engineering Professional (SEP) certifiers and those being certified	Primary	• Define body of knowledge for SEP certification • Form the basis of the SEP examination
SE Educators: Those who develop and teach SE courses, including universities and trainers	Primary	• Support structured, coherent, and comprehensive learning • Suggest relevant SE topics to trainers for their course content • Serve as a supplemental teaching aid
SE Tool Providers/Vendors: Those who provide tools and methods to support SE practitioners	Primary	• Suggest tools, methods, or other solutions to be developed that help practitioners in their work
Prospective SE Practitioner or Manager: Those who may be interested in pursuing a career in SE or who need to be aware of SE practices	Secondary	• Provide an entry level survey to understand what SE is about to someone who has a basic technical or engineering background
Interactors: Those who perform in disciplines that exchange (consume and/or produce) information with SE practitioners	Secondary	• Understand basic terminologies, scope, structure, and value of SE • Understand the role of the SE practitioner and their relationship to others in a project or an organization

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ORGANIZATION AND STRUCTURE

As shown in Figure 0.1, this handbook is organized into six major parts, plus appendices.

Systems Engineering Introduction (Part I) provides foundational SE concepts and principles that underpin all other parts. It includes the what and why of SE and why it is important, key definitions, systems science and systems thinking, and SE principles and concepts.

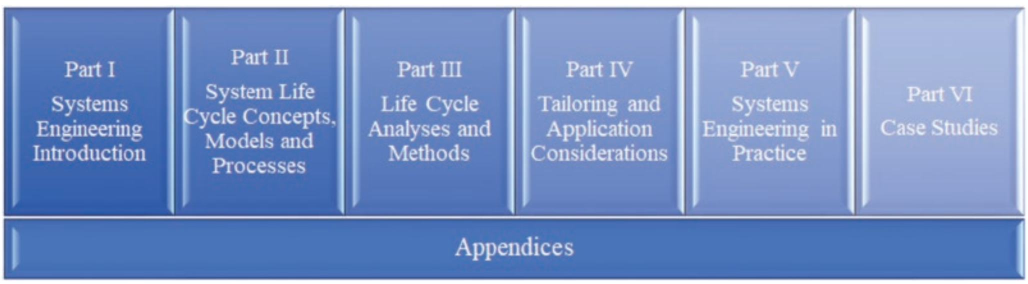

FIGURE 0.1 Handbook structure. INCOSE SEH original figure created by Mornas. Usage per the INCOSE Notices page. All other rights reserved.

System Life Cycle Concepts, Models, and Processes (Part II) describes an informative life cycle model with six stages: concept, development, production, utilization, support, and retirement. It also describes a set of life cycle processes to support SE consistent with the four process groups of ISO/IEC/IEEE 15288: Agreement Processes, Organizational Project Enabling Processes, Technical Management Processes, and Technical Processes.

Life Cycle Analyses and Methods (Part III) describes a set of quality characteristics approaches that need to be considered across the system life cycle. This part also describes methods that can apply across all processes, reflecting various aspects of the concurrent, iterative, and recursive nature of SE.

Tailoring and Application Considerations (Part IV) describes information on how to tailor (adapt and scale) the SE processes. It also introduces various considerations to view and apply SE: SE methodologies and approaches, system types, and project sectors and domains.

Systems Engineering in Practice (Part V) describes SE competencies, diversity, equity, and inclusion, SE relationship to other disciplines, SE transformation, and insight into the future of SE.

Case Studies (Part VI) describes several case studies that are used throughout the handbook to reinforce the SE principles and concepts.

Appendix A contains a list of references used in this handbook. Appendices B and C provide a list of acronyms and a glossary of SE terms and definitions, respectively. Appendix D provides an N^2 diagram of the SE life cycle processes showing an example of the dependencies that exist in the form of shared inputs or outputs. Appendix E provides a list of all the typical inputs/outputs identified for each SE life cycle process. Appendix F acknowledges the various contributors to this handbook. Errors, omissions, and other suggestions for this handbook can be submitted to the INCOSE using instructions found in Appendix G.

SYMBOLOGY

As described in Section 2.3.1.2, SE is a concurrent, iterative, and recursive process. The following symbology is used throughout this handbook to reinforce these concepts

Concurrency is indicated by the parallel lines.
Iteration is indicated by the circular arrows.

Recursion is indicated by the down and up arrows.

TERMINOLOGY

One of the SE practitioner's first and most important responsibilities on a project is to establish nomenclature and terminology that support clear, unambiguous communication and definition of the system and its elements, functions, operations, and associated processes. Further, to promote the advancement of the field of SE throughout the world, it is essential that common definitions and understandings be established regarding general methods and terminology that in turn support common processes. As more SE practitioners accept and use common terminology, SE will experience improvements in communications, understanding, and, ultimately, productivity.

The glossary of terms used throughout this book (see Appendix C) is based on the definitions found in ISO/IEC/IEEE 15288; ISO/IEC/IEEE 24765 (2017); and the SEBoK.

1

SYSTEMS ENGINEERING INTRODUCTION

1.1 WHAT IS SYSTEMS ENGINEERING?

Systems Engineering (SE)

Our world and the systems we engineer continue to become more complex and interrelated. SE is an integrative approach to help teams collaborate to understand and manage systems and their complexity and deliver successful systems. The SE perspective is based on systems thinking—a perspective that sharpens our awareness of wholes and how the parts within those wholes interrelate (incose.org, *About Systems Engineering*). SE aims to ensure the pieces work together to achieve the objectives of the whole. SE practitioners work within a project team and take a holistic, balanced, life cycle approach to support the successful completion of system projects (INCOSE Vision 2035, 2022). SE has the responsibility to realize systems that are *fit for purpose*, namely that systems accomplish their intended purposes and be resilient to effects in real-world operation, while minimizing unintended actions, side effects, and consequences (Griffin, 2010).

Definition of SE

INCOSE Definitions (2019) and ISO/IEC/IEEE 15288 (2023) define:

> **Systems Engineering** is a transdisciplinary and integrative approach to enable the successful realization, use, and retirement of engineered systems, using systems principles and concepts, and scientific, technological, and management methods.

INCOSE Definitions (2019) elaborates:
SE focuses on:

INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities, Fifth Edition.
Edited by David D. Walden, Thomas M. Shortell, Garry J. Roedler, Bernardo A. Delicado, Odile Mornas, Yip Yew-Seng, and David Endler.

- establishing, balancing and integrating stakeholders' goals, purpose and success criteria, and defining actual or anticipated stakeholder needs, operational concepts, and required functionality, starting early in the development cycle;
- establishing an appropriate life cycle model, process approach and governance structures, considering the levels of complexity, uncertainty, change, and variety;
- generating and evaluating alternative solution concepts and architectures;
- baselining and modeling requirements and selected solution architecture for each stage of the endeavor;
- performing design synthesis and system verification and validation;
- while considering both the problem and solution domains, taking into account necessary enabling systems and services, identifying the role that the parts and the relationships between the parts play with respect to the overall behavior and performance of the system, and determining how to balance all of these factors to achieve a satisfactory outcome.

SE provides facilitation, guidance, and leadership to integrate the relevant disciplines and specialty groups into a cohesive effort, forming an appropriately structured development process that proceeds from concept to development, production, utilization, support, and eventual retirement.

SE considers both the business and the technical needs of acquirers with the goal of providing a quality solution that meets the needs of users and other stakeholders, is fit for the intended purpose in real-world operation, and avoids or minimizes adverse unintended consequences.

The goal of all SE activities is to manage risk, including the risk of not delivering what the acquirer wants and needs, the risk of late delivery, the risk of excess cost, and the risk of negative unintended consequences. One measure of utility of SE activities is the degree to which such risk is reduced. Conversely, a measure of acceptability of absence of a SE activity is the level of excess risk incurred as a result.

Definitions of System

While the concepts of a *system* can generally be traced back to early Western philosophy and later to science, the concept most familiar to SE practitioners is often traced to Ludwig von Bertalanffy (1950, 1968) in which a system is regarded as a "whole" consisting of interacting "parts."

INCOSE Definitions (2019) and ISO/IEC/IEEE 15288 (2023) define:

> A **system** is an arrangement of parts or elements that together exhibit behavior or meaning that the individual constituents do not.

A system is sometimes considered as a product or as the services it provides.

In practice, the interpretation of its meaning is frequently clarified using an associative noun (e.g., medical system, aircraft system). Alternatively, the word "system" is substituted simply by a context-dependent synonym (e.g., pacemaker, aircraft), though this potentially obscures a system principles perspective.

A complete system includes all of the associated equipment, facilities, material, computer programs, firmware, technical documentation, services, and personnel required for operations and support to the degree necessary for self-sufficient use in its intended environment.

INCOSE Definitions (2019) elaborates:
Systems can be either physical or conceptual, or a combination of both. Systems in the physical universe are composed of matter and energy, may embody information encoded in matter-energy carriers, and exhibit observable behavior. Conceptual systems are abstract systems of pure information, and do not directly exhibit behavior, but exhibit "meaning." In both cases, the system's properties (as a whole) result, or emerge, from:

a) the parts or elements and their individual properties,
b) the relationships and interactions between and among the parts, the system, other external systems (including humans), and the environment.

SE practitioners are especially interested in systems which have or will be "systems engineered" for a purpose. Therefore, INCOSE Definitions (2019) defines:

> An **engineered system** is a system designed or adapted to interact with an anticipated operational environment to achieve one or more intended purposes while complying with applicable constraints.

"Engineered systems" may be composed of any or all of the following elements: people, products, services, information, processes, and/or natural elements.

Origins and Evolution of SE

Aspects of SE have been applied to technical endeavors throughout history. However, SE has only been formalized as an engineering discipline beginning in the early to middle of the twentieth century (INCOSE Vision 2035, 2022). The term "systems engineering" dates to Bell Telephone Laboratories in the early 1940s (Fagen, 1978; Hall, 1962; Schlager, 1956). Fagen (1978) traces the concepts of SE within the Bell System back to early 1900s and describes major applications of SE during World War II. The British used multidisciplinary teams to analyze their air defense system in the 1930s (Martin, 1996). The RAND Corporation was founded in 1946 by the United States Air Force and claims to have created "systems analysis." Hall (1962) asserts that the first attempt to teach SE as we know it today came in 1950 at MIT by Mr. Gilman, Director of Systems Engineering at Bell. TRW (now a part of Northrop Grumman) claims to have "invented" SE in the late 1950s to support work with ballistic missiles. Goode and Machol (1957) authored the first book on SE in 1957. In 1990, a professional society for SE, the National Council on Systems Engineering (NCOSE), was founded by representatives from several US corporations and organizations. As a result of growing involvement from SE practitioners outside of the US, the name of the organization was changed to the International Council on Systems Engineering (INCOSE) in 1995 (incose.org, *History of Systems Engineering*; Buede and Miller, 2016).

With the introduction of the international standard ISO/IEC 15288 in 2002, the discipline of SE was formally recognized as a preferred mechanism to establish agreement for the creation of products and services to be traded between two or more organizations—the supplier(s) and the acquirer(s). This handbook builds upon the concepts in the latest edition of ISO/IEC/IEEE 15288 (2023) by providing additional context, definitions, and practical applications. Table 1.1 provides a list of key SE standards and guides related to the content of this handbook.

TABLE 1.1 SE standards and guides

Reference	Title
ISO/IEC/IEEE 15026	Systems and software engineering—Systems and software assurance (Multi-part standard)
ISO/IEC/IEEE 15288	Systems and software engineering—System life cycle processes
IEEE/ISO/IEC 15289	Systems and software engineering—Content of life cycle information items (documentation)
ISO/IEC/IEEE 15939	Systems and software engineering—Measurement process
ISO/IEC/IEEE 16085	Systems and software engineering—Life cycle processes—Risk management
ISO/IEC/IEEE 16326	Systems and software engineering—Life cycle processes—Project management
ISO/IEC/IEEE 21839	Systems and software engineering—System of systems (SoS) considerations in life cycle stages of a system
ISO/IEC/IEEE 21840	Systems and software engineering—Guidelines for the utilization of ISO/IEC/IEEE 15288 in the context of system of systems (SoS)
ISO/IEC/IEEE 21841	Systems and software engineering—Taxonomy of systems of systems
ISO/IEC/IEEE 24641	Systems and software engineering—Methods and tools for model-based systems and software engineering

(Continued)

TABLE 1.1 (Continued)

Reference	Title
ISO/IEC/IEEE 24748–1	Systems and software engineering—Life cycle management—Part 1: Guidelines for life cycle management
ISO/IEC/IEEE 24748–2	Systems and software engineering—Life cycle management—Part 2: Guidelines for the application of ISO/IEC/IEEE 15288
ISO/IEC/IEEE 24748–4	Systems and software engineering—Life cycle management—Part 4: Systems engineering planning
ISO/IEC/IEEE 24748–6	Systems and software engineering—Life cycle management—Part 6: System integration engineering
ISO/IEC/IEEE 24748–7	Systems and software engineering—Life cycle management—Part 7: Application of systems engineering on defense programs
ISO/IEC/IEEE 24748–8 / IEEE 15288.2	Systems and software engineering—Life cycle management—Part 8: Technical reviews and audits on defense programs
ISO/IEC/IEEE 24765	Systems and software engineering—Vocabulary
ISO/IEC/IEEE 26550	Software and systems engineering—Reference model for product line engineering and management
ISO/IEC/IEEE 26580	Software and systems engineering—Methods and tools for the feature-based approach to software and systems product line engineering
ISO/IEC/IEEE 29148	Systems and software engineering—Life cycle processes—Requirements engineering
ISO/IEC/IEEE 42010	Systems and software engineering—Architecture description
ISO/IEC/IEEE 42020	Software, systems and enterprise—Architecture processes
ISO/IEC/IEEE 42030	Software, systems and enterprise—Architecture evaluation framework
ISO/IEC 29110	Systems and Software Engineering Standards and Guides for Very Small Entities (VSEs) (Multi-part set)
ISO/IEC 31000	Risk management
ISO/IEC 31010	Risk management—Risk assessment techniques
ISO/IEC 33060	Process assessment—Process assessment model for system life cycle processes
ISO/PAS 19450	Automation systems and integration—Object-Process Methodology (OPM)
ISO 10007	Quality management—Guidelines for configuration management
ISO 10303-233	Industrial automation systems and integration—Product data representation and exchange—Part 233: Application protocol: Systems engineering
NIST SP 800–160 Vol. 1	Systems Security Engineering: Considerations for a Multidisciplinary Approach in the Engineering of Trustworthy Secure Systems
NIST SP 800–160 Vol. 2	Developing Cyber-Resilient Systems: A Systems Security Engineering Approach
OMG SysML™	OMG Systems Modeling Language
SEBoK	Guide to the Systems Engineering Body of Knowledge (SEBoK)
SAE-EIA 649C	Configuration Management Standard
SAE 1001	Integrated Project Processes for Engineering a System (Note: Replaced ANSI/EIA 632)
ANSI/AIA.A G.043B	Guide to the Preparation of Operational Concept Documents
CMMI	CMMI® V2.0

1.2 WHY IS SYSTEMS ENGINEERING IMPORTANT?

The purpose of SE is to conceive, develop, produce, utilize, support, and retire the right product or service within budget and schedule constraints. Delivering the right product or service requires a common understanding of the current system state and a common vision of the system's future states, as well as a methodology to transform a set of stakeholder needs, expectations, and constraints into a solution. The right product or service is one that accomplishes the required service or mission. A common vision and understanding, shared by acquirers and suppliers, is achieved through application of proven methods that are based on standard approaches across people, processes, and tools. The application of these methods is continuous throughout the system's life cycle.

SE is particularly important in the presence of complexity (see Section 1.3.7). Most current systems are formed by integrating commercially available products or by integrating independently managed and operated systems to provide emergent capabilities which increase the level of complexity (see Sections 4.3.3 and 4.3.6). This increased reliance on off-the-shelf and systems of systems has significantly reduced the time from concept definition to market availability of products. Over the years between 1880 and 2000, average 25% market penetration has been reduced by more than a factor of four as illustrated in Figure 1.1.

In response to complexity and compressed timelines, SE methods and tools have become more adaptable and efficient. Introduction of agile methods (see Section 4.2.2) and SE modeling language standards such the Systems Modeling Language (SysML) have allowed SE practitioners to manage complexity and increase the implementation of a common system vision (see bottom of Figure 1.1). Model Based SE (MBSE) methods adoption continues to grow (see Section 4.2.1), particularly in the early conceptual design and requirements analysis (SEBOK, *Emerging Topics*). MBSE research literature continues to report on the increased productivity and quality of design and promises further progression toward a digital engineering (DE) approach, where data is transparent and cooperation optimized across all engineering disciplines. Standards organizations are updating or developing new approaches that take DE into consideration. SE will have to address this new digital representation of the system as DE becomes the way of doing business (see Section 5.4). The rapid evolution and introduction of Artificial Intelligence (AI) and Machine Learning (ML) into SE further increases complexity of verifiability, safety, and trust of self-learning and evolving systems.

The overall value of SE has been the subject of studies and papers from many organizations since the introduction of SE. A 2013 study was completed at the University of South Australia to quantify the return on investment (ROI) of SE activities on overall project cost and schedule (Honour, 2013). Figure 1.2 compares the total SE effort with cost

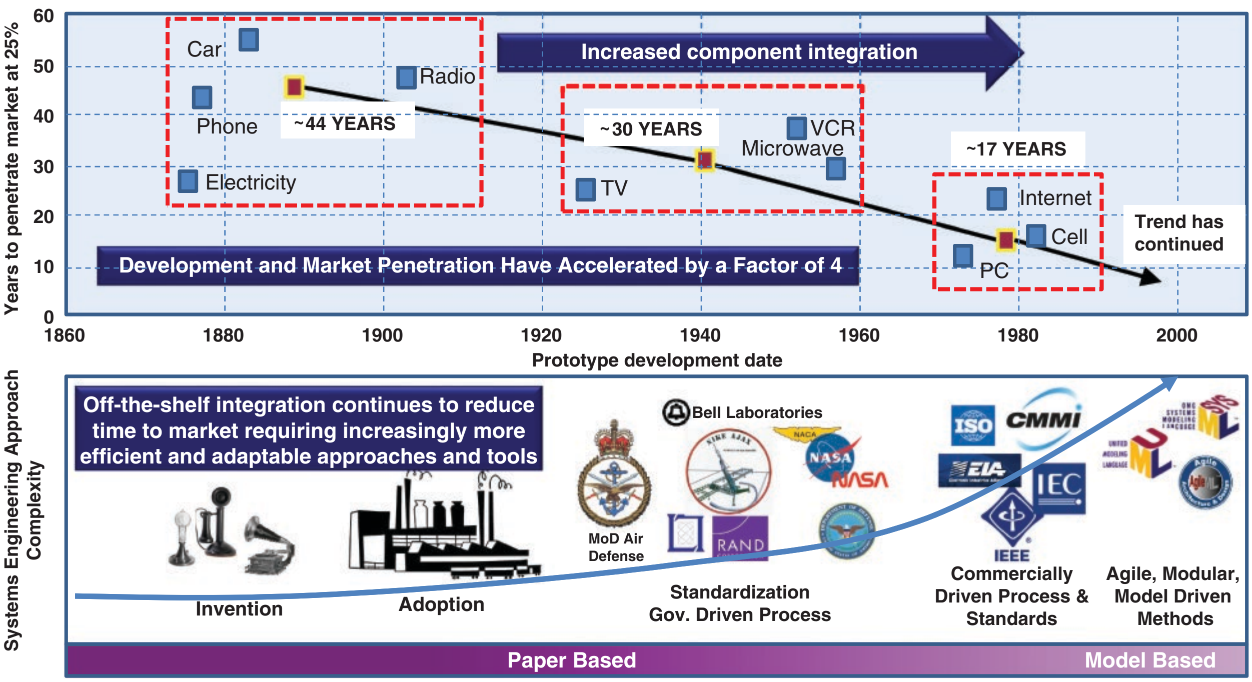

FIGURE 1.1 Acceleration of design to market life cycle has prompted development of more automated design methods and tools. INCOSE SEH original figure created by Amenabar. Usage per the INCOSE Notices page. All other rights reserved.

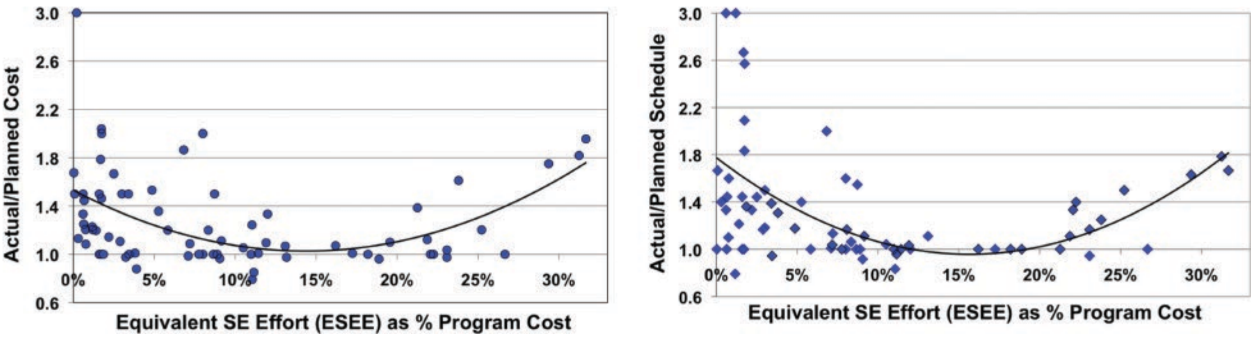

FIGURE 1.2 Cost and schedule overruns correlated with SE effort. From Honour (2013) with permission from University of South Wales. All other rights reserved.

compliance (left figure) and schedule performance (right figure). In both graphs, increasing the percentage of SE within the project results in better success up to an optimum level, above which SE ROI is diminished above those total program expenditure levels due to increased unwarranted processes. Study data shows that SE effort had a significant, quantifiable effect on project success, with correlation factors as high as 80%. Results show that the optimum level of SE effort for a normalized range of 10% to 14% of the total project cost.

The ROI of adding additional SE activities to a project is shown in Table 1.2, and it varies depending on the level of SE activities already in place. If the project is using no SE activities, then adding SE carries a 7:1 ROI; for each cost unit of additional SE, the project total cost will reduce by 7 cost units. At the median level of the projects interviewed, additional SE effort carries a 3.5:1 ROI.

A joint 2012 study by the National Defense Industrial Association (NDIA), the Institute of Electrical and Electronic Engineers (IEEE), and the Software Engineering Institute (SEI) of Carnegie Mellon University (CMU) surveyed 148 development projects and found clear and significant relationships between the application of SE activities and the performance of those projects as seen in Figure 1.3 (Elm and Goldenson, 2012). The study broke the projects by the maturity of their SE processes as measured by the quantity and quality of specific SE work products and considered the complexity of each project and the maturity of the technologies being implemented (n=number of projects). It also assessed the levels of project performance, as measured by satisfaction of budget, schedule, and technical requirements. The left column represents those projects deploying lower levels of SE expertise and capability. Among these projects, only 15% delivered higher levels of project performance and 52% delivered lower levels of project performance. The center column represents those projects deploying moderate levels of SE expertise and capability. Among these projects, the number delivering higher levels of project performance increased to 24% and those delivering lower levels decreased to 29%. The right column represents those projects deploying higher levels of SE expertise and capability. For these projects, the number delivering higher levels of project performance increased substantially

TABLE 1.2 SE return on investment

Current SE effort (% of program cost)	Average cost overrun (%)	ROI for additional SE effort (cost reduction $ per $ SE added)
0	53	7.0
5	24	4.6
7.2 (median of all programs)	15	3.5
10	7	2.1
15	3	–0.3
20	10	–2.8

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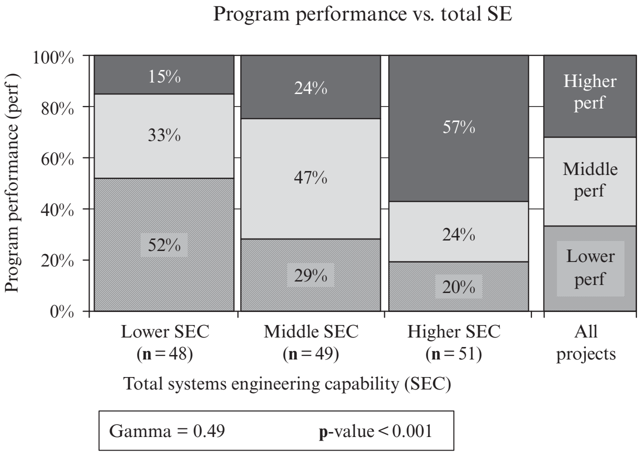

FIGURE 1.3 Project performance versus SE capability. From Elm and Goldenson (2012) with permission from Carnegie Mellon University. All other rights reserved.

to 57%, while those delivering lower levels decreased to 20%. As Figure 1.3 shows, well-applied SE increases the probability of successfully developing an engineered system.

A 1993 Defense Acquisition University (DAU) statistical analysis on US Department of Defense (DoD) projects examined spent and committed life cycle cost (LCC) over time (DAU, 1993). As illustrated notionally in Figure 1.4, an important result from this study is that by the time approximately 20% of the actual costs have been accrued, over 80% of the total LCC has already typically been committed. Figure 1.4 also shows that it is less costly to fix or address issues if they are identified early. Good SE practice is the means by which the issues are identified and ensures that the understanding obtained is applied as appropriate during the life cycle, thus reducing technical debt.

INCOSE maintains value proposition statements (INCOSE Value Strategic Initiative Report, 2021) as tailored to different areas and industries. Areas covered include individual INCOSE membership, organizational INCOSE membership, INCOSE SE certification, and the discipline of SE. Industries include commercial, government, and nonprofit organizations. A sample of these findings includes:

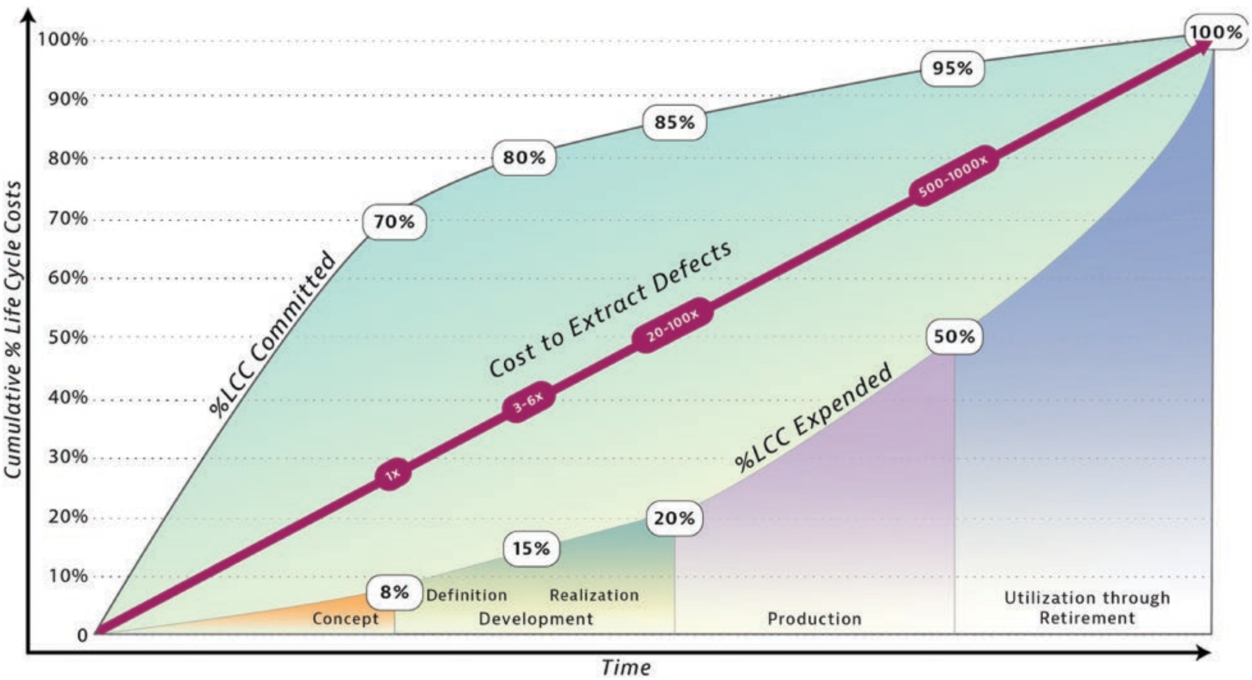

FIGURE 1.4 Life cycle costs and defect costs against time. INCOSE SEH original figure created by Walden derived from DAU (1993). Usage per the INCOSE Notices page. All other rights reserved.

- *Value of SE to the Commercial/Market-Driven Industry*: Companies and other enterprises in commercial industry will benefit from the internal practice of professional SE by having enhanced their capability for the development of innovative products and services for distribution in both mature and immature markets, in a more efficient and competitive manner.
- *Value of SE to Government/Infrastructure/Aerospace/Defense Industry*: SE provides a tailorable, systematic approach to all stages of a project, from concept to retirement. SE can accommodate different approaches including agile and sequential and facilitate commonality and open architectures to ensure lower acquisition, maintenance, and upgrade costs. By confirming correct and complete requirements and requirements allocations, the resulting design has fewer and less significant changes resulting in improved overall cost and schedule performance.
- *Value of SE to Nonprofit/Research Industry*: A nonprofit enterprise will benefit from the internal practice of professional SE by having enhanced their capability for the development of innovative client services in a more efficient and effective manner. An enterprise engaged in basic or applied research will benefit from the internal practice of SE by having enhanced its capabilities for discovery and invention that supports technology development in a more effective manner.

1.3 SYSTEMS CONCEPTS

Important system concepts include the system of interest (SoI), the system environment, and external systems. The boundaries between the system and the surrounding elements are important to understand. These boundaries separate the SoI, enabling systems, interoperating systems, and interfacing systems, supporting the SE practitioner in properly accounting for all the necessary elements which comprise the whole system context. Part of the system concept are the system's modes and states which are fundamental system behavior characteristics important to SE. Systems can be hierarchical in their structural organization, or they can be complex where hierarchy is not always present. The system concepts encompass all types of systems structures and support the SE practitioner with a framework in which to engineer a system.

1.3.1 System Boundary and the System of Interest (SoI)

General System Concepts An external view of a system must introduce elements that specifically do not belong to the system but do interact with the system. This collection of elements is called the *system environment or context* and can include the users (or operators) of the system. It is important to understand that the system environment or context is not limited to the operating environment, but also includes external systems that interface with or support the system at any time of the life cycle.

The internal and external views of a system give rise to the concept of a *system boundary*. In practice, the system boundary is a "line of demarcation" between the system under consideration, called the system of interest (SoI), and its greater context. It defines what belongs to the system and what does not. The system boundary is not to be confused with the subset of elements that interact with the environment.

The *functionality* of a system is typically expressed in terms of the interactions of the system with its operating environment, especially the users. When a system is considered as an integrated combination of interacting elements, the functionality of the system derives not just from the interactions of individual elements with the environmental elements but also from how these interactions are influenced by the organization (interrelations) of the system elements. This leads to the concept of *system architecture*, which ISO/IEC/IEEE 42020 (2019) defines as:

> Fundamental concepts or properties of an entity in its environment and governing principles for the realization and evolution of this entity and its related life cycle processes.

This definition speaks to both the internal and external views of the system and shares the concepts from the definitions of a system (see Section 1.1).

Scientific Terminology Related to System Concepts In general, *engineering* can be regarded as the practice of creating and sustaining systems, services, devices, machines, structures, processes, and products to improve the quality of life—getting things done effectively and efficiently. The repeatability of experiments demanded by science is critical for delivering practical engineering solutions that have commercial value. Engineering in general, and SE in particular, draw heavily from the terminology and concepts of science.

An *attribute* of a system (or system element) is an observable characteristic or property of the system (or system element). For example, among the various attributes of an aircraft is its air speed. Attributes are represented symbolically by variables. Specifically, a *variable* is a symbol or name that identifies an attribute. Every variable has a domain, which could be but is not necessarily measurable. A *measurement* is the outcome of a process in which the SoI interacts with an observation system under specified conditions. The outcome of a measurement is the assignment of a *value* to a variable. A system is in a *state* when the values assigned to its attributes remain constant or steady for a meaningful period of time (Kaposi and Myers, 2001). In SE and software engineering, the *system elements* (e.g., software objects) have *processes* (e.g., operations) in addition to attributes. These have the binary logical values of being either *idle* or *executing*. A complete description of a system state therefore requires values to be assigned to both attributes and processes. *Dynamic behavior* of a system is the time evolution of the system state. *Emergent behavior* is a behavior of the system that cannot be understood exclusively in terms of the behavior of the individual system elements. See Section 1.3.2 for further information on emergent behavior and Section 1.3.6 for more information on states and modes.

The key concept used for problem solving is the *black box/white box* (also known as *opaque box/transparent box*) system representation. The *black box (opaque box)* representation is based on an external view of the system (attributes). The *white box (transparent box)* representation is based on an internal view of the system (attributes and structure of the elements). Both representations are useful to the SE practitioner and there must be an understanding of the relationship between the two. A system, then, is represented by the external attributes of the system, its internal attributes and structure, and the interrelationships between these that are governed by the laws of science.

1.3.2 Emergence

Emergence describes the phenomenon that whole entities exhibit properties which are meaningful only when attributed to the whole, not to its elements. Every model of human activity system exhibits properties as a whole entity that derive from its element activities and their structure, but cannot be reduced to them (Checkland, 1999). Emergence is a fundamental property of all systems (Sillitto and Dori, 2017). According to Rousseau et al. (2018), emergence derives from the systems science concept of “properties the system has but the elements by themselves do not.”

System elements interact between themselves and can create desirable or undesirable phenomena called *emergent properties* such as inhibition, interference, resonance, or reinforcement of any property. Emergent properties can also result from the interaction between the system and its environment. Many engineering disciplines include emergence as a property. For example, system safety (Leveson, 1995) and resilience (Rasoulkahni, 2018) are examples of emergent properties of engineered systems (see Sections 3.1.11 and 3.1.9, respectively).

Definition of the architecture of the system includes an analysis of interactions between system elements in order to reinforce desirable and prevent undesirable emergent properties. According to Rousseau et al. (2019), the systemic virtue of emergent properties are used during systems architecture and design definition to highlight necessary derived functions and internal physical or environmental constraints (see Sections 2.3.5.4 and 2.3.5.5, respectively). Corresponding derived requirements should be added to system requirements baseline when they impact the SoI.

Calvo-Amodio and Rousseau (2019) explain how emergence applies to systems in which complexity is dominant. Complexity dominance, they say, encourages us to consider the significance of the difference between kinds of

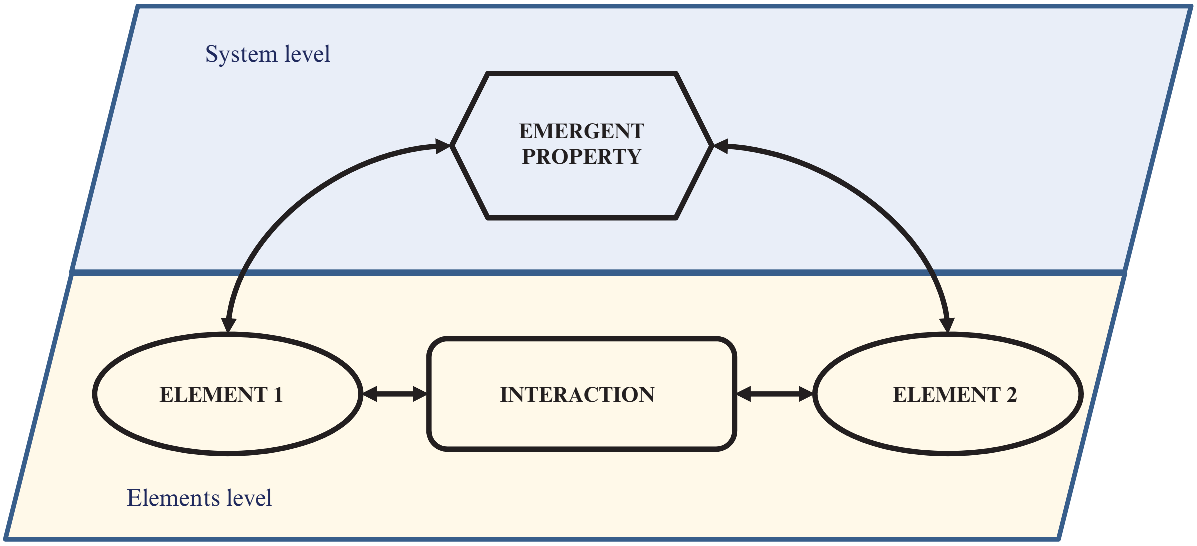

FIGURE 1.5 Emergence. INCOSE SEH original figure created by Jackson. Usage per the INCOSE Notices page. All other rights reserved.

complexity and degrees of complexity systems have. Doing so enables the SE practitioner to use variety engineering to manage complexity accordingly.

Figure 1.5 illustrates how the interaction between elements can result in emergent properties in any kind of system. This figure illustrates the basic rules of emergence. First, individual elements cannot exhibit higher-level system emergence. Second, two or more elements are required for emergence. Finally, emergence occurs at a level above the individual elements.

1.3.3 Interfacing Systems, Interoperating Systems, and Enabling Systems

External systems are systems beyond (or outside of) the SoI boundary. *Interfacing systems* are external systems that share an interface (e.g., physical, material, energy, data/information) with the SoI. Typically, humans also interface with the SoI throughout the SoI's life cycle stages. *Interoperating systems* are interfacing systems that interface with the SoI in its operational environment to perform a common function that supports the SoI's primary purpose. The set of SoI and interoperating systems can be seen as a system of systems (see Section 4.3.6). *Enabling systems* are external systems that facilitate the life cycle activities of the SoI but are not a direct element of the operational environment. The enabling systems provide services that are needed by the SoI during one or more life cycle stages. Some enabling systems share an interface with the SoI and some do not. Examples of enabling systems include collaboration development systems, production systems, and logistics support systems. Table 1.3 gives examples of these types of external systems.

During the life cycle stages for an SoI, it is necessary to concurrently consider interfacing, interoperating, and enabling systems along with the SoI. Otherwise, important requirements may not be identified, which will lead to significant costs in the further course of system development. Typical pitfalls include assuming that a new enabling system will come online in time to support the development of the SoI or that an existing enabling system will be available for the duration of the life cycle of the SoI. A delay in an enabling system coming online or the loss of an existing enabling system can lead to significant issues with the development and deployment of the SoI. In addition, horizontal and vertical integration considerations (see Section 2.3.5.8) may arise from the system context represented by interfacing, interoperating, and enabling systems.

TABLE 1.3 Examples for systems interacting with the SoI

SOI and External Systems	Interfacing System	Interoperating System	Enabling System
Aircraft			
Flight simulator	No	No	Yes
Fuel Truck	Yes	No	Yes
Remote Maintenance	Yes	Yes	Yes
Communication system	Yes	Yes	No
Runway	Yes	No	No
Automobile			
SE Tool	No	No	Yes
Car carrier	Yes	No	Yes
Diagnosis system	Yes	Yes	Yes
Parking assistant	Yes	Yes	No
Windshield snow cover	Yes	No	No

1.3.4 System Innovation Ecosystem

Sections 1.3.1 and 1.3.3 describe the system boundary and external systems in the overall context of the SoI. This section focuses on learning. Over single, and eventually multiple life cycles, engineered system innovation may be viewed as a form of group learning by "ecosystems" composed of individuals, teams, enterprises, supply chains, markets, and societies. Effective innovation requires effective learning and adaptation at a group level across these ecosystems and brings related challenges. To represent, plan, analyze, and improve such performance, the neutral descriptive System Innovation Ecosystem Pattern has been found to be useful (Schindel and Dove, 2016) (Schindel 2022b). Figure 1.6 provides a high-level view of that multiple-layered descriptive model, further discussed as a formal pattern in Section 3.2.6.

Figure 1.6 identifies three top-level system boundaries:

1. **System 1 – The Engineered System** may be a product developed for a market, a defense system created under contract, a service-providing system, or other system subject to SE life cycle management. It is shown in its larger environment, the Life Cycle Project Management System (System 2). System 1 examples include Medical Devices, Aircraft, Consumer Packaged Goods, and Gas Turbine Engines. This system is typically referred to as the engineered SoI in this handbook.
2. **System 2 – The Life Cycle Project Management System** provides the environment of System 1 over its life cycle, including the life cycle management processes responsible for System 1—described in Part II. System 2, a socio-technical system of people, processes, and facilities, is responsible to learn about System 1 and its environment, and to effectively apply that learning in the life cycle management by System 2. System 2 examples include System Requirements Definition Processes, Verification Processes, Product Manufacturing Processes, Product Distribution Processes, Product Sustainment Systems, Product Life Cycle Management (PLM) Information Systems, and Product Digital Twin Systems.
3. **System 3 – The Enterprise Process and Innovation System** contains System 2 and is responsible for learning about and improving System 2. In that sense, System 3 includes formal life cycle management for the processes of System 2. System 3 contains the "organizational change management" for advancing and adapting System 2 as a recognized formal system in its own right. System 3 examples include Product Life Cycle Management Processes, Program and Project Configuration and Tailoring Processes, Engineering Recruitment, Education, and Advancement Processes, Product Development Methodology Descriptions, Engineering Automation Tooling Acquisition and Development, Development Process Performance Analysis Systems, Regulatory Authorities, Engineering Professional Societies, and Engineering Facilities Construction and Acquisition.

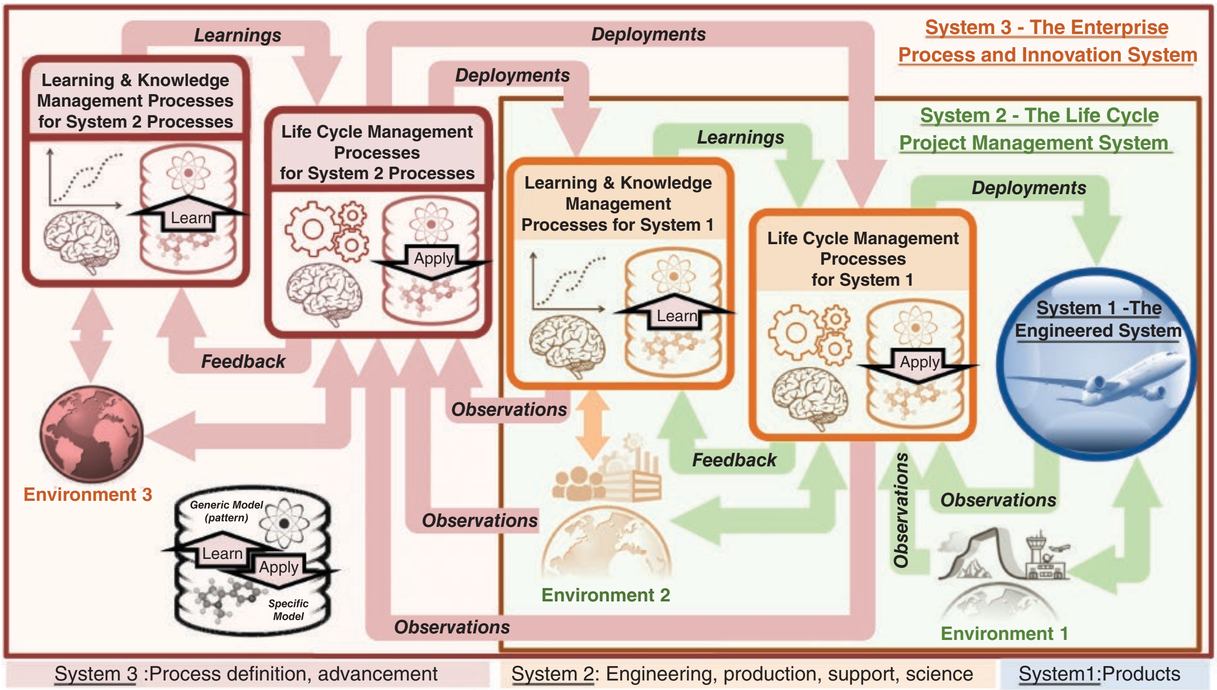

FIGURE 1.6 System innovation ecosystem pattern. From Schindel and Dove (2016) and Schindel (2022b). Used with permission.

The System Innovation Ecosystem Pattern emphasizes the learning and execution aspects of the enterprise ecosystem and directly integrates the SE life cycle processes described in Part II of this Handbook. Those processes are applied to two different managed SoIs (System 1 and System 2) and explicate the processes of learning versus application in each of the SE life cycle processes, along with how, and how effectively, execution is coupled with prior learning. The (configurable) System Innovation Ecosystem Pattern intentionally describes any engineering environment, whether effective in its learning and adaptation or not. It is intended as a descriptive, not prescriptive, reference model that can be used to plan and analyze any engineering and life cycle management ecosystem. So, while the "learned models" shown inside System 2 describe knowledge of System 1 (The Engineered System), the models shown inside System 3 describe knowledge of System 2 (The Life Cycle Project Management System).

The formal System Innovation Ecosystem Pattern includes the ability to be configured specific to a local enterprise, project, or supply chain, and for use to plan a series of migration increments representing advancing System 2 capabilities. For more details, refer to Section 3.2.6 and the INCOSE S*Patterns Primer (2022).

1.3.5 The Hierarchy within a System

As explained in Section 1.1, "A system is an arrangement of parts or elements." A *system element* is a member of a set of elements that constitute a system (ISO/IEC/IEEE 15288, 2023). A system element is a discrete part of a system that can be implemented to fulfil specified requirements. Hardware, software, data, humans, processes (e.g., processes for providing service to users), procedures (e.g., operator instructions), facilities, materials, and naturally occurring entities or any combination are examples of system elements.

In the ISO/IEC/IEEE 15288 (2023) usage of terminology, the system elements can be atomic (i.e., not further decomposed), or they can be systems on their own merit (i.e., decomposed into further subordinate system elements).

A system element that needs only a black box (also known as opaque box) representation (i.e., external view) to capture its requirements and confidently specify its real-world solution definition can be regarded as atomic. Decisions to make, buy, or reuse the element can be made with confidence without further specification of the element.

One of the challenges of system definition is to understand what level of detail is necessary to define each system element and the interrelations between elements. The integration of the system elements must establish the relationship between the effects that organizing the elements has on their interactions and how these effects enable the system to achieve its purpose. One approach to defining the elements of a system and their interrelations is to identify a complete set of distinct system elements with regard only to their relation to the whole (system) by suppressing details of their interactions and interrelations. These considerations lead to the concept of *hierarchy* within a system. This is referred to as a *partitioning* of the system and the end result is called a *Product Breakdown Structure* (PBS) (see Section 2.3.4.1). As stated above, each element of the PBS can be either atomic or it can be at a higher level that could be viewed as a system itself. At any given level, the elements are grouped into distinct subsets of elements subordinated to a higher-level system, as illustrated in Figure 1.7. Thus, hierarchy within a system is an organizational representation of system structure using a partitioning relation.

The art of defining a hierarchy within a system relies on the ability of the SE practitioner to strike a balance between clearly and simply defining span of control and resolving the structure of the SoI into a complete set of system elements that can be implemented with confidence. Urwick (1956) suggested a possible heuristic for span of control, recommending that decomposition of any object in a hierarchy be limited to no more than seven subordinate elements, plus or minus two (7 +/–2). Others have also found this heuristic to be useful in other contexts (Miller, 1956). A level of design with too few subordinate elements is unlikely to have a distinct design activity. In this case, both design and verification activities may contain redundancy. In case of too many subordinate elements, it may be difficult to manage all the interfaces between the subordinate elements. In practice, the nomenclature and depth of the hierarchy can and should be adjusted to fit the nature of the system and the community of interest.

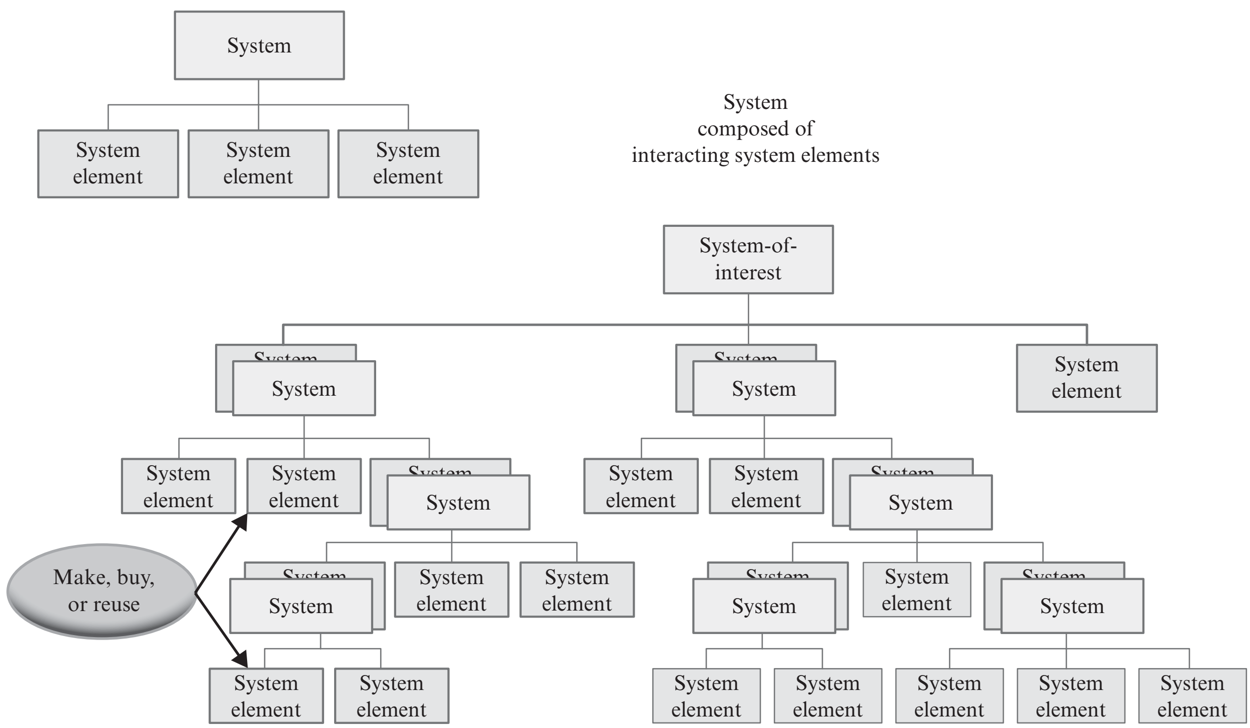

FIGURE 1.7 Hierarchy within a system. From ISO/IEC/IEEE 15288 (2023). Used with permission. All other rights reserved.

The interrelationships of system elements at a given architecture level of decomposition can be referred to as the horizontal view of the system. The horizontal view also includes requirements; integration, verification, or validation activities and results; various other related artifacts; and external elements. How the horizontal elements, activities, results, and artifacts are derived from or lead to higher-level systems and lower-level system element can be referred to as the vertical view of the system.

1.3.6 Systems States and Modes

States and modes are two related concepts that are used for defining and modeling system functional architectures and for modeling and managing system behaviors.

A *state* can be defined as:

> An observable and measurable … attribute used to characterize the current configuration, status, or performance-based condition of a System or Entity. (Wasson, 2016)

States are snapshots of a set of variables or measurements needed to describe fully the system's capabilities to perform the system's functions. *State variables* are the multidimensional list of variables that determine the state of the system. The list of variables does not change over time, but the values that these variables take do change over time (Buede and Miller, 2016). In control theory, the state of a dynamic system is a set of physical quantities, the specification of which (in Newtonian dynamics) completely determines the evolution of the system (Friedland, 2012). From the perspective of MBSE (see Section 4.2.1), "The state of the system is the most concise description of its past history."

The current system state and a sequence of subsequent inputs allow computation of the future states of the system. The state of a system contains all the information needed to calculate future responses without reference to the history of inputs and responses (Chapman, et al., 1992). Bonnet et al. (2017) states, "A state often directly reflects an operating condition or status on structural elements of the system (operational, failed, degraded, absent, etc.). States are also likely to represent the physical condition of a system element (full or empty fuel tank, charged or discharged battery, etc.). States can also be exploited to represent environment constraints (temperature, humidity, etc.)." If the system is transitioning from one state to another as time progresses, then time is one of the key attributes of the system. To monitor the system and manage it, the manager observes a state variable that is comprised of the appropriate collection of the system's attributes (Shafaat and Kenley, 2020).

A *mode* can be defined as:

> A distinct operating capability of the system during which some or all of the system's functions may be performed to a full or limited degree. (Buede and Miller, 2016)

For a personal computer, examples of modes are "off," "on," "waking up," "waiting," "reading from disk," "writing to disk," "computing," "printing," and, of course, "down" (Wymore, 1993). Modes are part of the system functional architecture and can be derived by affinity analysis of system use cases (Wasson, 2016). Various perspectives can be used to define the distinct operating modes of a system (Bonnet, et al., 2017), such as:

- the phases of mission operations (taxiing, taking-off, cruising, landing, etc.),
- the system operating conditions (connected, autonomous, etc.),
- the specific conditions in which the system is used (test, training, maintenance, etc.).

Transitioning from one mode to another is the result of decisions made by the system itself, its users, or external actors in order to adapt to new needs or new contexts (Bonnet, et al., 2017). Decisions that result in the system transitioning from one mode to another are typically based on the observed values of the state variables. When using models to depict system behavior, mode transitions are often based on triggering events that meet specified entry and exit criteria (Wasson, 2016).

1.3.7 Complexity

Systems engineering practitioners encounter a number of systems with simple, complicated, and complex characteristics. Many traditional systems engineering approaches and techniques work well for simple and complicated systems but do not handle complexity in systems (i.e., complex systems) well. Conversly, approaches and techniques that handle complexity well are also used in some complicated system contexts, especially when complex characteristics exist in some aspects of the system. Thus, care must be used to ensure the SE approaches and techniques for the SoI are appropriate and tailored for the type of system, especially with respect to its complexity. Complex systems are defined in the INCOSE publication "A Complexity Primer for Systems Engineers" (INCOSE Complexity Primer, 2021). A complex system has elements, the relationship between the states of which are weaved together so that they are not fully comprehended, leading to insufficient certainty between cause and effect. Complicated systems are less challenging. A complicated system has elements, the relationship between the states of which can be unfolded and comprehended, leading to sufficient certainty between cause and effect. Systems can also be simple. A simple system has elements, the relationship between the states of which, once observed, are readily comprehended. Complex systems can provide beneficial solutions yet also contain challenging characteristics. Complexity can result in positive behavior, such as self-organization and virtuous cycles of activity. However, intricate networks of evolving cause-and-effect relationships can lead to novel, nonlinear, and counterintuitive dynamics over time, resulting in suboptimal system operation, unintended consequences, and system obsolescence. The INCOSE Complexity Primer identifies 14 distinguishing characteristics that define complexity in a system. These characteristics provide insights into complexity, realizing that systems are not wholly complex: they are typically complex in some characteristics and complicated or even simple in others.

Traditional SE process for complicated systems takes a reductionist approach, whereby the problem is procedurally broken down into its parts (i.e., decomposition), solved, and reassembled to form the whole solution. This approach works well for complicated problems, where fixed, deterministic, or predictable patterns of behavior are required. However, these processes often do not perform well in complex environments, such as the challenges involved in designing autonomous vehicles or other socio-technical systems. A fundamentally different approach is required to understand the unexpected emergent interaction between the parts in the context of the whole through iterative exploration and adaptation (Snowden and Boone, 2007).

SE for complex systems requires a balance of linear, procedural methods for sorting through complicated and intricate tasks (e.g., systematic activity) and holistic, nonlinear, iterative methods for harnessing complexity (e.g., systems thinking). Complexity is not antithetical to simplicity, as even relatively simple systems can generate complex behavior. The INCOSE Complexity Primer provides guidance in the methods, approaches, and tools that may benefit complex systems engineering.

1.4 SYSTEMS ENGINEERING FOUNDATIONS

1.4.1 Uncertainty

There is uncertainty associated with much of the systems information and measurement data we use. This section provides a brief summary of the two major types of uncertainty, the sources of systems uncertainty, and decision making under uncertainty.

Types of Uncertainty. There are two types of uncertainties: epistemic and aleatory. In SE, *epistemic uncertainty* is due to our lack of knowledge about the potential demand for a new system and how a technology, system, or process will perform in the future, for example, the knowledge gap about key value attribute or about the acquirer's preferences. *Aleatory uncertainty* is uncertainty due to randomness. If a technology, system, or process can perform a function, there will be always some inherent randomness in every performance measurement. Our system requirements process, and development decisions focus on reducing epistemic uncertainty (overcoming our lack of knowledge), but we can never completely reduce aleatory uncertainty in our development or operational measurement of system performance.

TABLE 1.4 Sources of system uncertainty

Sources of Uncertainty	Major Questions	Potential Uncertainties
Business	Will political, economic, labor, social, technological, environmental, legal or, other factors adversely affect the business environment?	Changes in political viewpoint (e.g., elections) Economic disruptions (e.g., recession). Global disruptions (e.g., supply chain). Changes to laws and regulations. Disruptive technologies. Adverse publicity.
Market	Will there be a market if the product or service works?	User and consumer demand. Threats from competitors (quality and price) and adversaries (e.g., hackers and terrorists). Continuing stakeholder support.
Management	Does the organization have the people, processes, and culture to manage a major system?	Organization culture. SE and management experience and expertise. Mature baselining processes (technical, cost, schedule). Reliable cost estimating processes.
Performance (Technical)	Will the product or service meet the required desired performance?	Defining future requirements in dynamic environments. Understanding of the technical baseline. Technology maturity to meet performance. Adequate modeling, simulation, test, and evaluation capabilities to predict and evaluate performance. Availability of enabling systems needed to support use.
Schedule	Can the system that provides the product or service be delivered on time?	Concurrency in development. Impact of uncertain events on schedule. Time and budget to resolve technical and cost risks.
Development and Production Cost	Can the system be delivered within the budget? Will the cost be affordable?	Changes in missions. Technology maturity. Hardware and software development processes. Industrial/supply chain capabilities. Production facilities capabilities and processes.
Operations and Support Cost	Can the owner afford to operate and support the system? Will the cost be affordable?	Increasing operations and support (e.g., resource or environmental) costs. Resiliency of the design to new missions and tasks. Changes in maintenance or logistics strategy/needs.
Sustainability	Will the system provide sustainable future value?	Availability of future resources and impact on the natural environment.

INCOSE SEH original table created by Jackson and Parnell derived from Parnell (2016).

Sources of Uncertainty and Risk. There are many sources of epistemic uncertainty that impact SE in the system life cycle. Table 1.4 provides a partial list of some of the major uncertainties that confront project managers and SE practitioners and describes some of the implications for SE.

Decisions Under Uncertainty

As can be seen from Table 1.4, uncertainties impact every SE decision process. Taking decisions before having enough knowledge is potentially very risky. Key decisions that have a strong impact on the solution require reducing uncertainty by closing the knowledge gap to an appropriate level. However, SE practitioners must be able to make decisions under uncertainty and should record a corresponding risk with those decisions (see Sections 2.3.4.3 and 2.3.4.4).

1.4.2 Cognitive Bias

SE practitioners need to obtain information from stakeholders throughout the system life cycle. SE practitioners and stakeholders (individual or groups) are subject to cognitive biases when interpreting uncertain information. The best defense from cognitive biases is understanding what they are and how they can be avoided and setting up organizational projects to obtain unbiased assessments. Cognitive biases are mental errors in judgment under uncertainty caused by our simplified information processing strategies (sometimes called heuristics) and are consistent and predictable (Tversky and Kahneman, 1974). There are many lists of cognitive biases, including one that lists 50 sources (Hallman, 2022). Cognitive biases can affect both individual and teams of SE practitioners (McDermott, et al., 2020). Cognitive biases can contribute to incidents, failures, or disasters as a result of distorted decision making and can lead to undesirable outcomes. Cognitive biases are included in a field called Behavioral Decision-Making. Table 1.5 lists some of the most common cognitive biases.

For major systems decisions, more formal methods are required to avoid cognitive biases. Both Tversky and Kahneman (1974) and Thaler and Sunstein (2008) describe mitigation methods suitable to different environments. The most effective methods are external group methods. For example, NASA (2003) recommends the Independent Technical Authority (ITA) to warn decision makers of the potential for failure. The ITA must be both financially and organizationally independent of the project manager. Another method, adopted by the aviation industry, is called the Crew Resource Management (CRM) method. With the CRM method, all crew members, including the co-pilot, are responsible for warning the pilot of imminent danger.

1.4.3 Systems Engineering Principles

SE is a relatively young discipline. The emergence of a set of SE principles has occurred over the past 30 years within the discipline. In reviewing various published SE principles, a set of criteria emerged for SE principles. SE principles cover broad application within the practice; they are not constrained to a particular system type, to the system development or operational context, or to a particular life cycle stage. SE principles transcend these system characteristics and inform a worldview of the discipline. Thus, a SE principle:

- transcends a particular life cycle model or stage,
- transcends system types,
- transcends a system context,
- informs a world view on SE,
- is not a "how to" statement,
- is supported by literature or widely accepted by the community (i.e., has proven successful in practice across multiple organizations and multiple system types),
- is focused, concise, and clearly worded.

SE principles are a form of guidance proposition which provide guidance in application of the SE processes and a basis for the advancement of SE. SE has many kinds of guidance propositions that can be classified by their sources, e.g., heuristics (derived from practical experience as discussed in Section 1.4.4), conventions (derived from social agreements), values (derived from cultural perspectives), and models (based on theoretical mechanisms). Although these all support purposeful judgment or action in a context, they can vary greatly in scope, authority, and conferred capability. They can all be refined, and as they mature, they gain in their scope, authority, and capability, while the set becomes more compact. A key moment in their evolution occurs with gaining insight into why they work, at which point they become principles. Principles can have their origins associated in referring to them as "heuristic principles," "social principles," "cultural principles," and "scientific principles," although in practice it is usually sufficient to just refer to them as SE principles. SE principles are derived from principles of these various origins providing a diverse set of transcendent principles based on both practice and theory.

TABLE 1.5 Common cognitive biases

Cognitive Bias	Description	Implication for the SE Practitioner.
Framing	How we ask the question or describe the decision matters.	Carefully word questions and problem description to avoid influencing the response.
Representativeness	People draw conclusions based on representative characteristics and often ignore relevant facts or the base rates.	Discuss the relevant facts and data before requesting a judgment about an uncertainty or risk. Use Bayes Law to update our beliefs after we receive new data. Teams that reflect Diversity, Equity, and Inclusion principles can help reduce the bias for the team (see Section 5.2).
Availability	We place too much weight on vivid, striking, and recent events.	Ask about the relevant facts and data before requesting a judgment about an uncertainty or risk. Design systems to provide the relevant data.
Anchoring	The initial estimate affects the final estimates.	Never begin by asking about the expected outcome. Instead obtain information about the worst or best outcomes first to understand the range of outcomes.
Motivational	When making probability judgments, people have incentives to provide estimates that will benefit themselves	Understand the potential bias of an individual providing an assessment. For example, a technology developer has an incentive to overestimate technology readiness if a more conservative estimate could result in loss of funding.
Optimism	We overestimate the likelihood of good outcomes and underestimate the likelihood bad outcomes.	Seek data on similar bad outcomes. Obtain assessments from experts not involved in the decision.
Confirmation	We seek or put more weight on data that confirms our beliefs.	Actively seek data that would disprove our current belief in all tests and evaluations.
Group Think	A group of people make irrational or unsound decisions to suppress dissent and maintain group harmony.	Seek dissenting opinions inside the group and seek outside assessments.
Authority	We trust and are more often influenced by the opinions of people in positions of authority	Assess the opinion independent of the source.
Rankism	Assumption that person of higher rank is always correct in decisions	Seek to determine correct decision

In addition, SE principles differ from systems principles in important ways (Watson, et al., 2019). System principles address the behavior and properties of all kinds of systems, looking at the scientific basis for a system and characterizing this basis in a system context via specialized instances of a general set of system principles. SE principles build on systems principles that are general for all kinds of systems (Rousseau, 2018) (Watson, 2020) and for all kinds of human activity systems (Senge, 1990) (Calvo-Amodio and Rousseau, 2019).

INCOSE compiled an early list of principles consisting of 8 principles and 61 subprinciples in 1993 (Defoe, 1993). These early principles were important considerations recognized in practice for the success of system developments and ultimately became the basis for the SE processes. These early principles were focused on particular aspects of the

SE process and particular life cycle stages. The INCOSE work on SE principles considered these earlier sources and compiled a set of SE principles that are transcendent. The INCOSE SE Principles (2022) documents each SE principle with a description, evidence that supports the principle (e.g., observable evidence of the application, proof from scientific evidence), and implications in SE practice for application of the principle. There are presently 15 SE principles and 20 subprinciples as shown in Table 1.6.

TABLE 1.6 SE principles and subprinciples

1	SE in application is specific to stakeholder needs, solution space, resulting system solution(s), and context throughout the system life cycle.
2	SE has a holistic system view that includes the system elements and the interactions amongst themselves, the enabling systems, and the system environment.
3	SE influences and is influenced by internal and external resources, and political, economic, social, technological, environmental, and legal factors.
4	Both policy and law must be properly understood to not over-constrain or under-constrain the system implementation.
5	The real system is the perfect representation of the system.
6	A focus of SE is a progressively deeper understanding of the interactions, sensitivities, and behaviors of the system, stakeholder needs, and its operational environment.
	Sub-Principle 6(a): Mission context is defined based on the understanding of the stakeholder needs and constraints
	Sub-Principle 6(b): Requirements and models reflect the understanding of the system
	Sub-Principle 6(c): Requirements are specific, agreed to preferences within the developing organization
	Sub-Principle 6(d): Requirements and system design are progressively elaborated as the development progresses
	Sub-Principle 6(e): Modeling of systems must account for system interactions and couplings
	Sub-Principle 6(f): SE achieves an understanding of all the system functions and interactions in the operational environment
	Sub-Principle 6(g): SE achieves an understanding of the system's value to the system stakeholders
	Sub-Principle 6(h): Understanding of the system degrades during operations if system understanding is not maintained.
7	Stakeholder needs can change and must be accounted for over the system life cycle.
8	SE addresses stakeholder needs, taking into consideration budget, schedule, and technical needs, along with other expectations and constraints.
	Sub-Principle 8(a): SE seeks a best balance of functions and interactions within the system budget, schedule, technical, and other expectations and constraints.
9	SE decisions are made under uncertainty accounting for risk.
10	Decision quality depends on knowledge of the system, enabling system(s), and interoperating system(s) present in the decision making process.
11	SE spans the entire system life cycle.
	Sub-Principle 11(a): SE obtains an understanding of the system
	Sub-Principle 11(b): SE defines the mission context (system application)
	Sub-Principle 11(c): SE models the system
	Sub-Principle 11(d): SE designs and analyzes the system
	Sub-Principle 11(e): SE tests the system
	Sub-Principle 11(f): SE supports the production of the system
	Sub-Principle 11(g): SE supports operations, maintenance, and retirement
12	Complex systems are engineered by complex organizations.
13	SE integrates engineering and scientific disciplines in an effective manner.
14	SE is responsible for managing the discipline interactions within the organization.
15	SE is based on a middle range set of theories.
	Sub-Principle 15 (a): SE has a systems theory basis
	Sub-Principle 15 (b): SE has a physical logical basis specific to the system
	Sub-Principle 15 (c): SE has a mathematical basis
	Sub-Principle 15 (d): SE has a sociological basis specific to the organization

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These principles provide a start in defining a transcendent disciplinary basis for SE. Application of the principles aids in determining a system life cycle model, implementing SE processes, and defining organizational constructs to help the SE practitioner successfully develop and sustain the SoI.

1.4.4 Systems Engineering Heuristics

Summary Heuristics provide a way for an established profession to pass on its accumulated wisdom. This allows practitioners to gain insights from what has been found to work well in the past, and apply the lessons learned. Heuristics usually take the form of short expressions in natural language. These can be memorable phrases encapsulating shortcuts, "rules of thumb," or "words of the wise," giving general guidelines on professional conduct or rules, advice, or guidelines on how to act under specific circumstances. Heuristics usually do not express all there is to know, yet they can act as a useful entry point for learning more. At their best, heuristics can act as aids to decision making, value judgments, and assessments.

Interest in SE heuristics currently centers on their use in two contexts: (1) encapsulating engineering knowledge in an accessible form, where the underlying practice is widely accepted and the underlying science understood, and (2) overcoming the limitations of more analytical approaches, where the science is still of limited use. This is especially applicable as we extend the practice of SE to providing solutions to inherently complex, unbounded, ill-structured, or very difficult problems.

Background Engineering first emerged as a series of skills acquired while transforming the ancient world, principally through buildings, cities, infrastructure, and machines of war. Since then, mankind has sought to capture the knowledge of "how to" to allow each generation to learn from its predecessors, enabling more complex structures to be built with increasing confidence while avoiding repeated real-world failures. For example, early cathedral builders encapsulated their knowledge in a small number of "rules of thumb," such as "maintain a low center of gravity" and "put 80% of the mass in the pillars." Designs were conservative, with large margins. When the design margins were exceeded (e.g., out of a desire to build higher and more impressive structures), a high price was sometimes paid, with the collapse of a roof, a tower, or even a whole building. From such failures, new empirical rules emerged. Much of this took place before the science behind the strength of materials or building secure foundations was understood. Only in recent times have computer simulations revealed the contribution toward certain failures played by such dynamic effects as wind shear on tall structures.

Since then, engineering and applied sciences have co-evolved: with science providing the ability to predict and explain performance of engineered artefacts with greater assurance and engineering developing new and more complex systems, requiring new scientific explanations and driving research agendas. In the modern era, complex and adaptive systems are being built which challenge conventional engineering sciences, and we are turning to social and behavioral sciences, management sciences, and increasingly systems science to deal with some of the new forms of complexity involved and guide the profession accordingly.

Current Use Renewed interest in the application of heuristics to the field of SE stems from the seminal work of Maier and Rechtin (2009), and their book remains the best single published source of such knowledge. Their motivation was to provide guidance for the emerging role of system architect as the person (or team) responsible for coordinating engineering effort toward devising solutions to complex problems and overseeing their implementations. They observed that it was in many cases better to apply heuristics than attempt detailed analysis. The reason for this is the number of variables involved and the complexity of the interactions between stakeholders, internal dynamics of system solutions, and the organizations responsible for their realization. Some examples of SE heuristics are:

- ***Don't assume that the original statement of the problem is necessarily the best, or even the right one.*** This has to be handled with tact and respect for the user, but experience shows that failure to reach mutual understanding early on is a fundamental cause of failure, and strong relationships forged in the course of doing such coordination with stakeholders can pay off when solving more difficult issues which might arise later on.

- ***In the early stages of a project, unknowns are a bigger issue than known problems.*** Sometimes developing a clear understanding of the environment, all of the stakeholders, and the ramifications of possible solutions uncovers many unanticipated issues.
- ***Model before build, wherever possible.*** System Science postulates "The only complete model of the system in its environment is the system in its environment," which leads into using evolutionary life cycles, rapid deployment of prototypes, agile life cycles, and so on. This heuristic opens a door into twenty-first-century systems.

A repository of heuristics can act as a knowledge base, especially if media (such as video clips or training materials) or even interactive media (to encourage discussion and feedback) are included. A heuristics repository should link to other established knowledge sources and be tagged with other metadata to allow flexible retrieval. It should be organized to reflect accepted areas of SE competency and allow users to assemble a personal set of heuristics most meaningful to them, being relevant to their professional or personal sphere of activity.

1.5 SYSTEM SCIENCE AND SYSTEMS THINKING

This section considers the nature and relationship between systems science and systems thinking and describes how they relate to SE.

Relationship between Systems Science, Systems Thinking, and SE

The association of concepts such as system, boundary, relationships, environment/context, hierarchy, emergence, communication, and control, among others, when interrelated with purpose, gives rise to a *systems worldview* (Rousseau, et al., 2018). Interrelating concepts with purpose changes how we investigate and reason about things, producing *systems thinking*. Systems thinking enables us to recognize systems patterns across different phenomena, problem contexts, and disciplines. Studying these patterns has produced the systems sciences of General System Theory, Cybernetics, and Complexity Theory and their related systems methodologies, models, and methods. The application of systems thinking and systems science concepts, principles, methodologies, models, and methods in engineering is one of the bases for the practice of SE. Applying SE, and reflecting on the results, help us improve systems science and systems thinking, further enhancing our ability to design and intervene in complex systems—a virtuous cycle. Through this virtuous cycle, we develop principles to better our SE applications (Rousseau, et al., 2022).

Figure 1.8 depicts this virtuous cycle as a multifaceted and purposeful activity to deliver elegant solutions to complex problems, supported by principles that guide why, what, and how we do SE. To connect our purpose to our actions, we adopt a systemic approach, because complexity and elegance are both systems phenomena. Our systemic approach is of course guided by our systems principles. The kinds and relationships of principles, as well as how they inform and are informed by SE practice, is depicted in Figure 1.8. We select and organize these based on our intentions as expressed by our motivational principles. We use our transdisciplinary principles to select and organize our technique principles. In this way, the systemic relationships between our principles support how our principles guide the systemic relationships between our purpose, approach, and practice. The systemic roles our principles play in our discipline thus support the systematic evolution of our value in society.

The success of SE applications reinforces the credibility of the systems worldview which in turn enhances the SE practitioner's ability to conceptualize why a solution is needed, how a solution can be conceptualized, and what tools and/or methods to use to solve complex problems and achieve elegant solutions.

Systems Science

Questions about the nature of systems, organization, and complexity are not specific to the modern age. As Warfield (2006) put it:

> Virtually every important concept that backs up the key ideas emergent in systems literature is found in ancient literature and in the centuries that follow.

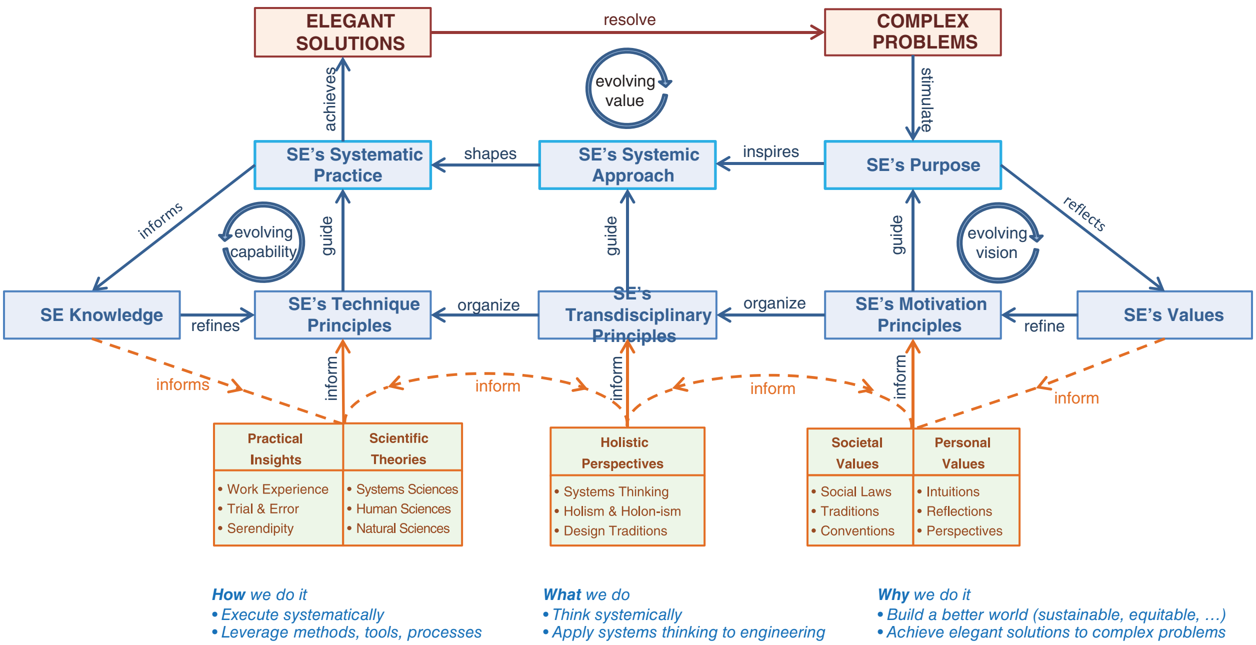

FIGURE 1.8 An architectural framework for the evolving the SE discipline. From Rousseau, et al. (2022). Used with permission.

Systems science can be defined as a transdisciplinary approach interested in understanding all aspects of systems with the goals of (1) identifying, exploring, and understanding patterns of behavior crossing disciplinary fields and areas of application, and (2) establishing a general theory applicable to all types of systems whether physical, natural, engineered, or social. Attempts to establish a systems science have taken on both reductionist and holistic forms, and both are valuable. For instance, a clock is a system, but its workings can be explained through reductionism. On the other hand, a holistic approach helps us understand why we need clocks, how clocks exist (operate/sustain/degrade) in their environment throughout their life cycle, and how the esthetics of their design evolve over time. Both the reductionist and the holistic approaches to explanation involve systemic arguments but each starts from different directions—bottom-up and outside-in. Complexity Theory has had some success developing a science of systems using a reductionist approach. Agent-based modeling, pioneered at the Santa Fe Institute, works from the "bottom-up" and seeks to explain the behavior of whole systems in terms of the rules of interaction of the "agents" that constitute the system.

Where reductionist (traditional) methods prove unsuccessful, systems science relies on the holistic approach. A holistic approach is adept at connecting and contextualizing systems, system elements, and their environments to understand difficult to explain patterns of organized complexity. This was the approach taken by Ludwig von Bertalanffy in developing General System Theory and Norbert Wiener in developing Cybernetics. The biologist von Bertalanffy was one of the first to argue for a general science of systems. He explained the scientific need for systems-based research as an alternative to traditional analytical procedures in science. This alternative method would overcome the limitations that result from explaining a system by breaking it down to its constituent parts and then being reconstituted from its parts, either materially or conceptually:

> This is the basic principle of *classical* science, which can be circumscribed in different ways: resolution into isolable causal trains or seeking for *atomic* units in the various fields of science, etc. (von Bertalanffy, 1969)

This makes it impossible to account for the emergent properties that systems display as a result of the interrelationships between their parts (see Section 1.3.2). Instead, von Bertalanffy promoted an alternative worldview concerned with the laws that apply to systems behavior in general. Such a General System Theory was possible, von Bertalanffy thought, and would be particularly valuable, because of the large number of parallelisms that appear across systems independent of the types and quantities of system elements in the systems:

> Thus, there exist models, principles, and laws that apply to generalized systems or their subclasses, irrespective of their particular kind, the nature of their component elements, and the relations or 'forces' between them. It seems legitimate to ask for a theory, not of systems of a more or less special kind, but of universal principles applying to systems in general. In this way we postulate a new discipline called General System Theory. (von Bertalanffy, 1971)

The study of general systems was to focus on such principles as:

> growth, regulation, hierarchical order, equifinality, progressive differentiation, progressive mechanization, progressive centralization, closed and open systems, competition, evolution toward higher organization, teleology, and goal-directedness. (Hammond, 2003)

The systems sciences, including General Systems Theory, Cybernetics, and Complexity Theory, seek to provide key foundational concepts to build a common language and intellectual foundations to make rigorous systems theories and tools accessible to practitioners. Where they succeed, they can serve as the foundation for a meta-discipline such as SE, transdisciplinary in nature, that unifies scientific and engineering practices. SE, informed by systems science, would be in a powerful position to enhance its theory and practice in ways that would make it applicable to the most complex of systems.

Finally, identifying SE's principles and heuristics can offer a useful approach to categorize systems-related knowledge and to focus research efforts. Systems principles and heuristics are special cases of guiding propositions. A guiding proposition provides guidance for purposeful judgment or action in a context and offers a wider perspective to that of a principle or a heuristic. Guiding propositions vary in (1) scope—the range of SE contexts they work, (2) authority—how compelling they are, and (3) capability—how predictable the outcomes of applying them are (Rousseau, et al., 2022). Readers can consult details on SE Principles in Section 1.4.3 and details on SE Heuristics in Section 1.4.4.

Systems Thinking Divergences

Systems thinking is a key enabler of SE. It is one of the core competencies defined in INCOSE SE Competency Framework (2018). Systems thinking applies the properties, concepts, and principles of systems to the given situation as a framework for curiosity—to get insight and understanding about the situation.

There needs to be a balance between the being systematic with the application of SE processes (as described in Part II) and being systemic, applying systems thinking to drive these processes. As SE practitioners, it is vital to possess the knowledge and skills necessary to perform holistic analysis and guide systemic intervention. Systems thinking lacks a unified definition; however, the following captures the nature of systems thinking and some key ideas:

> Systems thinking is a field characterized by a baffling array of methods and approaches. We posit that underlying all, however, are four universal rules called DSRP (distinctions, systems, relationships, and perspectives). We make distinctions between and among things and ideas, each implying the existence of another. We identify systems, which are composed of parts and wholes. We recognize relationships composed of actions and reactions. We take perspectives consisting of a point (from which we see) and a view (that which is seen). (Cabrera, et al., 2015)

This definition incorporates aspects of complex problem situations, such as "distinctions" and "perspectives," which it is essential to take account of, but which systems science may never be able to incorporate into its scientific models.

Based on the pioneering work of Ludwig von Bertalanffy in General System Theory, Norbert Wiener in Cybernetics, Jay Forrester in System Dynamics, Peter Checkland in Soft Systems Thinking, and others, a variety of systems methodologies, models, and methods have been formalized to perform systemic analyses and interventions. The SE practitioner can take advantage of this diversity providing they are aware of what the different methodologies, models, and methods do well, and what they are less good at. To assist systems thinking practitioners in selecting the most appropriate systems approaches, Jackson and Keys (1984) offered an initial classification of systems methodologies, the Systems of Systems Methodologies (SOSM), according to their strengths in addressing the complexity of systems and in reconciling divergences among stakeholder viewpoints. Jackson (2019) has since updated the SOSM, to reflect developments in Complexity Theory, by incorporating lessons from the Cynefin framework (Kurtz and Snowden, 2003). This use of different systems approaches in informed combinations, according to their strengths and weaknesses and the nature of the problem situation, is called Critical Systems Thinking (CST) (Jackson, 2003, 2019). CST is a multi-perspectival, multi-methodological, and multi-method approach.

While most of the prominent systems thinking approaches are rooted and/or contextualized within the management sciences, these approaches apply equally to SE practice. This is because the problems faced by SE practitioners, such as the need to incorporate cultural, social, political, and project management perspectives into systems models and other SE tasks, are common to the management sciences.

According to Jackson (2019), systems methodologies translate hypotheses about the nature of problem situations, and how they can be improved, into practical action. There are a number of systems methodologies available, for example, system dynamics, the viable system model, soft systems methodology, and critical systems heuristics. Each is based upon different assumptions about the world and how best to intervene in it. Together, these methodologies can recognize and respond to the range of issues encountered during the exploration of complex problem situations. These systems approaches can then be used, individually or in combination, in the problem situation. When the systems approaches are used in combination, the weighting of each system approach in the hybrid solution will be tailored based on the technical, organizational, cultural, and political factors within the organization and the relative dominance of those factors. According to systems thinkers, if SE can embrace the full range of systems methodologies, models, and methods, it will be in a much better position to tackle the hyper-complexity plaguing projects, organizations, and society in the contemporary world.

2

SYSTEM LIFE CYCLE CONCEPTS, MODELS, AND PROCESSES

2.1 LIFE CYCLE TERMS AND CONCEPTS

The overall purpose of Systems Engineering (SE) is to enable successful realization of the system while optimizing among competing stakeholder objectives. One way in which realization is managed is by breaking the overall effort into transformational steps, or stages, then checking for satisfactory fulfillment of system characteristics at the end of each stage, as well as checking whether risk is acceptable and the system is ready to enter other stages. Stages do not need to be executed sequentially or singularly. They can be executed multiple times as needed, and often in parallel. The critical feature of this approach is that progress is gated by specific decision points, generally called decision gates. By analogy with the stages that living things go through, called a life cycle, the set of stages for a system is termed a system life cycle. In summary, engineered systems progress in some manner through a set of stages, conceptually forming a system life cycle, with decision gates determining the completion of one stage and start of another. This part of the SE Handbook gives details for each of these parts of the system life cycle concept, as well as pointing out the role of the SE practitioner throughout a system's life cycle. Further details can be found in ISO/IEC/IEEE 24748–1 (2018).

2.1.1 Life Cycle Characteristics

As the introduction states, life cycles are defined in terms of the stages that mark progress in achieving the system characteristics. A commonly encountered set of life cycle stages is shown in Figure 2.1. These stages are also shown in ISO/IEC/IEEE 15288 (2023) and in ISO/IEC/IEEE 24748–1 (2018).

System life cycle stages can be entered based on the needs of the SoI or any system element. Stages can be entered into as many times as needed. Stages often are not sequential and can occur concurrently or as needed. Stages can overlap and stages can be entered at any point in the life cycle. The retirement stage does not require the entire SoI to be retired, it can be any system element, and retirement does not need to be in the order the systems are delivered.

INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities, Fifth Edition.
Edited by David D. Walden, Thomas M. Shortell, Garry J. Roedler, Bernardo A. Delicado, Odile Mornas, Yip Yew-Seng, and David Endler.

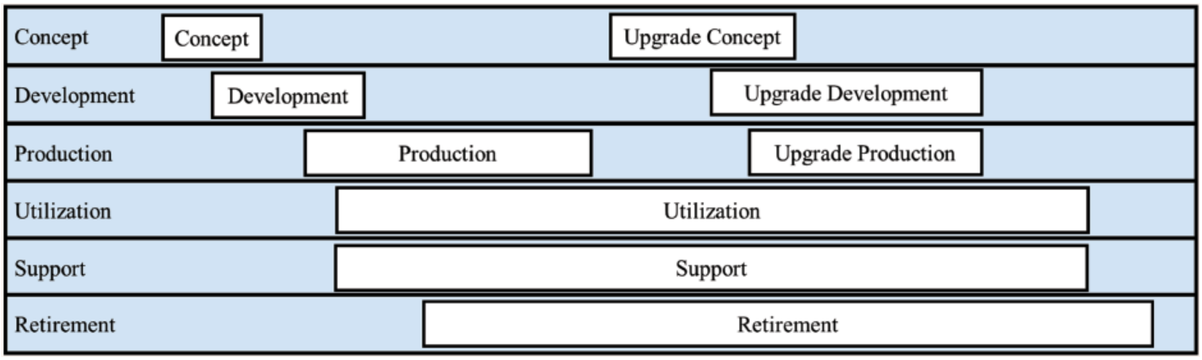

FIGURE 2.1 System life cycle stages. INCOSE SEH original figure created by Yokell. Usage per the INCOSE Notices page. All other rights reserved.

Typically, life cycle stages have both entry and exit decision gates. The entry decision gate is intended to help ensure that the entry criteria are met and the resources needed for the stage are available. The exit decision gate is intended to help ensure that the objectives of the stage have been achieved and the risk of going forward is acceptable. Decision gates are discussed in more detail in Section 2.1.3.

Figure 2.2 compares the generic life cycle stages to other life cycle viewpoints. Typical decision gates are represented along the bottom.

Major system elements may have their own life cycles. These life cycles have to be managed so that an integrated SoI is achieved and used over a span of time. When the SoI is, or is part of, an SoS (see Section 4.3.6), the influences from the evolution of the SoS need to be considered in the life cycle of the SoI. Each constituent system of the SoS has its own life cycle. Further, enabling systems (see Section 1.3.3) also have their own life cycles, which must be integrated with that of the SoI.

Requirements must be flowed down to the elements to be integrated and the decision gates should support progressive integration into the final SoI in a timely manner to help ensure that the elements can be progressively integrated. The decision gates associated with the various life cycle models should be synchronized, whatever the types of system element or parts of the life cycle are involved to support progressive integration into the final SoI.

Note that the above figures are notional and do not attempt to scale the relative time spans of the stages. For example, a system could move from initial concept to a fielded system in a few years, then remain in utilization, being supported and possibly upgraded, for decades (e.g., jet aircraft, nuclear power facility, day care nursery). A different system could have a series of development efforts, each resulting in relatively short periods of utilization and retirement (e.g., mobile phone, consumer electronics). While that is of interest from a programmatic viewpoint, it is secondary to the rationale of breaking the life cycle into stages to allow decisions to be made at key points.

2.1.2 Typical Life Cycle Stages

As shown in Figure 2.1, a system progresses through various life cycle stages that span the conception, development, production, utilization, support, and retirement of the SoI. This section highlights specific characteristics of each life cycle stage. Note that other life cycle models use different names for the stages and the associated characteristics of the stage. For other types of stages, such as those illustrated in Figure 2.2, the discussion here needs to be adapted as appropriate. Additional discussion of life cycle stage characteristics is in ISO/IEC/IEEE 24748–1 (2018).

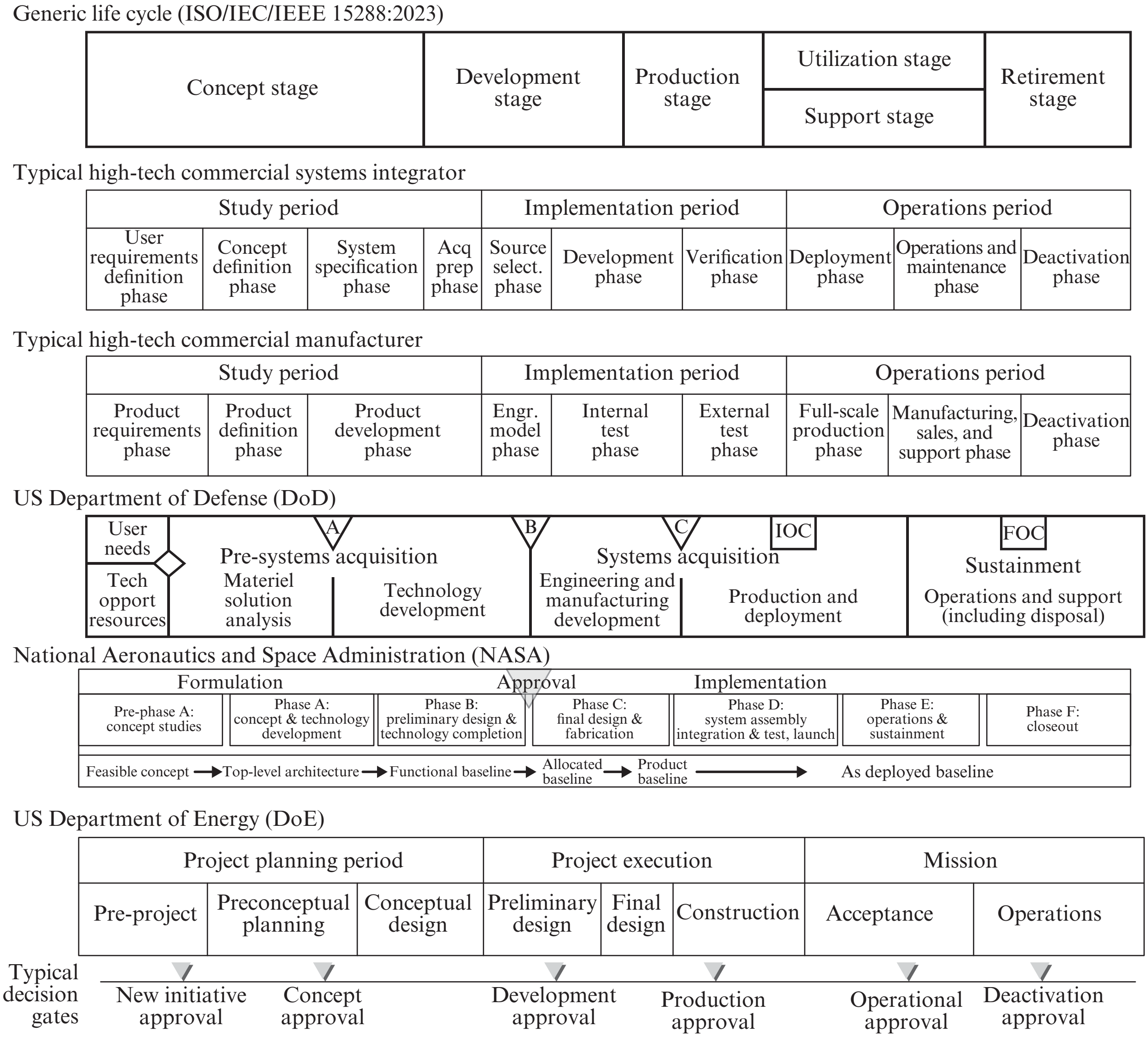

FIGURE 2.2 Generic life cycle stages compared to other life cycle viewpoints. Derived from Forsberg, et al. (2005) with permission from John Wiley & Sons. All other rights reserved.

Concept Stage The concept stage can include exploratory research and begins with recognition of a need for a new or modified mission or business capability. Unless the solution is immediately at hand, which is the first thing to analyze, new potential solutions will need to be sought. Exploration needs to address both short- and long-range factors, including technical, economic, market, and resource considerations, including human resources. Surveys, trade-off studies, business or mission analyses, and other means of exploring the solution space are used. It is key that the problem space is clearly defined (existing issue or new opportunity), the solution space is characterized, business or mission requirements, and stakeholder needs and requirements are identified. From this, an estimate of the cost, schedule, and performance across the life cycle can be derived. Throughout the concept stage, it is critical to perform ongoing and robust assessment and management of risks. Getting feedback from current and potential stakeholders (e.g., customers, users, suppliers) significantly aids in developing solution concepts. The maturity and availability of enabling systems over the system life cycle must also be considered.

Typical outputs from the concept stage include preliminary concept artifacts (e.g., Operational Concept (OpsCon), Support Concept), SE methodology approach considerations, feasibility assessments (e.g., models, simulations, prototypes), preliminary architecture solutions, and stakeholder requirements. The concept stage could create key preliminary system requirements and could outline design solutions and acquisition strategies. Enabling systems are also addressed, as are first estimates of cost and schedule over the whole life cycle.

The concept stage is a particularly critical part of the system life cycle because the decisions made during the stage will shape, with increasing difficulty to change, the possibilities for all the remaining stages. It is difficult to project the possibilities for as-yet untried solutions, though these may provide the greatest long-term benefit. At the same time, it is easy to fall into the trap of projecting incremental changes to what has worked in the past and is used now, which can significantly limit the future possibilities.

Development Stage The development stage defines an SoI that meets its agreed-to stakeholder needs and requirements and can be produced, utilized, supported, and retired. System analyses, including trade-off analysis, as well as further modeling, simulation, and prototyping are performed to achieve system balance and to optimize the design for key parameters.

The main aspect of the development stage is to mature the system concepts and stakeholder needs and requirements into an engineering baseline that can be produced, utilized, and supported over the desired span of its useful life, and finally retired in a responsible manner. The goal is not perfection, but rather to adequately meet the stakeholder needs and requirements in a manner that is supportable. The engineering baseline includes system requirements, architecture, design, documentation, and plans for subsequent stages. Outputs can include an SoI prototype, enabling system requirements (or the enabling systems themselves), plans for integration, verification, validation, transition, acquisition, logistics support, risk management, staffing and training, and detailed cost estimates and schedules for future stages. These outputs can occur incrementally, supporting a phased realization of the SoI, especially for complex systems.

Production Stage The production stage begins with approval to translate the baselines of the development stage into an actual system, or those parts of the SoI where approval is given (which is not uncommon for a complex system). The approval includes the enabling systems and must address all areas of the baseline. In this stage, the SoI becomes reality, is qualified for use, and is ready for installation and transition under the utilization stage. The outputs of this stage are the realized portions of the SoI (with its enabling systems) as well as the documentation that will go forward for use in the utilization, support, and retirement stages.

Utilization Stage The utilization stage begins with the transition to use of a system, or the parts of a system approved for use. This includes any enabling systems that will support use of the system being used in its intended environment to provide its intended capabilities. Product modifications are often introduced throughout the utilization stage, which generally is much longer than the other stages. Such changes can remedy deficiencies, enhance the capabilities, or extend the life of the system. Throughout, it is critical to maintain documentation from prior stages, as well as to ensure that Technical Management Processes, such as Configuration Management and Risk Management, and SE support remain in place and are robustly applied. The utilization stage proceeds in parallel with the support stage and ends, possibly by steps for different parts of the SoI, with the retirement stage.

Support Stage The support stage begins with provisioning of support for the SoI's utilization. Planning and acquisition actions for the system support are often taken before utilization is allowed to start. In this stage, deficiencies and failures are noted and used as the basis for either remediation of the problems, or to build a case for evolutionary modification. Modifications may be proposed to resolve supportability problems, to reduce operational costs, or to extend the life of a system. These changes require SE assessments to avoid loss of system capabilities while under operation, or violation of non-performance related requirements. The support stage ends when a decision is made that the system is at the end of its useful life or that it should no longer be supported.

Retirement Stage The retirement stage is where the system or a system element and its related services are removed from operation. SE activities in this stage are primarily focused on ensuring that disposal requirements, which can be extensive, are satisfied. However, it is often of value to ensure that documentation generated during at least the utilization and support stages is archived. That information can be invaluable when belated recognition arises that there is a need for new system.

Planning for retirement is part of the system definition during the concept and development stages. Experience has repeatedly demonstrated the consequences when system retirement is not considered from the outset. Early in the twenty-first century, many countries have changed their laws to hold the developer of a SoI accountable for proper end-of-life disposal of the system.

2.1.3 Decision Gates

It is good practice to have risk-managing decision points that occur at the beginning and end of each stage. This approach ensures that progress is gated by specific decision points that are clearly visible. These decision points help ensure the readiness to proceed with a stage and that the stage accomplishes is objective as it finishes. They often take place within the context of "project milestones," "project reviews," or "milestone reviews." Key is to help ensure that decisions are clearly made and documented and that they relate directly to the criteria established to begin or end a particular stage of a system's life cycle. Note that some approaches, such as agile (see Section 4.2.2), accomplish their decision points in a different cadence and tend to avoid the terms "milestones" and "decision gates." In agile development, frequent interaction with stakeholders can change the frequency (more frequent) and scope (smaller scope), and formality (less formal) of decision gates.

Typical goals of decision gates are to confirm that:

- increase in system maturity is within the defined threshold;
- the project deliverables satisfy the business case;
- the resources are sufficient to for the stage and subsequent stages;
- unresolved issues that need to be addressed in that stage are addressed; and
- overall risk for proceeding forward in the system life cycle is acceptable.

As shown in Figure 2.3, decision criteria can also include stage entry/exit criteria, entry/exit criteria from other stages, and risk assessment. Figure 2.3 shows the following cases:

- the entry criteria are met, but the start of the stage is delayed;
- when the entry criteria are met, the decision to start the stage is made;
- although the entry criteria are not met, the stage is started;
- although the exit criteria are met, the decision to end the stage is delayed;
- when the exit criteria are met, the decision to end the stage is made;
- the decision to end the stage is made before the exit criteria are met.

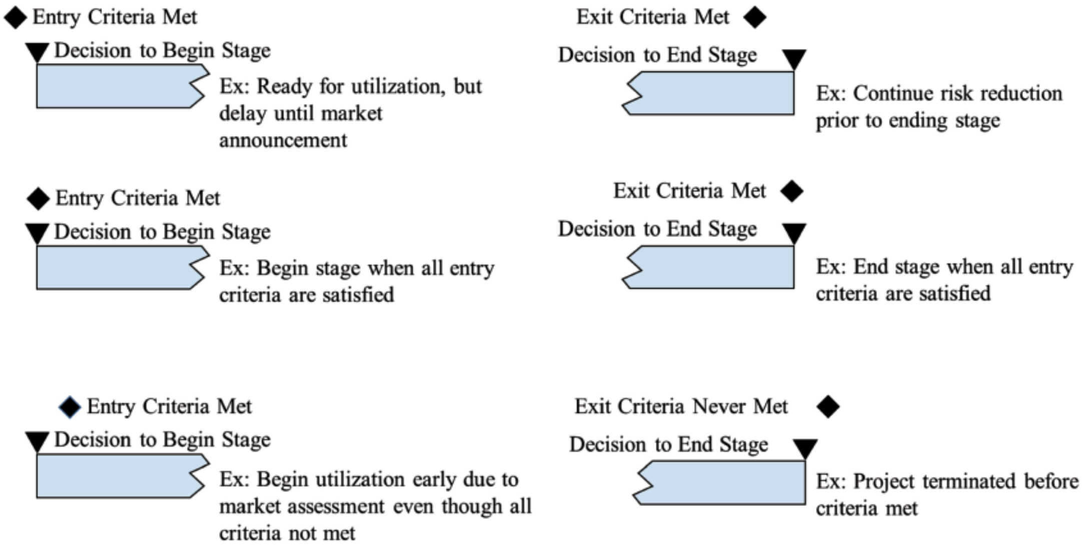

FIGURE 2.3 Criteria for decision gates. INCOSE SEH original figure created by Yokell. Usage per the INCOSE Notices page. All other rights reserved.

At each decision gate, the options can include:

- Begin subsequent stage or stages;
- Continue this stage, possibly after some reformulation;
- Go to or restart a preceding stage;
- Hold the project activity;
- Terminate the project.

The option selected depends on the quality of the results of the effort so far (based on the answers to the exit questions) plus the risk of moving forward (based on the answers to the entrance questions). Stages do not need to be sequential. Transitions often occur, but it is common to have stages occurring concurrently. In complex systems, the decision can also be more differentiated. For example: move part of the effort forward; hold on some; and terminate or reform others.

Decision gate approval follows review by qualified experts, involved stakeholders, and management. Approval should be based on evidence of compliance to the criteria of the review. Balancing the formality and frequency of decision gates is seen as a critical success factor for all SE process areas. The consequences of conducting a superficial review, omitting a critical discipline, or skipping a decision gate are usually long-term and costly.

It is important to note that there may be significant changes in the project's environment. This may impact the project's business case, system scope, or resources needed. Consequently, the related decision criteria should be updated and evaluated at every decision gate. Inadequate consideration can set up subsequent failures—usually a major factor in cost overruns and delays.

Upon successful completion of a decision gate, some artifacts (e.g., documents, analysis results, models, or other products of a system life cycle stage) will have been approved as the basis upon which future work must build. These artifacts are placed under configuration management along with the decisions made and the associated rationale and assumptions (see Section 2.3.4.5).

2.1.4 Technical Reviews and Audits

Technical reviews and audits are used to assess technical progress, coordinate activities, and determine the technical status of a system of interest (SoI). According to ISO/IEC 24748–8 / IEEE 15288.2 (2014):

> A technical review is "a series of systems engineering activities conducted at logical transition points in a system life cycle, by which the progress of a [project] is assessed relative to its technical requirements using a mutually agreed-upon set of criteria" and
>
> An audit is "a detailed review of processes, product definition information, documented verification of compliance with requirements, and an inspection of products to confirm that products have achieved their required attributes or conform to released product configuration definition information."

The technical reviews and audits to be performed occur throughout the system life cycle and should be captured in the project's Systems Engineering Management Plan (SEMP) and reflected in the project's schedule (see Section 2.3.4.1). They may be part of a decision gate review (see Section 2.1.3). A representative set of technical reviews and audits are listed in Table 2.1. They should be tailored for the needs of the project and the methodologies being used. ISO/IEC/IEEE 24748–1 (2018), Annex F and ISO/IEC 24748–8 / IEEE 15288.2 (2014) provide useful guidance for the planning and tailoring of reviews to the needs of the project and its stakeholders.

Figure 2.4 depicts the relationship between these reviews and audits identified in ISO/IEC 24748–8 / IEEE 15288–2 (2014) and the typical technical baselines across the system life cycle applicable for a sequential life cycle model. This depiction will vary significantly for incremental life cycle models.

TABLE 2.1 Representative technical reviews and audits

Defense Projects per ISO/IEC/IEEE 24748-8/IEEE 15288.2 (2014)	Space Projects per NASA (2007b)	Incremental Commitment Spiral Model per Boehm, et al. (2014)
Alternative Systems Review (ASR)	Mission Concept Review (MCR)	Exploration Commitment Review (ECR)
System Requirements Review (SRR)	System Requirements Review (SRR)	Valuation Commitment Review (VCR)
System Functional Review (SFR)	Mission Definition Review (MDR)	Foundation Commitment Review (FCR)
Preliminary Design Review (PDR)	System Definition Review (SDR)	Development Commitment Review$_n$ (DCR$_n$)
Critical Design Review (CDR)	Preliminary Design Review (PDR)	Operations Commitment Review$_n$ (OCR$_n$)
Test Readiness Review (TRR)	Critical Design Review (CDR)	
Functional Configuration Audit (FCA)	System Integration Review (SIR)	
System Verification Review (SVR)	Operational Readiness Review (ORR)	
Production Readiness Review (PRR)	Flight Readiness Review (FRR)	
Physical Configuration Audit (PCA)	Mission Readiness Review (MRR)	
	Post-Launch Assessment Review (PLAR)	
	Critical Events Readiness Review (CERR)	
	Post-Flight Assessment Review (PFAR)	
	Decommissioning Review (DR)	
	Disposal Readiness Review (DRR)	

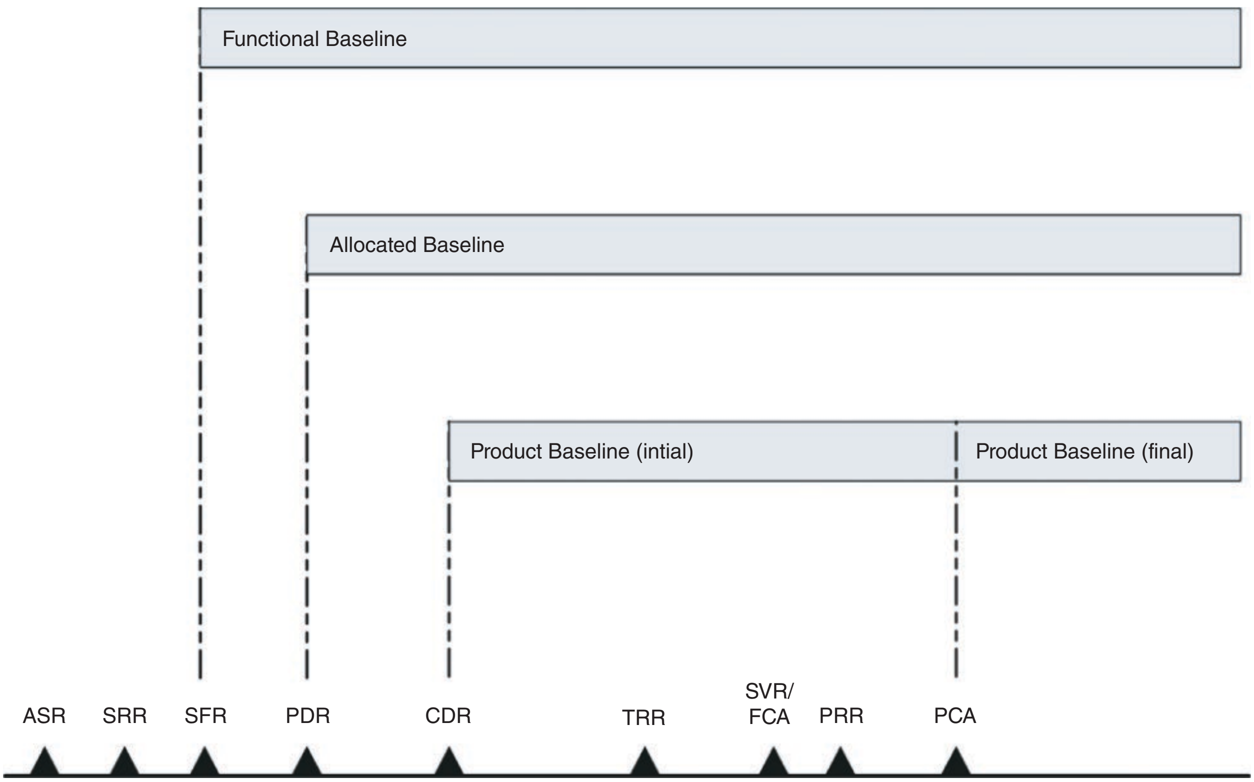

FIGURE 2.4 Relationship between technical reviews and audits and the technical baselines. From ISO/IEC 24748–8 / IEEE 15288.2 (2014). Used with permission. All other rights reserved.

Technical reviews and audits provide an opportunity to assess the following:

- The SoI is meeting its requirements
- The SoI is meeting stakeholder expectations, internal and external
- The SoI will have acceptable quality characteristics (QCs)
- The SoI is at an appropriate level of maturity
- The SoI is at an acceptable level of risk
- There is a clear path toward verifying and validating the SoI and its elements

Good practices for technical reviews and audits include:

- Plan the review or audit, including getting concurrence on a mutually agreeable location and date
- Application of multiple instances of the reviews and audits, both at multiple levels of the systems hierarchy and during each increment or iteration
- Elimination of unnecessary reviews or audits
- Establish clear entry and exit criteria for each review and audit
- Establish roles and responsibilities for the preparation, conduct, and acceptance of each review

- Make the reviews be risk-driven (risk is at an acceptable level) or event-driven (the entry criteria has been satisfied), not schedule-driven (must happen on a certain date)
- Consider "dry-runs" to make the review as efficient as possible
- Include subject matter experts (SMEs) and independent reviewers
- Include all members of the team, including acquirers and suppliers
- Capture clear actions, with ownership and due dates, for all issues that arise
- Follow up on actions that were raised

Each technical review or audit should include knowledgeable participants as well as participants with sufficient objectivity to assess satisfaction of the pre-established review criteria. Based on the purpose and level of the review, the participants may include representatives from the acquirer or supplier organizations, or both. A list of possible participants is provided below:

- Project Manager
- Lead SE Practitioner / Chief Engineer / Lead Engineer
- Review or Audit Chair
- Recorder (person charged with capturing the results of the review or audit)
- Acquirer Representative(s)
- Supplier Representative(s)
- Project Verification and Validation Lead
- Other Technical Leads

2.2 LIFE CYCLE MODEL APPROACHES

Section 2.1 introduces the concept of life cycle stages. The life cycle models are thus the framework within which the individual life cycle stages and transitions between them are planned and implemented. There are many different life cycle models, each suitable for different situations. A common way to differentiate them is to divide the life cycle model approaches into three groups: sequential, incremental, and evolutionary. Figure 2.5 provides the general concept for each of these approaches, and Table 2.2 summarizes their distinguishing characteristics.

ISO/IEC/IEEE 24748–1 (2018) provides further information on sequential (identified as "once-through"), incremental, and evolutionary life cycle model approaches. Sections 2.2.1 to 2.2.3 elaborate on how these approaches can be applied to manage the work within each life cycle stage.

There are many factors that help determine which life cycle models are suitable for a specific system or project. Clause 6 of ISO/IEC/IEEE 24748–1 (2018) provides informational considerations that may influence the selection and adaptation of life cycle model, including:

a) stability of, and variety in, operational environments;
b) risks, commercial or performance, to the concern of stakeholders;
c) novelty, size, and complexity;
d) starting date and duration of utilization;
e) integrity issues such as safety, security, privacy, usability, availability;
f) emerging technology opportunities;
g) profile of budget and organizational resources available;
h) availability of the services of enabling systems;

i) roles, responsibilities, accountabilities, and authorities in the overall life cycle of the system;
j) the need to conform to other standards.

Other sources define characteristics and tailoring factors that can be used to guide tailoring. As an example, Project Management Institute (PMI) has published their Situation Context Framework (SCF) (2022) that "defines how to select and tailor a situation-dependent strategy for software development. The SCF is used to provide context for organizing your people, process, and tools for a software-based solution delivery team." Seven dimensions (team size,

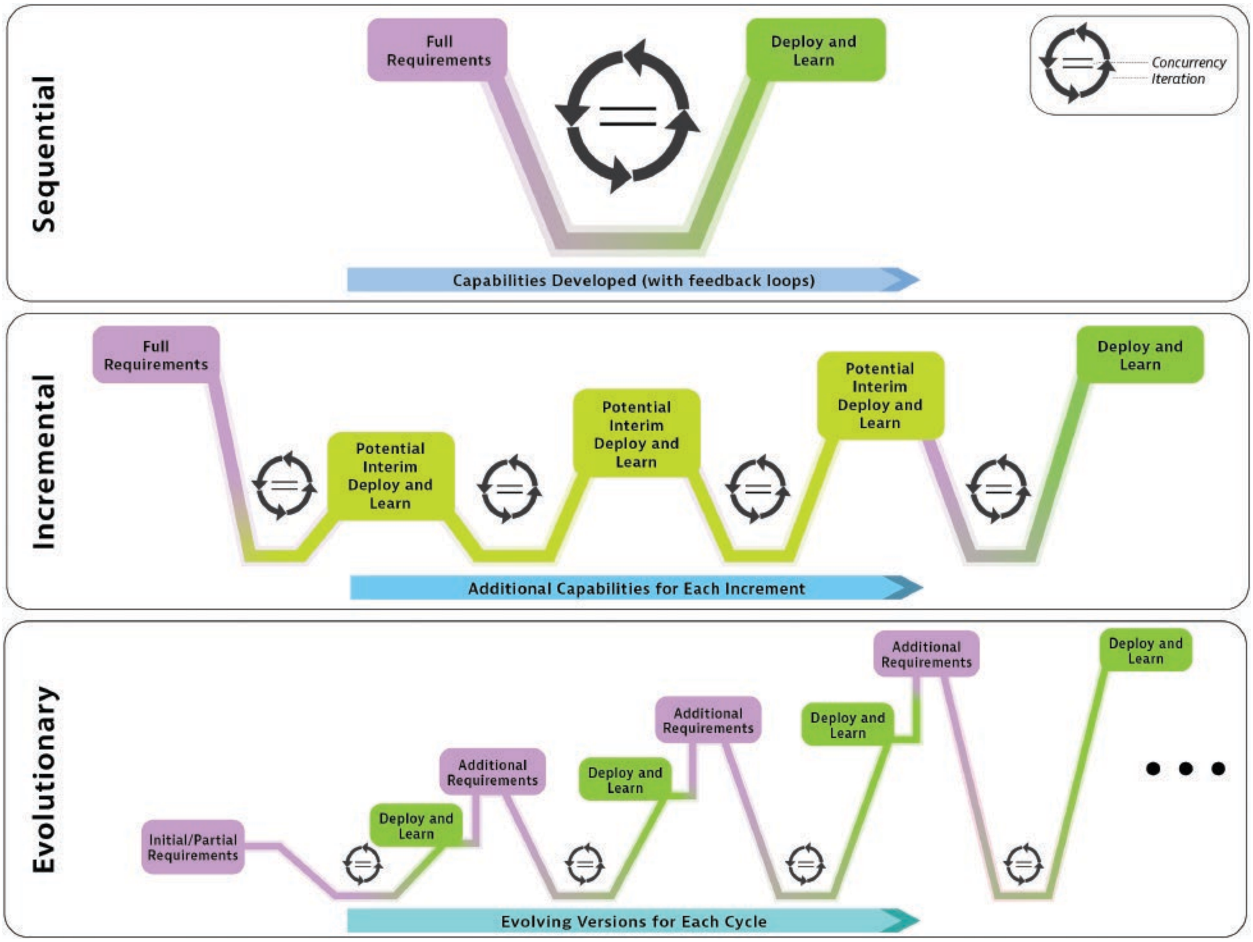

FIGURE 2.5 Concepts for the three life cycle model approaches. INCOSE SEH original figure created by Endler. Usage per the INCOSE NOTICES page. All other rights reserved.

TABLE 2.2 Life cycle model approach characteristics

Life cycle approach	Requirement set at start	Planned iterations	Multiple deployments
Sequential	Full	Single	No
Incremental	Full	Multiple	Potential
Evolutionary	Partial	Multiple	Typically

INCOSE SEH original table created by Endler derived from ISO/IEC/IEEE 24748-1 (2018). Usage per the INCOSE Notices page. All other rights reserved.

geographic distribution, organizational distribution, skill availability, compliance, domain complexity, and solution complexity) with scaling factors in each dimension are defined within this framework.

As there is no universal approach, it is recommended that each organization continuously questions itself as to which approach, or combination of approaches, is most suitable. Part IV of this handbook addresses tailoring and application considerations in more detail.

2.2.1 Sequential Methods

The sequential approach is focused on the general flow of the processes with feedback loops, but a single delivery. Sequential life cycle models break down SE activities into linear sequential stages, where each stage depends on the deliverables of the previous stages, along with feedback from subsequent stages.

On projects where it is necessary to coordinate large teams of people working in multiple companies, sequential approaches provide an underlying framework to provide discipline to the life cycle processes. Sequential life cycle models are characterized by a systematic approach that adheres to specified processes as the system moves through a series of representations from requirements through design to finished product. Specific attention is given to the completeness of documentation, traceability from requirements, and verification of each representation after the fact.

The strengths of sequential life cycle models are predictability, stability, repeatability, and high assurance. Process improvement focuses on increasing process capability through standardization, measurement, and control. These models rely on "master plans" to anchor their processes and provide project-wide communication. Historical data is usually carefully collected and maintained as inputs to future planning to make projections more accurate (Boehm and Turner, 2004).

The waterfall model, introduced by Royce (1970), was used to characterize the advantages and disadvantages of sequential approaches. The waterfall model has been used successfully in the manufacturing and construction industries, where the highly structured physical environments meant that design changes became prohibitively expensive much sooner in the development process. In addition, safety-critical products, such as the Therac-25 medical equipment (see the case study in Section 6.1), can only meet modern certification standards by following a thorough, documented set of plans and specifications. Such standards mandate strict adherence to process and specified documentation to achieve safety or security.

The SE Vee model (named due to its shape representing the letter "V" in the English language), introduced in Forsberg and Mooz (1991), described in Forsberg, et al. (2005), and shown in Figure 2.6, is another example of a sequential approach used to visualize key areas for SE focus, associating each development stage with a corresponding testing stage. The Vee highlights the need for continuous validation with the stakeholders, the need to define verification plans during requirements development, and the importance of continuous risk and opportunity assessment.

There are several variations of the Vee model. Typically, the "left" side of the Vee is called system definition and the "bottom" and "right" side of the Vee are called system realization. In the Vee model, time and system maturity conceptually proceed from left to right (down the left side of the Vee and up the right side of the Vee). However, all the system life cycle processes are performed concurrently and iteratively at each level of the system hierarchy and all the system life cycle processes are recursively applied at each level of the system hierarchy (see Section 2.3.1.2). One of the strengths of the Vee model is its depiction of the relationships between the left and right sides of the Vee.

The left side of the Vee depicts the evolving baseline from stakeholder requirements, to system requirements, to the identification of a system architecture, to definition of elements that will comprise the final system. The development team then can move from the highest level of the system requirements down to the lowest level of detail. Risk and opportunity management investigations address development options to provide assurance that the baseline performance being considered can indeed be achieved and to initiate alternate concept studies at the lower levels of detail to determine the best approach. Stakeholder discussions (in-process validation) occur to ensure that the proposed baselines are acceptable to the organization, customer, user, and other stakeholders. Changes to enhance system performance or to reduce risk or cost are welcome for consideration, but after baselining these must go through formal change control, since others may be building on previously defined and released design decisions. The bottom of the Vee depicts either the recursive application of the systems life cycle processes at the next level of the system hierarchy or the implementation of atomic system elements (see Section 1.3.5). The broadening at the base of the figure shows the growth in the

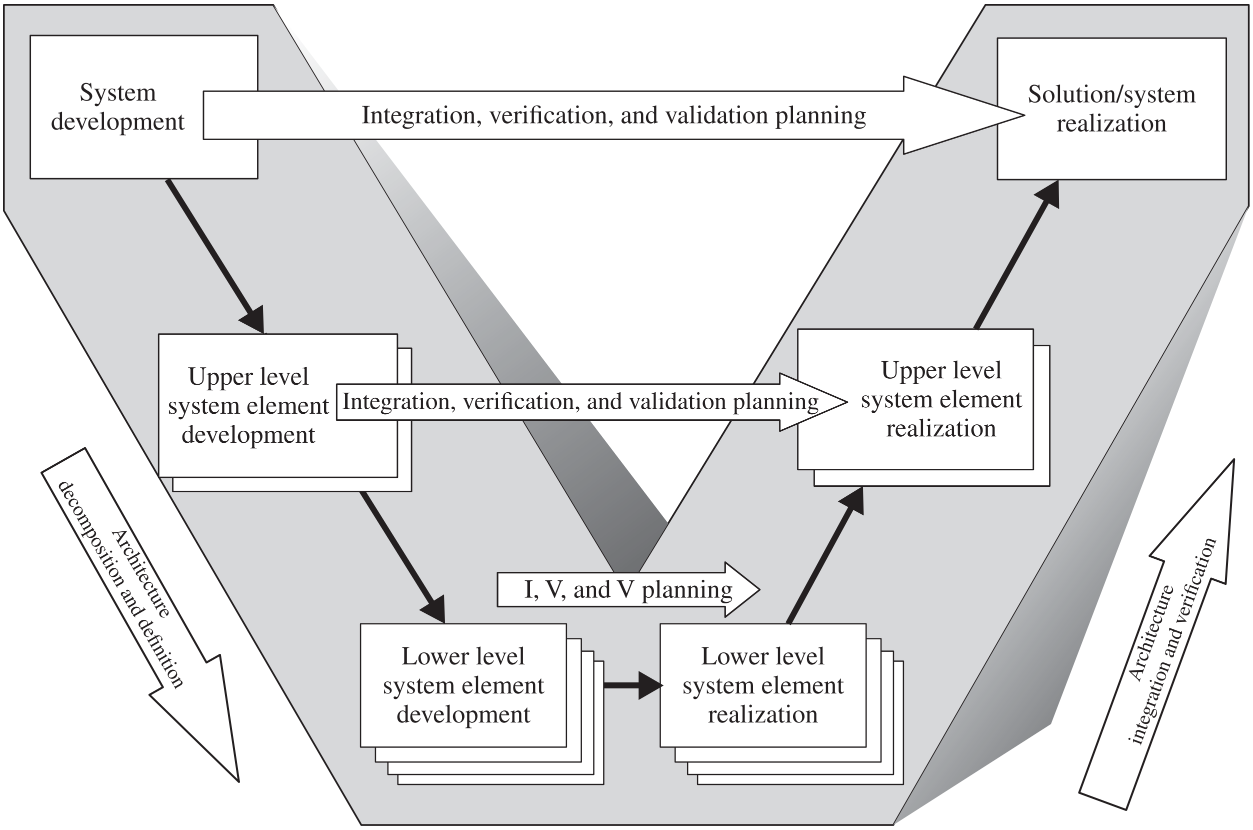

FIGURE 2.6 The SE Vee model. From Forsberg, et al. (2005) with permission from John Wiley & Sons. All other rights reserved.

number of system elements. Note that system elements can also be bought or reused. The right side of the Vee depicts the evolving baseline of system elements that are implemented, integrated, verified, and validated. In each stage of the system life cycle, the SE processes iterate to ensure that a concept or design is feasible and that the stakeholders remain supportive of the solution as it evolves.

ISO/IEC/IEEE 24748–2 (2018), Clause 6.4.3.1 provides further details on sequential life cycle models, including typical applicable systems as well as risks and opportunities associated with these models.

2.2.2 Incremental Methods

Incremental approaches have been in use since the 1960s (Larman and Basili, 2003). They represent a practical and useful approach that allows a project to provide an initial capability (or a limited set of capabilities) followed by successive deliveries to reach the desired SoI. The goal of an incremental approach is to provide rapid value and responsiveness. Generally, each increment adds capabilities intended to converge on a stakeholder satisfying result for the increment. Based on a set of requirements, a candidate set of increments is defined and the initial increment is initiated. Subsequent increments are initiated, and the process is repeated, until a complete system is deployed or until the organization decides to terminate the effort. Intermediate increments can potentially be deployed to support learning.

An incremental approach works well when an organization intends to market new versions of a product at planned intervals. Typically, the capabilities of the final delivery are known at the beginning. However, as there is significant technical risk, the development of the capabilities is performed incrementally to allow for the latest technology insertion or potential changes in needs or requirements. A core part of the planning process for an incremental approach establishes the cycle times for increments. Increments are beneficially timed in development projects to accommodate coordinated events such as integration testing and evaluation, capability deployment, experimental deployment, or release to production. Iteration cycles are beneficially timed to minimize rework cost as a project learns experimentally and empirically. Project planning and management often benefit from a constant cadence among increments.

One example of an incremental approach is the Incremental Commitment Spiral Model (ICSM) (Boehm, et al., 2014), which extends the classic Spiral Model for software introduced in Boehm (1987) for SE. A view of the ICSM is shown in Figure 2.7. In the ICSM, each increment addresses requirements and solutions concurrently, rather than sequentially. ICSM also considers products and processes; hardware, software, and human aspects; and business case analyses of alternative product configurations or product line investments. The stakeholders consider the risks and risk mitigation plans and decide on a course of action. If the risks are acceptable and covered by risk mitigation plans, the project proceeds into the next spiral (increment).

ISO/IEC/IEEE 24748–2 (2018), Clause 6.4.3.2 provides further details on incremental life cycle models, including typical applicable systems as well as risks and opportunities associated with these models.

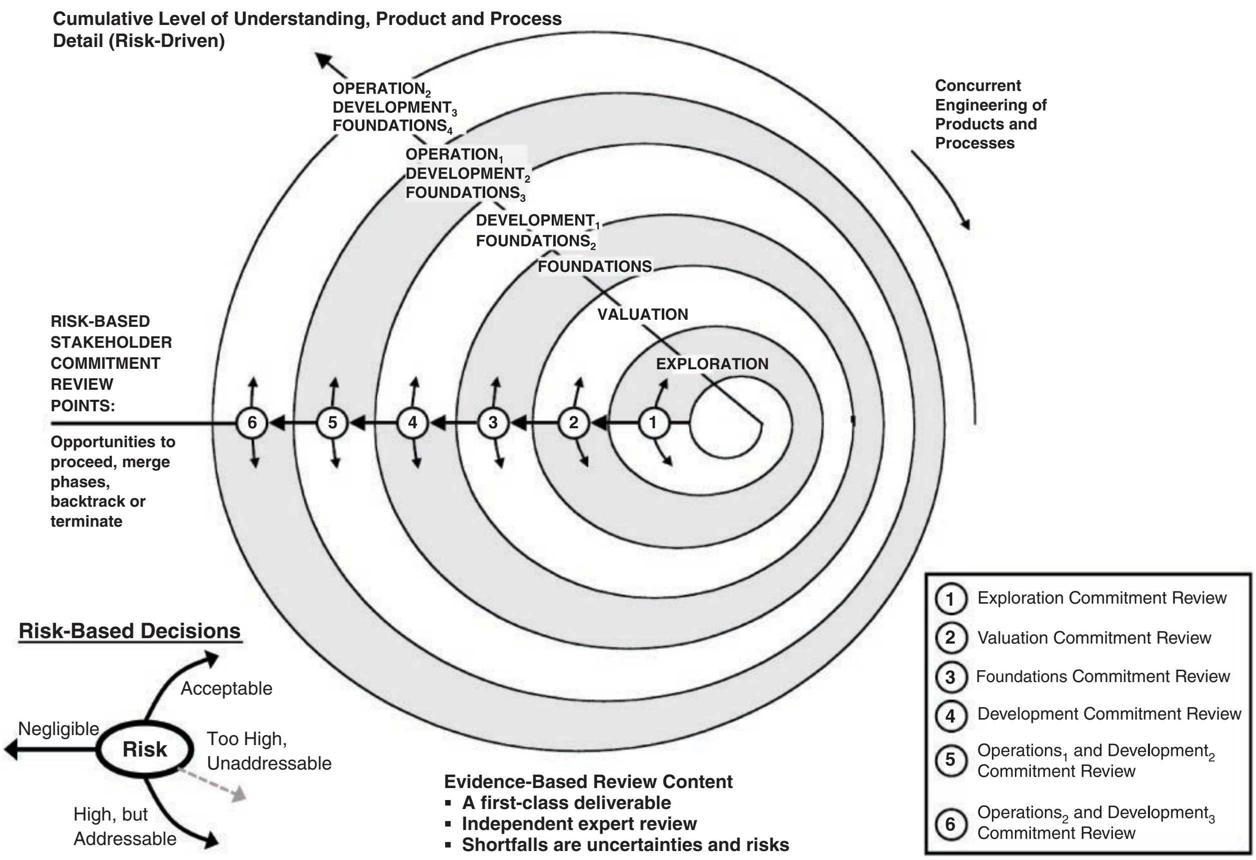

FIGURE 2.7 The Incremental Commitment Spiral Model (ICSM). From Boehm, et al. (2014) with permission from Pearson Education. All other rights reserved.

2.2.3 Evolutionary Methods

In the sequential and incremental approaches described previously, the full set of required capabilities of the final system is assumed to be mostly known at the start of the effort. In some cases, especially in novel systems, the final system requirements may be unknown or only partially known. An evolutionary approach provides the adaptability and flexibility needed for the development in these situations. For example, the high-temperature tiles of the NASA space shuttle were developed using an evolutionary approach (Forsberg, 1995). An evolutionary approach is often used in research and development (R&D) projects and SoS developments. Software development efforts are increasingly using agile methods, which are a type of evolutionary development.

In evolutionary approaches, cycles are typically planned on a regular periodic basis, each resulting in a deployable version. The requirements for the SoI are typically only partially known and are increasingly refined with each cycle. At the beginning, the goal of each cycle may be more or less unknown. Therefore, it is particularly important that the experience gained with the earlier cycles is taken into account for the subsequent cycles. Similar to the incremental approach, versions may be developed sequentially or in parallel. This is a particular challenge for those involved in the project, since new capabilities are typically assigned to exactly one version. If this assignment is lost or becomes unclear, this leads to confusion and negatively impacts the schedule and cost targets. Thus, a well-functioning configuration control is essential, also since multiple versions can be operated and supported simultaneously (see Section 2.3.4.5). Aspects to be considered include operating manuals, maintenance instructions, spare parts, disposal instructions, etc.

The evolutionary approach offers significant advantages if it is possible to obtain steady and high-quality feedback from relevant stakeholders. For example, the first versions can be used to demonstrate basic feasibility, such as a minimal viable product (MVP), and facilitate market entry. Likewise, emerging technical innovations can be planned for later versions.

When developing subsequent versions, it is recommended to carefully examine whether the previous versions should be completely replaced by newer ones. Alternatively, subsequent versions can be developed such that a partial or even complete upgrade of the previous versions to the new version is possible. For this, it is necessary that these things are considered during the early cycles. Criteria such as adaptability, flexibility, and modularity should be carefully considered to enable the long-term evolution of the system. Decisions should be made in the context of life cycle cost (see Section 3.1.2).

An example of an evolutionary approach is DevOps (a blend of the terms and concepts for "development" and "operations"). The goal of DevOps is to provide continuous integration of the system and continuous delivery of capabilities. DevOps is typically characterized by three key principles: shared ownership, workflow automation, and rapid feedback. DevSecOps (a blend of "development," "security," and "operations"), shown in Figure 2.8, integrates security practices into DevOps. In DevSecOps, each delivery team is responsible and empowered to pick appropriate security means.

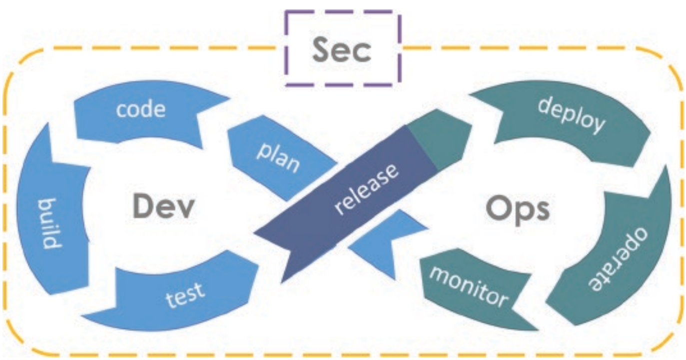

FIGURE 2.8 DevSecOps INCOSE SEH original figure created by D'Souza derived from Banach (2019) and Anx (2021). Usage per the INCOSE Notices page. All other rights reserved.

ISO/IEC/IEEE 24748–2 (2018), Clause 6.4.3.4 provides further details on evolutionary life cycle models, including typical applicable systems as well as risks and opportunities associated with these models.

Figure 2.9 is an example of a mixed approach (both incremental and evolutionary). This figure shows the agile mixed-discipline approach employed by Rockwell Collins in the development of military radios (Dove, et al., 2017). Teams working on electronic-board hardware, firmware, and software have different timings for hardware increments and firmware and software epics

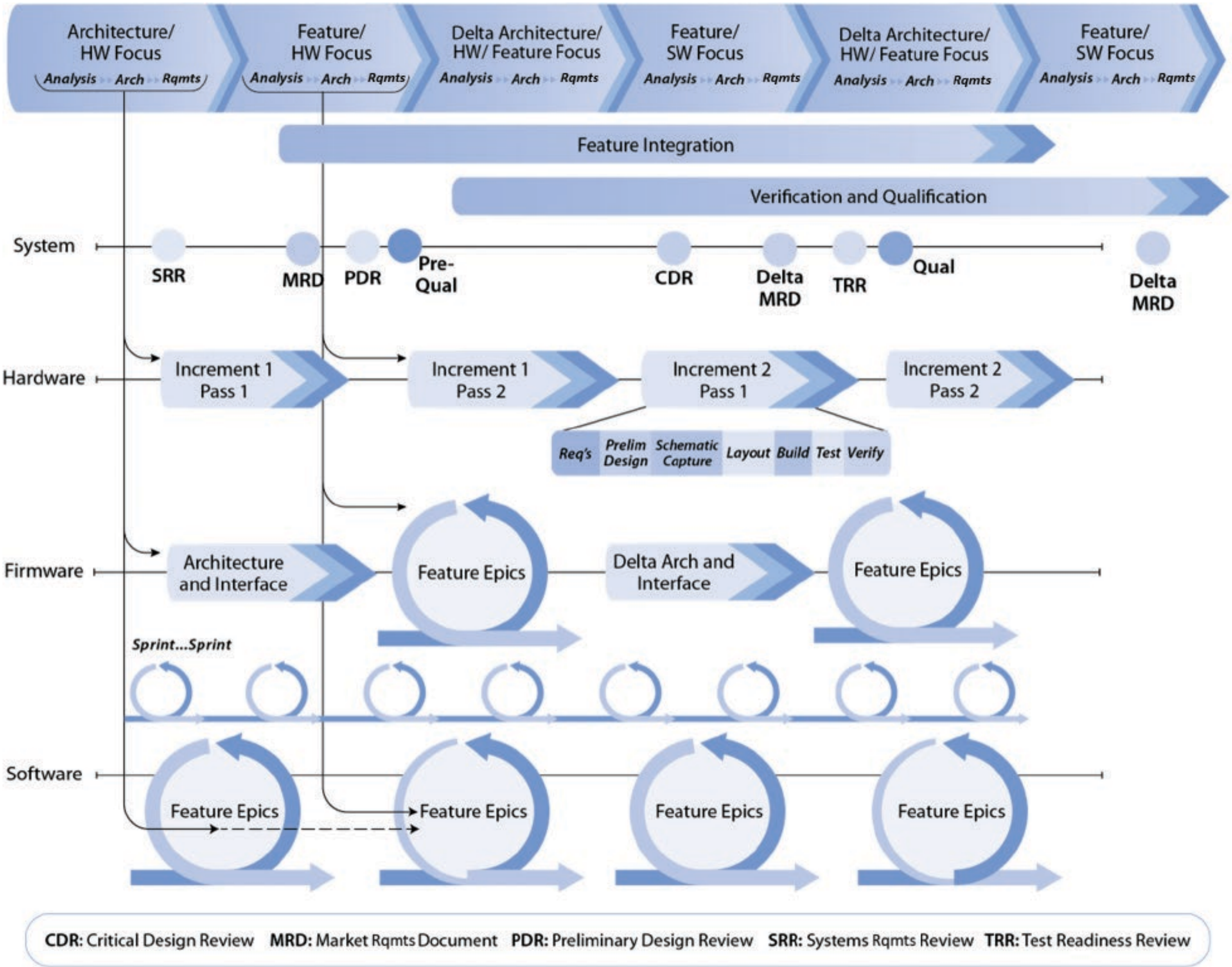

FIGURE 2.9 Asynchronous iterations and increments across agile mixed discipline engineering. From Dove, et al. (2017). Used with permission. All other rights reserved.

(versions). The teams accomplish integrated work-in-process testing with the latest increments and versions from each of the disciplines.

All these life cycle approaches are supported by the processes defined in ISO/IEC/IEEE 15288 (2023) and this handbook. The life cycle model should be chosen so that it is sufficiently adaptable and flexible. Section 4.1 provides more information on tailoring life cycle models.

2.3 SYSTEM LIFE CYCLE PROCESSES

2.3.1 Introduction to the System Life Cycle Processes

A process is a series of activities and tasks performed to achieve one or more outcomes for a stated purpose. In SE, the system life cycle processes are one of the enablers to help manage a system solution across the life cycle stages. The processes are intended to be applied concurrently, iteratively, and recursively with other enablers (e.g., tools, technology) throughout the stages of the life cycle (see Section 2.3.1.2).

ISO/IEC/IEEE 15288 (2023) identifies four process groups for the system life cycle, providing "a common process framework for describing the life cycle of engineered systems, adopting a Systems Engineering approach." Each of these process groups is the subject of a section within Part 2. A graphical overview of these processes is given in Figure 2.10:

Agreement Processes (Section 2.3.2) include Acquisition and Supply.

Organizational Project-Enabling Processes (Section 2.3.3) include Life Cycle Model Management, Infrastructure Management, Portfolio Management, Human Resource Management, Quality Management, and Knowledge Management.

Technical Management Processes (Section 2.3.4) include Project Planning, Project Assessment and Control, Decision Management, Risk Management, Configuration Management, Information Management, Measurement, and Quality Assurance.

Technical Processes (Section 2.3.5) include Business or Mission Analysis, Stakeholder Needs and Requirements Definition, System Requirements Definition, System Architecture Definition, Design Definition, System Analysis, Implementation, Integration, Verification, Transition, Validation, Operation, Maintenance, and Disposal.

The application of these system life cycle processes is supported by SE practitioners having the relevant competencies. The competencies are defined in the INCOSE Systems Engineering Competency Framework (SECF) (2018). Note that the professional competencies (see Section 5.1.2) generally apply to all the processes.

Note: Acronyms for the process names are provided in Appendix D.

2.3.1.1 Format and Conventions A common section structure has been applied to describe the system life cycle processes in this handbook. The following structure provides consistency in the discussion of these processes:

- Process overview
  - Purpose
  - Description
  - Inputs/outputs
  - Process activities
  - Common approaches and tips
- Process elaboration

To ensure consistency with ISO/IEC/IEEE 15288, the purpose statements from the standard are included verbatim for each process described herein. The titles of the process activities listed in each section are also consistent with ISO/IEC/IEEE 15288. The process activities describe "what" should be done, not "how" to do it. In some cases, additional items have been included to provide summary-level information regarding industry good practices and evolutions in the application of SE processes. Process elaborations provide additional details on topics that are unique to the specific life cycle process. See Part III for topics that cross-cut multiple life cycle processes.

In addition, each system life cycle process is illustrated by an input–process–output (IPO) diagram showing typical inputs, process activities, and typical outputs. A sample is shown in Figure 2.11. To understand a given process, readers are encouraged to study the complete information provided in the combination of figures and text and not rely solely on the IPO diagrams.

Typical inputs and outputs are listed by name within the respective IPO diagrams with which they are associated. The typical inputs and outputs are consistent with ISO/IEC 33060 (2020) when possible. Note that the IPO diagrams throughout this handbook represent "a" way that the SE processes can be performed, but not necessarily "the" way that they must be performed. The system life cycle processes produce "results" that are often captured in "documents" or "artifacts" or "models," rather than producing "documents" simply because they are identified as outputs. A complete list of all inputs and outputs with their respective descriptions appears in Appendix E.

The controls and enablers shown in Figure 2.11 govern all processes described herein and, as such, are not repeated on the subsequent IPO diagrams. Typically, IPO diagrams do not include controls and enablers, but since they are not

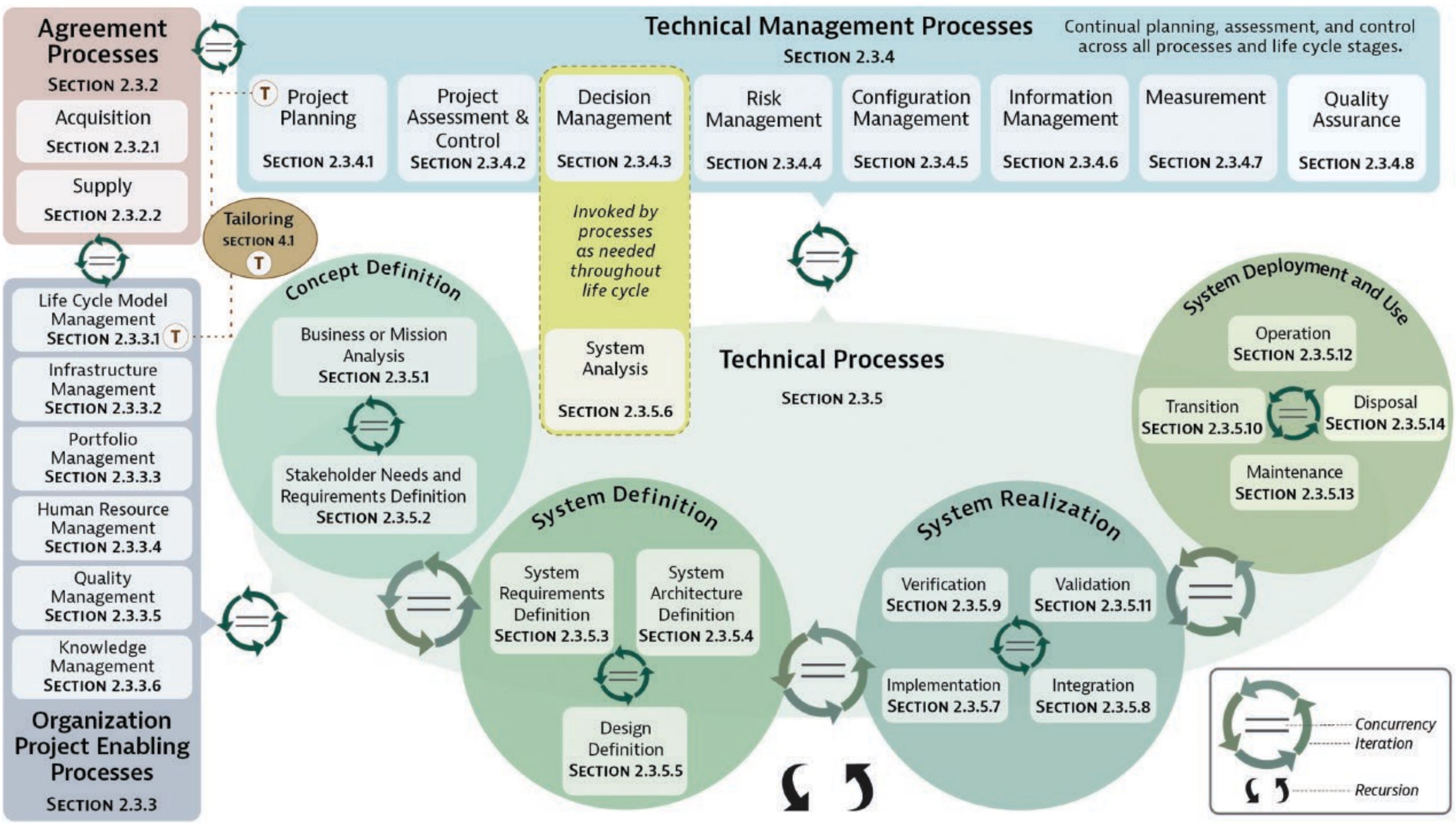

FIGURE 2.10 System life cycle processes per ISO/IEC/IEEE 15288. INCOSE SEH original figure created by Roedler and Walden. Usage per the INCOSE Notices page. All other rights reserved.

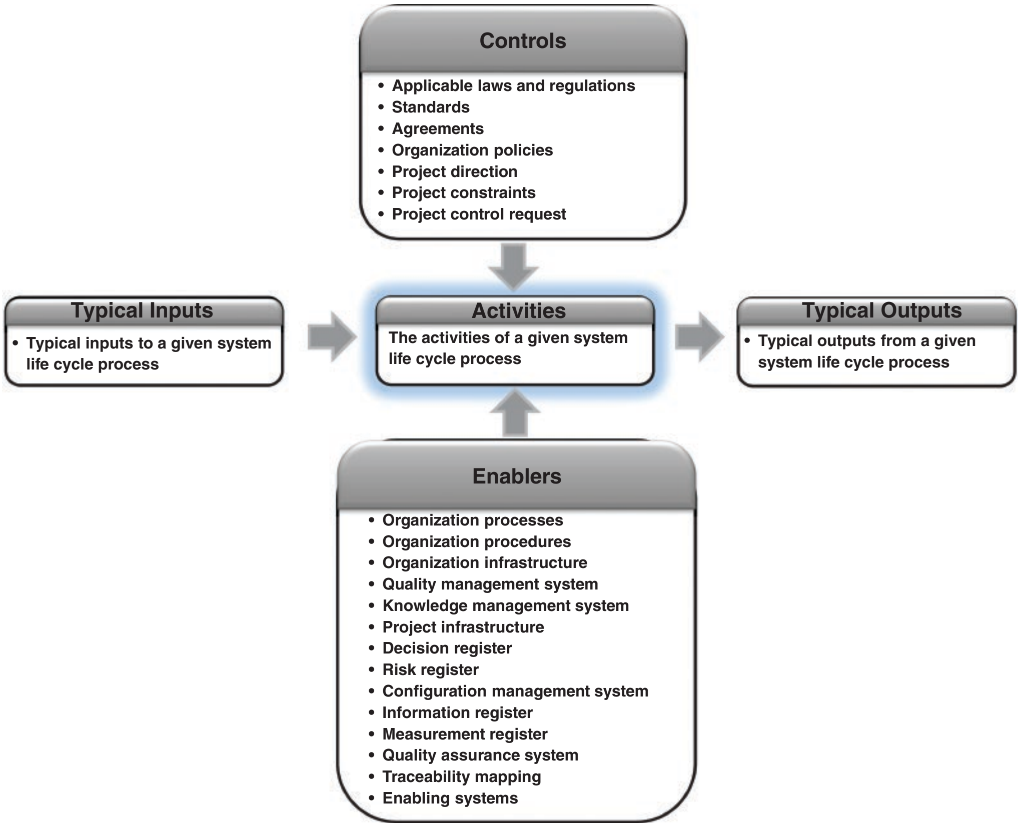

FIGURE 2.11 Sample IPO diagram for SE processes. INCOSE SEH original figure created by Walden, Shortell, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

repeated in the IPO diagrams throughout the rest of the handbook, we have chosen to label them IPO diagrams. The enablers work together with the inputs to be transformed by the process into the outputs under the direction of the controls. A complete list of all controls and enablers with their respective descriptions appears in Appendix E.

2.3.1.2 Concurrency, Iteration, and Recursion Too often, the system life cycle processes are viewed as being applied in a sequential, linear manner at a single level of the system hierarchy. However, valuable information and insight need to be exchanged between the processes in order to ensure a good system definition that effectively and efficiently meets the stakeholder needs and requirements. The application of concurrency, iteration, and recursion to the system life cycle processes helps to ensure communication that accounts for ongoing learning and decisions. This facilitates the incorporation of learning from further analysis and process application as the technical solution evolves. Figure 2.12 shows an illustration of the concurrent, iterative, and recursive nature of the system life cycle processes.

Concurrency (indicated by the parallel lines in the figure) is the parallel application of two or more processes at a given level in the system hierarchy. Concurrent work is likely to happen on any project and the system life cycle processes can likewise be performed in a concurrent manner. It is not necessary for processes to be performed serially, especially when one process is not dependent on another for information or results. For example, the Risk Management process and Measurement process usually are performed in a continual and concurrent manner. This is illustrated in Figure 2.34, in which both of these processes occur concurrently, yet provide information to one another. Additionally, the system architecture should enable concurrency through modularization, encapsulation, commonality/reuse, and other design methods.

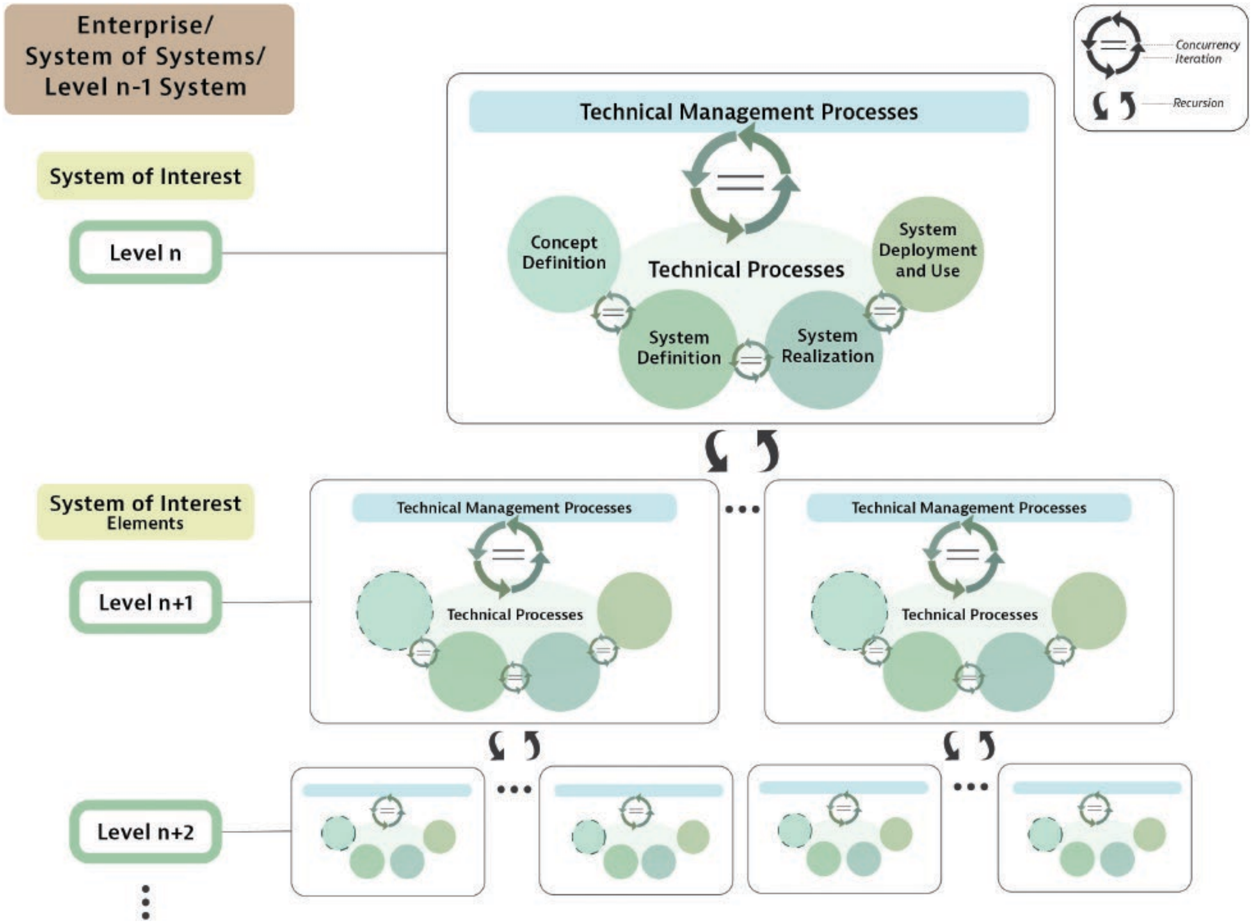

FIGURE 2.12 Concurrency, iteration, and recursion. INCOSE SEH original figure created by Roedler and Walden. Usage per the INCOSE Notices page. All other rights reserved.

Iteration (indicated by the circular arrows in the figure) is the repeated application of and interaction between two or more processes at a given level in the system hierarchy. Iteration is needed to accommodate stakeholder decisions and evolving understanding, account for architectural decisions or constraints, and resolve trade-offs for affordability, adaptability, feasibility, resilience, etc. There can be iteration between any of the processes. For example, there is often iteration between System Requirements Definition and System Architecture Definition processes. The system architecture will reflect these design iterations through identification of functions, their allocation to system elements, assignment to logical and physical interfaces, and verification as intended in the design. In this case, there is a concurrent application of the processes with iteration between them, where the evolving system requirements help to shape the architecture through identified constraints and functional and quality requirements. For example, the system architecture may need to be changed due to detailed electrical modeling indicating that a particular system element's load exceeds its allocated power budget and forces a design change or reallocation of the network power assignments in the overall system. The tradeoffs between candidate architectures or elements of the architecture, in turn, may identify requirements that are not feasible, driving further requirements analysis with trade-offs that change some requirements. Likewise, the Design Definition process could identify the need to reconsider decisions and trade-offs in the

System Requirements Definition or System Architecture Definition processes. Any of these can invoke additional application of the System Analysis and Decision Management processes.

Recursion (indicated by the down and up arrows in the figure) is the repeated application of the set of life cycle processes, tailored as appropriate, at successive levels in the system hierarchy. The Technical Management and Technical Processes are expected to be recursively applied for each successive level of the system hierarchy until the level is reached where the decision is made to make, buy, or reuse a system element (see Section 1.3.5). During the recursive application of the processes, the outputs at one level become inputs for the next successive level (below for system definition, above for system realization).

Horizontal integration ensures completeness before diving deeper and *vertical integration* ensures consistency between levels in this concurrent, iterative, and recursive environment (see Section 2.3.5.8). For example, one may have to define functions, their inputs and outputs, their associated performance and conditions of operations *before* writing the associated requirement. Then one can define the related verification (method, conditions, criteria), from which one can postulate the next-lower-level architecture to assess feasibility and perform the related function, performance, or requirements decomposition and allocation. However, teams may need to go down multiples levels to validate that the functions or elements at the lower levels are going to be suitable solutions for the SoI and its stakeholders.

2.3.2 Agreement Processes

The initiation of a project begins with the identification of a problem or opportunity to be addressed, which results in the development of needs to be satisfied. Once a need is identified and resources are committed to establish a project, it is possible to define the terms and conditions of an acquisition and supply relationship through the Agreement Processes, which are defined in ISO/IEC/IEEE 15288 as follows:

> [5.7.2] [Agreement] Processes define the activities necessary to establish an agreement between two organizations.

The Agreement Processes in this handbook and in ISO/IEC/IEEE 15288 are focused on the acquisition and supply of systems, system elements, products, or services, although agreements could be established for other objectives. With respect to the acquisition and supply relationship, the acquirer and supplier could be two independent organizations (i.e., no common parent organization or enterprise) or two organizations from the same parent organization or enterprise.

The Agreement Processes are utilized under many conditions, including when:

- an organization cannot satisfy a defined need itself,
- a supplier can satisfy a defined need in a more economical or timely manner,
- a higher authority has directed the use of a specific supplier, and
- an organization needs materials or specialized services.

An overall objective of Agreement Processes is to identify the interfaces between the acquirer and supplier(s) and establish the terms and conditions of these relationships, including identifying the inputs required and the outputs that will be provided.

Agreement negotiations are handled in various ways depending on the specific organizations and the formality of the agreement. In a formal agreement, there is usually a contract negotiation activity to refine the contract terms and conditions. Note also that the Agreement Processes can be used for coordinating within an organization between different business units or functions. In this case, the agreement will usually be more informal, not requiring a formal or specific contract.

An important contribution of ISO/IEC/IEEE 15288 is the recognition that SE practitioners are relevant contributors to the Agreement Processes (Arnold and Lawson, 2003). The SE practitioner is usually in a supporting role to the project management practitioner during negotiations and is responsible for impact assessments for changes, trade studies on alternatives, risk assessments, and other technical input needed for decisions.

Acceptance criteria are critical elements to each party because they protect both sides of the business relationship—the acquirer from being coerced into accepting a product with poor quality and the supplier from the unpredictable actions of an indecisive acquirer. It is important to note that the acceptance criteria are negotiated during the Agreement Processes. During negotiations, it is also critical that both parties are able to track progress toward an agreement. Identifying where further work toward achieving consensus in the documents and clauses is vital.

Two Agreement Processes are identified by ISO/IEC/IEEE 15288: the Acquisition process and the Supply process. These processes, subject of Sections 2.3.2.1 and 2.3.2.2, respectively, are two sides of the same coin. They conduct the essential business of the organization related to the SoI. They establish the relationships between organizations relevant to the acquisition and supply of products and services, regardless of whether the agreement is formal (as in a contract) or informal. Each process establishes the context and constraints of the agreement under which the other system life cycle processes belonging to the project scope are performed. Note that an organization can be both a supplier and acquirer for a given system. For example, an organization may be contracted to supply a system to an end customer. However, that organization may choose to acquire some of the system elements, materials, or services for developing or producing the system. So, that organization is the supplier to the end customer of the system and is the acquirer to those organizations providing it system elements, materials, or services.

Changes may happen during the execution of an agreement including acquirer change requests, deviations and waivers from the supply chain, or changes in the context of the project that were foreseen in risk analysis or not. Upon decision of the parties, this may lead to modifications to the initial state of the agreement. For that purpose, a statement of compliance may be initiated and updated all along the project describing the agreed changes and can include requirements impacted by a modification, the reference of modification, compliance verification by the supplier, and compliance validation by the acquirer.

2.3.2.1 Acquisition Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

> [6.1.1.1] The purpose of the Acquisition process is to obtain a product or service in accordance with the acquirer's requirements.

The Acquisition process is invoked to establish an agreement between two organizations under which one party acquires products and/or services from the other. The acquirer experiences a need for an operational system, for services in support of an operational system, for elements of a system being developed by a project, or for services in support of project activities.

Description This section is written from the perspective of the acquirer organization. An acquiring organization applies due diligence in the selection of a supplier to avoid costly failures and impacts to the organization's budgets and schedules and other issues. Therefore, the role of the acquirer demands familiarity with the Technical, Technical Management, and Organizational Project-Enabling Processes, as it is through them that the supplier will execute the agreement.

Inputs/Outputs Inputs and outputs for the Acquisition process are listed in Figure 2.13. Descriptions of each input and output are provided in Appendix E.

Process Activities The Acquisition process includes the following activities:

- *Prepare for the acquisition*.
  - Develop and maintain acquisition policies, plans, and procedures to meet the organization strategies, goals, and objectives as well as the needs of the project management and SE organizations.

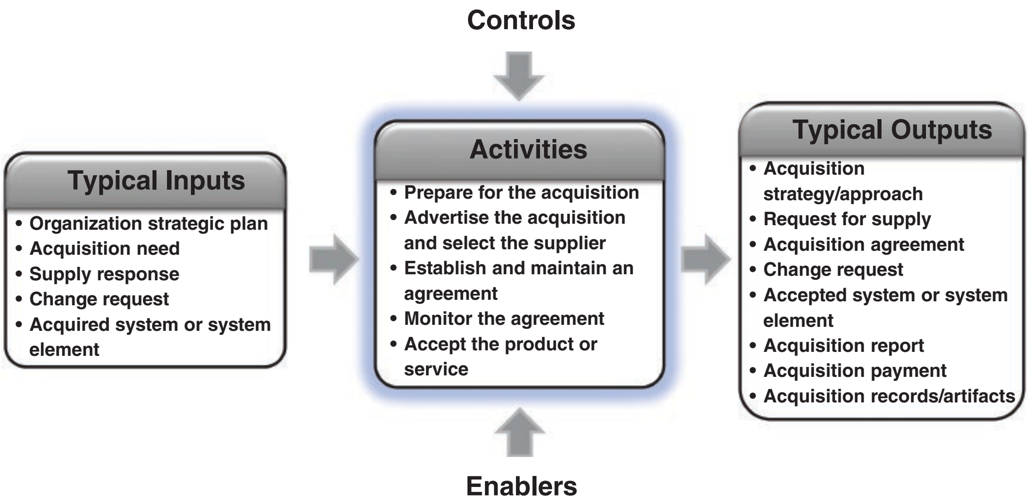

FIGURE 2.13 IPO diagram for the Acquisition process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

- Collect needs in a request for supply—such as a Request for Proposal (RFP) or Request for Quotation (RFQ) or some other mechanism—to obtain the supply of the service and/or product. Through the use of the Technical Processes, the acquiring organization produces a set of requirements and models that will form the basis for the technical information of the agreement.
- Identify a list of potential suppliers—suppliers may be internal or external to the acquirer organization.

• *Advertise the acquisition and select the supplier*.

- Distribute the request for supply and select appropriate suppliers—using selection criteria, rank suppliers by their suitability to meet the overall need and establish supplier preferences and corresponding justifications. Viable suppliers should be willing to conduct ethical negotiations, able to meet obligations, and willing to maintain open communications throughout the Acquisition process. Note that the approach may be less formal when a function within the organization is a candidate for the supply need.
- Evaluate supplier responses to the request for supply—ensure the offered product and/or service can meet acquirer needs and complies with industry and other standards. Assessments from the Project Portfolio Management (see Section 2.3.3.3) and Quality Management (see Section 2.3.3.5) processes and review results from the requesting organization are necessary to determine the suitability of each response and the ability of the supplier to meet the stated commitments. Record results from the evaluation of responses to the request for supply. This can range from formal documentation to less formal interorganizational interactions (e.g., between design engineering and marketing).
- Select the preferred supplier(s) based on acquisition criteria.

• *Establish and maintain an agreement*.

- Establish an agreement. Ensure an understanding of expectations, including acceptance criteria.
- This agreement ranges in formality from a written contract to a verbal agreement. Appropriate to the level of formality, the agreement establishes requirements, development and delivery milestones, verification, validation and acceptance conditions, process requirements (e.g., configuration management, risk management, measurements), exception-handling procedures, agreement change management procedures, payment schedules, and handling of data rights and intellectual property so that both parties understand the basis for executing the agreement. For a written contract, this occurs when the contract is signed.

  – Identify the necessary changes to the agreement and evaluate the related impacts on the agreement.
  – Update the agreement with the supplier as necessary.
- *Monitor the agreement*.
  – Manage Acquisition process activities, including decision making for agreements, relationship building and maintenance, interaction with organization management, responsibility for the development of plans and schedules, and final approval authority for deliveries accepted from the supplier.
  – Maintain communications with supplier, stakeholders, and other organizations regarding the project.
  – Status progress against the agreed-to schedule to identify risks and issues, to measure progress toward mitigation of risks and adequacy of progress toward delivery and cost and schedule performance, and to determine potential undesirable outcomes for the organization. The Project Assessment and Control process (see Section 2.3.4.2) provides necessary evaluation information regarding cost, schedule, and performance.
  – Amend agreements when impacts on schedule, budget, or performance are identified.
- *Accept the product or service*.
  – Accept delivery of products and services—in accordance with all agreements and relevant laws and regulations.
  – Render payment—or other agreed consideration in accordance with agreed payment schedules.
  – Accept responsibility in accordance with all agreements and relevant laws and regulations.
  – When an Acquisition process cycle concludes, a final review of performance is conducted to extract lessons learned for continued process performance.
  – Retire the agreement.

Note: The project is closed through the Portfolio Management process (see Section 2.3.3.3), *which manages the full set of projects of the organization*.

Common approaches and tips:

- Establish acquisition guidance and procedures that inform acquisition planning, including recommended milestones, standards, assessment criteria, and decision gates. Include approaches for identifying, evaluating, choosing, negotiating, managing, and terminating suppliers.
- Establish a technical point of responsibility within the organization for monitoring and controlling individual agreements. This person maintains communication with the supplier and is part of the decision-making team to assess technical development and progress in the execution of the agreement.

Note: There can be multiple points of responsibility for an agreement that focus on technical, programmatic, marketing, etc.

- Define and track measures that indicate progress on agreements. Avoid measures that are not focused on the true information needs. Leading indicators are preferable (see Section 2.3.4.7).
- Include technical representation in the selection of the suppliers to critically assess the capability of the supplier to perform the required task.
- Past performance of the supplier is highly important, but changes to key supplier personnel should be identified and evaluated to understand any impact with respect to the current request for supply.
- Communicate clearly with the supplier about priorities and avoid conflicting statements or making frequent changes in the statement of need that introduce risk into the process.
- Maintain traceability between the supplier's responses to the acquirer's solicitation. This can reduce the risk of contract modifications, cancellations, or follow-on contracts to fix the product or service.

- Smart contracts can be used to establish and maintain an agreement. A smart contract is a transaction protocol intended to execute automatically and control or document legally relevant events and actions according to the terms of a contract or an agreement (Tapscott and Tapscott, 2018). The objectives of smart contracts are the reduction of need in trusted intermediaries, arbitrations and enforcement costs, fraud losses, and the reduction of malicious and accidental exceptions (Fries and Paal, 2019).

Elaboration:
The Project Manager's role is to define, execute, and manage the acquisition. This is focused on the project needs to deliver the system, system elements, products, or services that meet the end user requirements and achieve the acquisition milestones. This is done in collaboration with the SE practitioners and the selected contractor to ensure the technical expectations and key performance parameters are achieved. The team needs to define plans and methods collectively, and refine them as more is learned about the nature and challenges inherent in the system or capability being built and its intended operating environment. For more information on PM-SE integration, see 5.3.3.

When the acquisition involves systems or system elements where technology or a system capability is not mature enough, it is necessary to account for uncertainty and the need for additional risk management actions in the planning. This includes allowing additional margin in the development/production timeframe, such as ample lead time in anticipation of inherent challenges, especially when technology maturation is required. These challenges may also include limited availability of adequate resources for the supplier (skilled labor and/or technologies), a need for customization of supplier products or equipment, poor or early understanding of interface requirements, integration challenges, and required verification and/or validation of the development. If there is no flexibility in the delivery date, then trade-offs may be needed to provide the system capabilities in an incremental manner.

Technical supplier management is about ensuring the supplier meets the allocated project requirements and that the supplier is effectively managed. This is usually achieved through the Statement of Work (SOW) and a set of requirements. The SOW is a mechanism to ensure progress is being made and describes the necessary work, quality, standards, designs, models, evidence, reviews, timescales, and meetings, etc. that the supplier is expected to provide contingent on the contract. To prove the system/system element functional, performance, and operational requirements are met, the supplier will also need to provide compliance matrices and verification and validation evidence.

2.3.2.2 Supply Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

> [6.1.2.1] The purpose of the Supply process is to provide an acquirer with a product or service that meets agreed requirements.

The Supply process is invoked to establish an agreement between two organizations under which one party supplies products or services to the other. Within the supplier organization, a project is conducted according to the recommendations of this handbook with the objective of providing a product or service to the acquirer that meets the agreed requirements. In the case of a mass-produced commercial product or service, a marketing, or similar, function may represent the acquirer and establish stakeholder expectations.

Description This section is written from the perspective of the supplier organization. The Supply process is highly dependent upon the Technical, Technical Management, and Organizational Project-Enabling Processes as it is through them that the work of executing the agreement is accomplished. This means that the Supply process is the larger context in which the other processes are applied under the agreement.

Inputs/Outputs Inputs and outputs for the Supply process are listed in Figure 2.14. Descriptions of each input and output are provided in Appendix E.

Process Activities The Supply process includes the following activities:

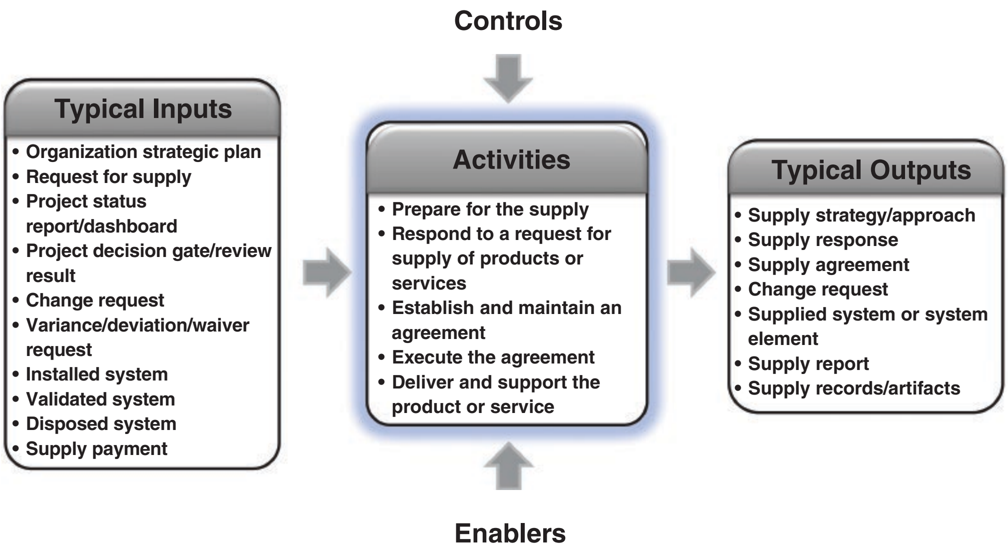

FIGURE 2.14 IPO diagram for the Supply process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

- *Prepare for the supply*.
  - Develop and maintain strategic plans, policies, and procedures to meet the needs of potential acquirer organizations, as well as internal organization goals and objectives including the needs of the project management and technical SE organizations.
  - Identify opportunities.
- *Respond to a request for supply of products or services*.
  - Select appropriate acquirers willing to conduct ethical negotiations, able to meet financial obligations, and willing to maintain open communications throughout the Supply process.
  - Evaluate the acquirer requests and propose a product or service that meets acquirer needs and complies with industry and other standards. Assessments from the Portfolio Management, Human Resource Management, Quality Management, and Business or Mission Analysis processes are necessary to determine the suitability of this response and the ability of the organization to meet these commitments.
- *Establish and maintain an agreement*.
  - Establish an agreement. Ensure an understanding of expectations, including acceptance criteria.
  - Identify the necessary changes to the agreement and evaluate the related impacts on the agreement.
  - Update the agreement with the acquirer as necessary.
- *Execute the agreement*.
  - Start the project and invoke the other processes defined in this handbook.
  - Manage the Supply process and related activities including the development of plans and schedules, decision making for agreements, relationship building and maintenance, interaction with organization management, and final approval authority for deliveries made to acquirer.
  - Maintain communications with acquirers, suppliers, stakeholders, and other organizations regarding the agreement.

  - Carefully evaluate the terms of the agreement to identify risks and issues, progress toward mitigation of risks, and adequacy of progress toward delivery. Also evaluate cost and schedule performance and determine potential undesirable outcomes for the organization.
- *Deliver and support the product or service*.
  - After acceptance and transfer of the final products and/or services, the acquirer will provide payment or other consideration in accordance with all agreements, schedules, and relevant laws and regulations. A support agreement is often ongoing after the transfer of products and/or services.
  - When a Supply process cycle concludes, a final review of performance is conducted to extract lessons learned for continued process performance.
  - Retire the agreement.

Note: The agreement is closed through the Portfolio Management process (see Section 2.3.3.3), *which manages the full set of systems and projects of the organization. When the project is closed, action is taken to close the agreement*.

Common approaches and tips:

- Relationship building and trust between the parties is a nonquantifiable quality that, while not a substitute for good processes, makes human interactions agreeable.
- Develop technology white papers or similar artifacts to demonstrate and describe to the (potential) acquirer the range of capabilities in areas of interest. Use traditional marketing approaches to encourage acquisition of mass-produced products.
- When expertise is not available within the organization (legal and other governmental regulations, laws, etc.), retain subject matter experts to provide information and specify requirements related to agreements.
- Invest sufficient time and effort into understanding acquirer needs before the agreement. This can improve the estimations for cost and schedule and positively affect agreement execution. Evaluate any technical specifications for the product or service for clarity, completeness, and consistency.
- Involve personnel who will be responsible for agreement execution to participate in the evaluation of and response to the acquirer's request. This reduces the start-up time once the project is initiated, which in turn is one way to recapture the cost of writing the response.
- Make a critical assessment of the ability of the organization to execute the agreement; otherwise, the high risk of failure and its associated costs, delivery delays, and increased resource commitment needs will reflect negatively on the reputation of the entire organization.

Elaboration:
Agreements fall into a large range, from formal to very informal based on verbal understanding (e.g., from a written contract to a verbal agreement). Agreements may call for a fixed price, cost plus fixed fee, incentives for early delivery, penalties for late deliveries, and other financial motivators. Appropriate to the level of formality, the agreement establishes requirements, development and delivery milestones, verification, validation and acceptance conditions, process requirements (e.g., configuration management, risk management, measurements), exception-handling procedures, agreement change management procedures, payment schedules, and handling of data rights and intellectual property so that both parties understand the basis for executing the agreement. For a written contract, this occurs when the contract is signed.

2.3.3 Organizational Project-Enabling Processes

The Organizational Project-Enabling Processes are defined in ISO/IEC/IEEE 15288 as follows:

> [5.7.3] The Organizational Project-Enabling Processes are concerned with providing the resources needed to enable the project to meet the needs and expectations of the organization's stakeholders. The Organizational Project-Enabling Processes are typically concerned at a strategic level with the management and improvement of the organization's undertaking, with the provision and deployment of resources and assets, and with its management of risks in competitive or uncertain situations.… The Organizational Project-Enabling Processes establish the environment in which projects are conducted.

This section focuses on the capabilities of an organization relevant to enabling the system life cycle; they are not intended to address general business management objectives, although sometimes the two overlap. Six Organizational Project-Enabling Processes are identified by ISO/IEC/IEEE 15288. They are Life Cycle Model Management, Infrastructure Management, Portfolio Management, Human Resource Management, Quality Management, and Knowledge Management. As defined in ISO/IEC/IEEE 15288 and this handbook, these processes provide the resources and organizational support to enable the projects that are focused on the system life cycle. The organization will tailor these processes and their interfaces to meet specific strategic and tactical objectives in support of the organization's projects (see Section 4.1).

2.3.3.1 Life Cycle Model Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

> [6.2.1.1] The purpose of the Life Cycle Model Management process is to define, maintain, and help ensure availability of policies, life cycle processes, life cycle models, and procedures for use by the organization with respect to the scope of ISO/IEC/IEEE 15288.

Description This process (i) establishes and maintains a set of policies and procedures at the organization level that support the organization's ability to acquire and supply products and services and (ii) provides integrated system life cycle models necessary to meet the organization's strategic plans, policies, goals, and objectives for all projects and all system life cycle stages. The processes are defined, adapted, and maintained to support the requirements of the organization, SE organizational units, individual projects, and personnel. The Life Cycle Model Management process is supplemented by recommended methods and tools. The resulting guidelines in the form of organization policies and procedures are still subject to tailoring by projects (see Section 4.1).

Inputs/Outputs Inputs and outputs for the Life Cycle Model Management process are listed in Figure 2.15. Descriptions of each input and output are provided in Appendix E.

Process Activities The Life Cycle Model Management process includes the following activities:

- *Establish the life cycle process.*
  - Establish policies and procedures for managing and deploying life cycle processes.
  - Establish the life cycle processes with process performance metrics to assess effectiveness and efficiency.
  - Define roles, responsibilities, accountabilities, and authorities to enable the implementation of the life cycle processes.
  - Establish entrance and exit criteria for decision gates.
  - Define an appropriate set of life cycle models that are comprised of stages.
  - Establish tailoring guidance for projects

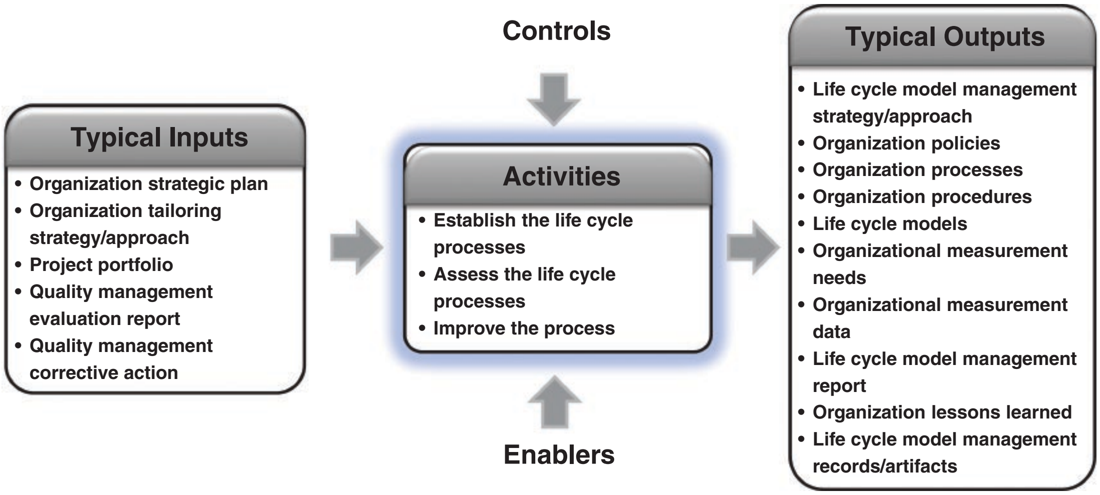

FIGURE 2.15 IPO diagram for Life Cycle Model Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

- *Assess the life cycle process.*
  - Use assessments and reviews of the life cycle models' performance to confirm the adequacy and effectiveness of the Life Cycle Model Management process.
  - Identify opportunities to improve the organization life cycle model management guidelines on a continuing basis based on individual project assessments, individual feedback, metrics, and changes in the organization strategic plan.
- *Improve the process.*
  - Prioritize and implement the identified improvement opportunities.
  - Communicate with all relevant organizations regarding the creation of and changes in the life cycle model management guideline.

NOTE: ISO/IEC/IEEE 15288 provides more details for the Life Cycle Model Management process that are aligned with the activities listed above.

Common approaches and tips:

- Base the policies and procedures on an organization-level strategic and business area plan that provides a comprehensive understanding of the organization's goals, objectives, stakeholders, competitors, future business, and technology trends.
- Ensure that policy and procedure compliance review is included as part of the business decision gate criteria.
- Develop a Life Cycle Model Management process information database with essential information that provides an effective mechanism for disseminating consistent guidelines and providing announcements about organization-related topics, as well as industry trends, research findings, and other relevant information. This provides a single point of contact for continuous communication regarding the life cycle model management guidelines and encourages the collection of valuable feedback, metrics, and the identification of organization trends.
- Establish an organization center of excellence for the Life Cycle Model Management process. This organization can become the focal point for the collection of relevant information, dissemination of guidelines, and analysis

of assessments, performance, and feedback. They can also develop checklists and other templates to support project assessments to ensure that the predefined measures and criteria are used for evaluation.

- Manage the network of external relationships by assigning personnel to identify standards, industry and academia research, and other sources of organization management information and concepts needed by the organization. The network of relationships includes government, industry, and academia. Each of these external interfaces provides unique and essential information for the organization to succeed in business and meet the continued need and demand for improved and effective systems and products for its stakeholders. It is up to the Life Cycle Model Management process to fully define and utilize these external entities and interfaces (i.e., their value, importance, and capabilities that are required by the organization):
  - Legislative, regulatory, and other government requirements
  - Industry SE and management-related standards, training, and capability maturity models
  - Academic education, research results, future concepts and perspectives, and requests for financial support
- Establish an organization communication plan for the policies and procedures. Most of the processes in this handbook include dissemination activities. An effective set of communication methods is needed to ensure that all stakeholders are well informed.
- Include stakeholders, such as engineering and project management organizations, as participants in developing the life cycle model management guidelines. This increases their commitment to the recommendations and incorporates a valuable source of organizational experience.
- Develop alternative life cycle models based on the type, scope, complexity, and risk of a project. This decreases the need for tailoring by engineering and project organizations.

Elaboration

Value Proposition for Organizational Processes. The value propositions to be achieved by instituting organization-wide processes for use by projects are as follows:

- Provide repeatable/predictable performance across the projects in the organization (this helps the organization in planning and estimating future projects and in demonstrating reliability to stakeholders)
- Leverage practices that have been proven successful by certain projects and instill those in other projects across the organization (where applicable)
- Enable process improvement across the organization
- Improve ability to efficiently transfer staff across projects as roles are defined and performed consistently
- Enable leveraging lessons that are learned from one project for future projects to improve performance and avoid issues
- Improve startup of new projects (less reinventing the wheel)

In addition, the standardization across projects may enable cost savings through economies of scale for support activities (tool support, process documentation, etc.).

Benchmarking. SE benchmarking involves comparing an organization's system life cycle processes and practices to those of other entities that are considered as good performers, internally or externally, or comparing to industry standards or good practices. SE benchmarking results and comparisons can be used to generate ideas for driving process improvement to maximize efficiency and effectiveness.

Standard SE Processes. An organization engaged in SE provides the requirements for establishing, maintaining, and improving the standard SE processes and the policies, practices, and supporting functional processes necessary to meet the needs throughout the organization. Further, it defines the process for tailoring the standard SE processes for use on projects addressing the specific needs of the project and for making improvements to the project-tailored SE processes.

Analysis of Process Performance. A high-performing organization also reviews the process (as well as work products), conducts assessments and audits (e.g., assessments based on CMMI (2018), ISO/IEC 33060 (2020), and ISO 9000 (2015) audits), retains corporate memory through the understanding of lessons learned, and establishes how benchmarked processes and practices of related organizations can affect the organization. Successful organizations should analyze their process performance, its effectiveness and compliance to organizational and higher directed standards, and the associated benefits and costs and then develop targeted improvements.

The basic requirements for standard and project-tailored SE process control, based on CMMI (2018), ISO/IEC 33060 (2020), or other resources, are as follows:

- Process responsibilities for projects:
  - Identify SE processes.
  - Document the implementation and maintenance of SE processes.
  - Use a defined set of standard methods and techniques to support the SE processes.
  - Apply accepted tailoring guidelines to the standard SE processes to meet project-specific needs.
- Good process definition includes:
  - Inputs and outputs
  - Entrance and exit criteria
- Process responsibilities for organizations and projects:
  - Assess strengths and weaknesses in the SE processes.
  - Compare the SE processes to benchmark processes used by other organizations.
  - Institute SE process reviews and audits of the SE processes.
  - Institute a means to capture and act on lessons learned from SE process implementation on projects.
  - Institute a means to analyze potential changes for improvement to the SE processes.
  - Institute measures that provide insight into the performance and effectiveness of the SE processes.
  - Analyze the process measures and other information to determine the effectiveness of the SE processes.

Although it should be encouraged to identify and capture lessons learned throughout the performance of every project, the SE organization must plan and follow through to collect lessons learned at predefined milestones in the system life cycle. The SE organization should periodically review lessons learned together with the measures and other information to analyze and improve SE processes and practices. The results need to be communicated and incorporated into training. It should also establish good practices and capture them in an easy-to-retrieve form.

For more information on process definition, assessment, and improvement, see the resources in the bibliography, including the CMMI and ISO/IEC TS 33060.

2.3.3.2 Infrastructure Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

> [6.2.2.1] The purpose of the Infrastructure Management process is to provide the infrastructure and services to projects to support organization and project objectives throughout the life cycle.

Description The work of the organization is accomplished through projects, which are conducted within the context of the infrastructure environment. This infrastructure needs to be defined and understood within the organization and the project to ensure alignment of the working units and achievement of overall organization strategic objectives. This process exists to establish, communicate, and continuously improve the system life cycle process environment.

Infrastructure Management is an organizational project-enabling process and foundational to all SE process management and improvement. Effective infrastructure management is imperative to an organization's ability to change and for that change to be positive, durable, and impactful. Each element of infrastructure is a SoI and both Technical Management and Technical Processes, as stated in ISO/IEC/IEEE 15288 apply to the establishment and maintenance of the infrastructure. Additionally, the Infrastructure Management process includes the physical, political, and process improvement infrastructures.

Inputs/Outputs Inputs and outputs for the Infrastructure Management process are listed in Figure 2.16. Descriptions of each input and output are provided in Appendix E.

Process Activities The Infrastructure Management process includes the following activities:

- *Establish the infrastructure*.
  - Define infrastructure requirements.
  - Define infrastructure elements (e.g., facilities, tools, hardware, software, services, and standards)
  - Define, gather, and negotiate infrastructure resource needs with the organization and projects.
  - Identify, obtain, and provide the infrastructure resources and services to ensure organization goals and objectives are met.
  - Control the infrastructure elements, resources, and services.
  - Conduct inventory management to include enumeration, lists, storage to establish ownership, accessibility, and expectations.
  - Manage resource and service conflicts and shortfalls with steps for resolution.
  - Conduct infrastructure management inventories including identification, status, type, location, access, and condition.
- *Maintain the infrastructure*.
  - Continue to assess whether the project infrastructure needs are met.

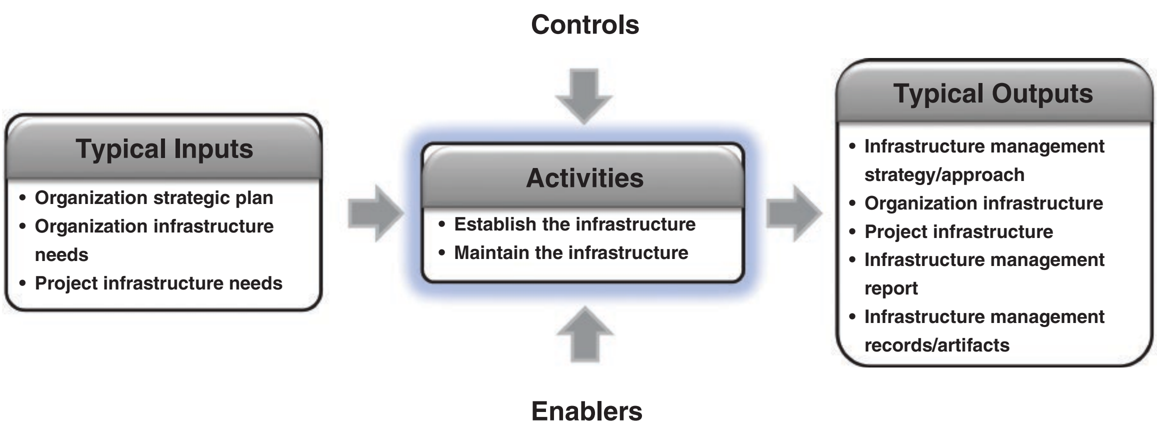

FIGURE 2.16 IPO diagram for Infrastructure Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

- Identify and provide improvements or changes to the infrastructure resources as the project requirements change.
- Manage infrastructure resource availability to ensure organization goals and objectives are met. Conflicts and resource shortfalls are managed with steps for resolution.
- Allocate infrastructure resources and services to support all projects.
- Evaluate the condition of the infrastructure.
- Perform cost analysis toward the cost of infrastructure management.
- Control multi-project infrastructure resource management communications to effectively allocate resources throughout the organization; and identify potential future or existing conflict issues and problems with recommendations for resolution.
- Provide change control for the infrastructure management.
- Conduct risk analysis regarding infrastructure management.
- Evaluate infrastructure management alternatives through analysis of alternatives. This evaluation and analysis compliments risk management and cost reduction activities.

Common approaches and tips:

- Qualified resources may be leased (insourced or outsourced) or licensed in accordance with the investment strategy.
- Establish an organization infrastructure architecture. Integrating the infrastructure of the organization can make the execution of routine business activities more efficient.
- Establish a resource management information system with enabling support systems and services to maintain, track, allocate, and improve the resources for present and future organization needs. Computer-based equipment tracking, facilities allocation, and other systems are recommended for organizations with over 50 people.
- Attend to physical factors, including facilities and human factors, such as ambient noise level and computer access to specific tools and applications.
- Begin planning in early life cycle stages of all system development efforts to address utilization and support resource requirements for system transition, facilities, infrastructure, information/data storage, and management. Enabling resources should also be identified and integrated into the organization's infrastructure.
- Engage project management, risk management, and business management processes to fully integrate Infrastructure Management processes to ensure organizational adoption.

Elaboration

Infrastructure Management Concepts. Projects all need resources to meet their objectives. Project planners determine the resources needed by the project and attempt to anticipate both current and future needs. The Infrastructure Management process provides the mechanisms whereby the organization infrastructure is made aware of project needs and the resources are scheduled to be in place when requested. While this can be simply stated, it is less simply executed. Conflicts must be negotiated and resolved, equipment must be obtained and sometimes repaired, buildings need to be refurbished, and information technology services are in a state of constant change. The infrastructure management organization collects the needs, negotiates to remove conflicts, and is responsible for providing the enabling organization infrastructure without which nothing else can be accomplished. Since resources are not free, their costs are also factored into investment decisions. Financial resources are addressed under the Portfolio Management process (see Section 2.3.3.3), but all other resources, except for human resources which are addressed under the Human Resource Management process (see Section 2.3.3.4), are addressed under this process.

Infrastructure management is complicated by the number of sources for requests, the need to balance the skills of the labor pool against the other infrastructure elements (e.g., computer-based tools), the need to maintain a balance between the budgets of individual projects and the cost of resources, the need to keep apprised of new or modified policies and procedures that might influence the skills inventory, and myriad unknowns.

Resources are allocated based on requests. Infrastructure management collects the needs of all the projects in the active portfolio and schedules or acquires nonhuman assets, as needed. Additionally, the infrastructure management process maintains and manages the facilities, hardware, and support tools required by the portfolio of organization projects. Infrastructure management is the efficient and effective deployment of an organization's resources when and where they are needed. Such resources may include inventory, production resources, or information technology. The goal is to provide materials and services to a project when they are needed to keep the project on target and on budget. A balance should be found between efficiency and robustness. Infrastructure management relies heavily on forecasts into the future of the demand and supply of various resources.

The organization environment and subsequent investment decisions are built on the existing organization infrastructure, including facilities, equipment, personnel, and knowledge. Efficient use of these resources is achieved by exploiting opportunities to share enabling systems or to use a common system element on more than one project. These opportunities are enabled by good communications within the organization. Integration and interoperability of supporting systems, such as financial, human resources (see Section 2.3.3.4), and training, is critically important to executing organization strategic objectives. Feedback from active projects is used to refine and continuously improve the infrastructure.

Further, trends in the market may suggest changes in the supporting environment. Assessment of the availability and suitability of the organization infrastructure and associated resources provides feedback for improvement and reward mechanisms. All organization processes require mandatory compliance with government and corporate laws and regulations. Decision making is governed by the organization strategic plan.

Infrastructure Management Process Maturity. The Infrastructure Management process primarily focuses on the establishment and deployment of infrastructure rather than the construction or actual use of the infrastructure. Since the quality of a product is related to the structure and use of the infrastructure employed, the maturity and quality of the process employed toward management of the infrastructure can help provide higher quality process inputs, outputs, and outcomes.

2.3.3.3 Portfolio Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

> [6.2.3.1] The purpose of the Portfolio Management process is to initiate and sustain necessary, sufficient, and suitable projects to meet the strategic objectives of the organization.

Portfolio management also provides organizational output regarding the set of projects, systems, and technical investments of the organization to external stakeholders, such as parent organizations, investors/funding sources, and governance bodies.

Description Projects create the products or services that meet the objectives and generate revenue for an organization. Thus, the conduct of successful projects requires an adequate allocation of funding and resources and the authority to deploy them to meet project objectives. Most business entities manage the commitment of financial resources using well-defined and closely monitored processes.

The Portfolio Management process also performs ongoing evaluation of the projects and systems in its portfolio. Based on periodic assessments, projects are determined to justify continued investment if they have the following characteristics:

- Contribute to the organization strategy
- Progress toward achieving established goals

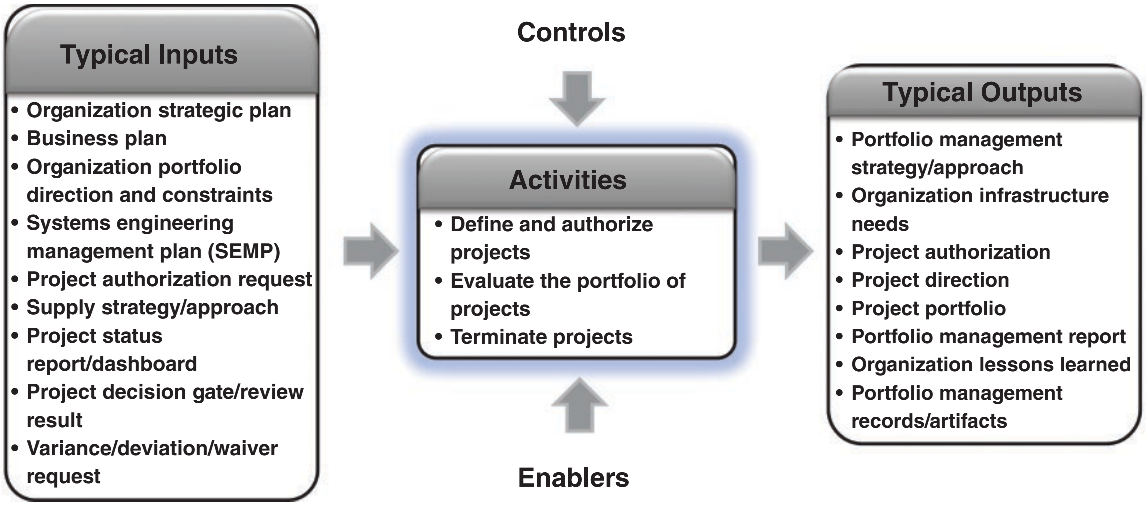

FIGURE 2.17 IPO diagram for Portfolio Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

- Comply with project directives from the organization
- Are conducted according to an approved plan
- Provide a service or product that is still needed and providing acceptable investment returns

Otherwise, projects may be redirected or, in extreme instances, terminated.

Inputs/Outputs Inputs and outputs for the Portfolio Management process are listed in Figure 2.17. Descriptions of each input and output are provided in Appendix E.

Process Activities The Portfolio Management process includes the following activities:

- *Define and authorize projects.*
  – Obtain business area plans and organization strategic plans—use the strategic objectives to identify candidate projects to fulfill them.
  – Identify, assess, prioritize, and select investment opportunities consistent with the organization strategic plan.
  – Establish project scope, define project management accountabilities and authorities, and identify expected project outcomes.
  – Establish the domain area of the product line defined by its main features and their suitable variability.
  – Allocate adequate funding and other resources to selected projects.
  – Identify interfaces and opportunities for multi-project synergies.
  – Specify the project governance process including organizational status reporting and reviews.
  – Authorize project execution.
- *Evaluate the portfolio of projects.*
  – Evaluate ongoing projects to provide rationale for continuation, redirection, or termination.
  – Provide direction and supporting actions for continuation or redirection, as applicable for successful completion.

- *Terminate projects*.
  - Close, cancel, or suspend projects that are completed or designated for termination.

Common approaches and tips:

- Logic modeling techniques that capture how an organization works can aid development or evaluation of business area plans at multiple levels of interest, ranging from the project- to portfolio-level plans (see, for example, PMBOK® (2021) *S*ection 4.2 for a list of commonly used models). The logic models typically describe the fundamental theory/assumptions, planned work (resources, inputs and activities) linked with intended results (outputs, outcomes, and impact).
- When investment opportunities present themselves, prioritize them based on measurable criteria such that projects can be objectively evaluated against a threshold of acceptable performance. This assessment is done in the context of the business area planning to focus resources to best meet present and future objectives.
- Expected project outcomes should be based on clearly defined, measurable criteria to ensure that an objective assessment of progress can be determined. Specify the investment information that will be assessed for each milestone. Initiation should be a formal milestone that does not occur until all resources are in place as identified in the project plan.
- Establish organizational coordination mechanisms to manage the synergies between active projects in the organization portfolio. Complex and large organization architectures require the management and coordination of multiple interfaces and make additional demands on investment decisions. These interactions occur within and between the projects.
- Use a product line engineering approach (see Section 4.2.4) when stakeholders need the same or similar systems (e.g., common features), with some customizations (e.g., variants). The goal is to manage a product line as one product definition with planned variants as opposed to multiple separate products managed individually, thereby streamlining and simplifying the management effort.
- Include risk assessments (see Section 2.3.4.4) in the evaluation of ongoing projects. Projects that contain risks that may pose a challenge in the future might require redirection. Cancel or suspend projects whose disadvantages or risks to the organization outweigh the investment.
- Include opportunity assessments (see Section 2.3.4.4) in the evaluation of ongoing projects. Addressing project challenges may represent a positive investment opportunity for the organization. Avoid pursuing opportunities that are inconsistent with the capabilities of the organization and its strategic goals and objectives or contain unacceptably high technical risk, resource demands, or uncertainty.
- Allocate resources based on the requirements of the projects; otherwise, the risk of cost and schedule overruns may have a negative impact on quality and performance of the project.
- Establish effective governance processes that directly support investment decision making and communications with project management.

Elaboration

Define the Business Cases and Assess Against Business Area Plans. Portfolio management tries to maximize the benefit obtained by the organization from the use of financial assets and other resources within the organization. Thus, business cases for potential projects are evaluated for cost-benefit and the business need before a project is approved for the proposed SoI. Each decision gate reviews the business case as the project matures. The result is reverification or perhaps restatement of the business case.

The business case may be validated in a variety of ways. For large projects, sophisticated engineering models, or even prototypes of key system elements, help prove that the objectives of the business case can be met, and that the system will work as envisioned prior to committing large amounts of resources to full-scale engineering and

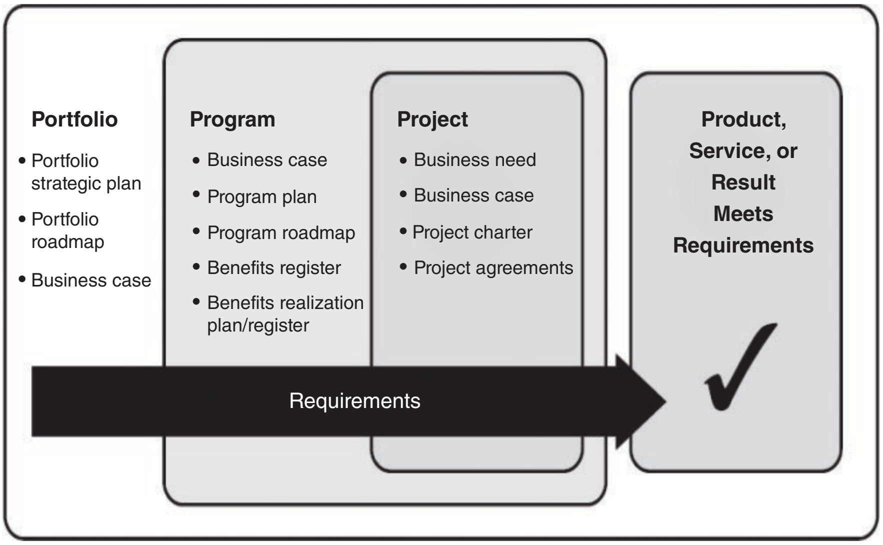

FIGURE 2.18 Requirements across the portfolio, program, and project domains. From PMI (2016). Used with permission. All other rights reserved.

manufacturing development. For smaller projects, when the total investment is modest, proof-of-concept models may be constructed during the concept stage to prove the validity of business case assumptions.

Investment opportunities are not all equal, and organizations are limited in the number of projects that can be conducted concurrently. Further, some investments are not well aligned with the overall strategic plan of the organization. For these reasons, opportunities are evaluated against the portfolio of existing agreements and ongoing projects, taking into consideration the attainability of the stakeholders' requirements.

Project Management and SE considerations. Portfolios may have multiple projects. As previously stated, projects are added to the portfolio after the candidate project can show that it is both feasible and meets organizational business needs. In many organizations projects with defined scope are organized in programs focused on a set of objectives that are part of the organization's strategic plan. As stated in the PMI (2017), the focus of portfolio management is "doing the right work" as opposed to program or project management which is more concerned with "doing work right."

The disciplines of project management and SE have overlapping responsibilities regarding portfolio management. To save time, share knowledge, facilitate the accomplishment of shared objectives, and achieve success, a strong partnership should exist between each of these disciplines (see Section 5.3.3).

At the portfolio level, the scope is extensive with consideration external to the organization and internal across the organization's enterprise. At the other end of the spectrum, the focus is internal to the project with consideration for the context of the product/service/result. An example of this is to look at the range in scope in requirements development, as shown in Figure 2.18.

At the portfolio level, the portfolio's strategic plan and roadmap address business and mission needs and provides direction and organizational focus, and plans/actions to realize the direction. Requirements often start at the concept or portfolio level as a high-level view associated with investment or business opportunities.

2.3.3.4 Human Resource Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

> [6.2.4.1] The purpose of the Human Resource Management process is to provide the organization with necessary human resources and to maintain their competencies, consistent with strategic needs.

Description Projects all need resources to meet their objectives. This process deals with human resources. Nonhuman resources, including tools, databases, communication systems, financial systems, and information technology, are addressed using the Infrastructure Management process (see Section 2.3.3.2).

Project planners determine the resources needed for the project by anticipating both current and future needs. The Human Resource Management process provides the mechanisms whereby the organization management is made aware of project needs and personnel are scheduled to be in place when requested. While this can be simply stated, it is less simply executed. Conflicts must be resolved, personnel must be trained, and employees are entitled to vacations and time away from the job.

The human resource management organization collects the needs, negotiates to remove conflicts, and is responsible for providing the personnel, without which nothing else can be accomplished. Since qualified personnel are not free, their costs are also factored into investment decisions.

Inputs/Outputs Inputs and outputs for the Human Resource Management process are listed in Figure 2.19. Descriptions of each input and output are provided in Appendix E.

Processes Activities The Human Resource Management process includes the following activities:

- *Identify skills.*
  - Identify and record skills of existing personnel to establish a "skills inventory."
  - Review current and anticipated projects to determine and record the skill needs across the portfolio of projects. The INCOSE Systems Engineering Competency Framework (SECF) (2018) and Systems Engineering Competency Assessment Guide (SECAG) (2023) can be used as resources to identify SE skills.
  - Evaluate skill needs against available personnel with the prerequisite skills to determine if training, hiring, or other skill acquisition activities are indicated.
- *Develop skills.*
  - Establish a strategy/approach for skills development.
  - Plan for the skill development per the strategy.
  - Obtain (or develop) and deliver training, education, and mentoring to close identified gaps of project personnel.

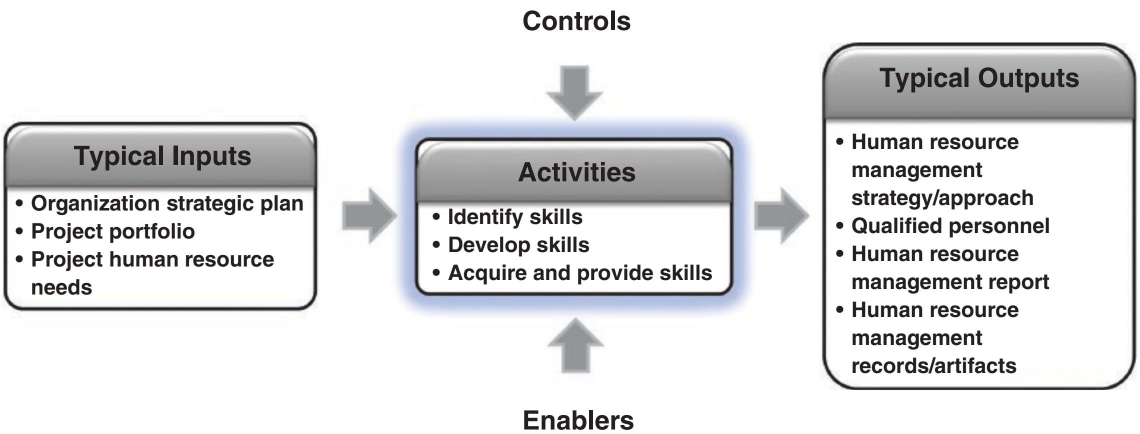

FIGURE 2.19 IPO diagram for Human Resource Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

- Identify skills, abilities, and behaviors needed for competencies. The INCOSE Systems Engineering Assessment Guide is a recommended resource for this.Identify training and development resources to match desired skills, abilities, and behaviors development. The INCOSE Professional Development Portal can help identify potential resources.
- Identify assignments that lead toward career progression.
- Create succession plans to ensure that the desired skill set and flow of skill development through the organization is sustained into the future.
- Create and maintain skill development records.

- *Acquire and provide skills*.
  - Provide human resources to support all projects.
  - Train or hire qualified personnel when gaps indicate that skill needs cannot be met with existing personnel.
  - Maintain and manage a skilled personnel pool to staff ongoing projects.
  - Assign personnel to projects based on personnel development and project needs.
  - Create and maintain staff assignment records.
  - Motivate personnel by providing career development and reward programs.
  - Resolve personnel conflicts between or within projects.
  - Maintain communication across projects to effectively allocate human resources throughout the organization and identify potential future or existing conflicts and problems with recommendations for resolution.
  - Schedule other related assets or, if necessary, acquire them.
- *Develop and Manage Competencies*.
  - Create and maintain job role definitions related to competencies required.
  - Identify organization competency gaps.
  - Align organization competencies with strategic objectives.
  - Maintain organization-level competency definitions and frameworks.

Common approaches and tips:

- The availability and suitability of personnel is one of the critical project assessments and provides feedback for improvement and reward mechanisms.
- Consider using an IPDT environment as a means to reduce the frequency of project rotation, recognize progress and accomplishments and reward success, and establish apprentice and mentoring programs for newly hired employees and students.
- Maintain both a listing of skill needs and the paths to obtain the necessary expertise, including a pipeline of candidates, training provisions, consultants, temporary outsourcing, reassignments, etc.
- Personnel are allocated based on requests and conflicts are negotiated. The goal is to provide personnel to a project when they are needed to keep the project on target and on budget.
- Try to avoid the overcommitment of project personnel, especially people with specialized skills.
- Skills inventory and career development plans are important documentation that can be validated by engineering and project management. The INCOSE SECF and SECAG are comprehensive resources of skills that can be used to develop career development plans.
- Maintain an organization career development program that is not sidetracked by project demands. Develop a policy that all personnel receive training or educational benefits on a regular cycle. This includes both undergraduate and graduate studies, in-house training courses, certifications, tutorials, workshops, and conferences.

- Remember to provide training on organization policies and procedures and system life cycle processes.
- Establish a resource management information infrastructure with enabling support systems and services to maintain, track, allocate, and improve the resources for present and future organization needs.
- Use the slack time in the beginning of a project to provide training to ensure necessary skills.
- Career development plans should be managed and aligned to the objectives of both the employee and the organization. Career development plans should be reviewed, tracked, and refined to provide a mechanism to help manage the employee's career within the organization.

Elaboration

Human Resource Management Concepts. The Human Resource Management process maintains and manages the people required by the portfolio of organization projects. Human resource management is the efficient and effective deployment of qualified personnel when and where they are needed. A balance should be found between efficiency and robustness. Human resource management relies heavily on forecasts into the future of the demand and supply of various resources.

The primary objective of this process is to provide a pool of qualified personnel to the organization. This is complicated by the number of sources for requests, the need to balance the skills of the labor pool against the other infrastructure elements (e.g., computer-based tools), the need to maintain a balance between the budgets of individual projects and the cost of resources, the need to keep apprised of new or modified policies and procedures that might influence the skills inventory, and myriad unknowns.

Project managers face their resource challenges competing for scarce talent in the larger organization pool. They must balance access to the experts they need for special studies with stability in the project team with its tacit knowledge and project memory. Today's projects depend on teamwork and optimally multidisciplinary teams. Such teams are able to resolve project issues quickly through direct communication between team members. Such intrateam communication shortens the decision-making cycle and is more likely to result in improved decisions because the multidisciplinary perspectives are captured early in the process.

2.3.3.5 Quality Management Process

Overview

Purpose As stated in ISO/IEC/IEEE 15288,

> [6.2.5.1] The purpose of the Quality Management process is to assure that products, services, and implementations of the Quality Management process meet organizational and project quality objectives and achieve customer satisfaction.

Description The overarching process for achieving quality goals is the Quality Management (QM) process and its supporting methods, values, and subprocesses. Properly communicated, through policy and procedure, it makes visible the goals of the organization to achieve customer satisfaction. These goals, when supported by measurable activities, provide feedback for maintaining consistency in work processes and delivering quality outcomes. Since primary drivers in any project are time, cost, and quality, inclusion of a comprehensive QM process and its subprocesses is essential to every organization and must be sustained by a work culture that is disciplined in the proper execution of QM foundational principles and values. System life cycle processes are concerned with quality issues, and this is sufficient justification for spending the time, money, and energy into establishing QM fundamentals in an organization, its processes, and its people.

The QM process for SE ensures that all SE processes are deployed consistently by capable staff that can then produce systems designs that fulfill the stakeholder's requirements and lead to development and build processes that are aligned to produce high levels of performance throughout the organization.

Inputs/Outputs Inputs and outputs for the Quality Management process are listed in Figure 2.20. Descriptions of each input and output are provided in Appendix E.

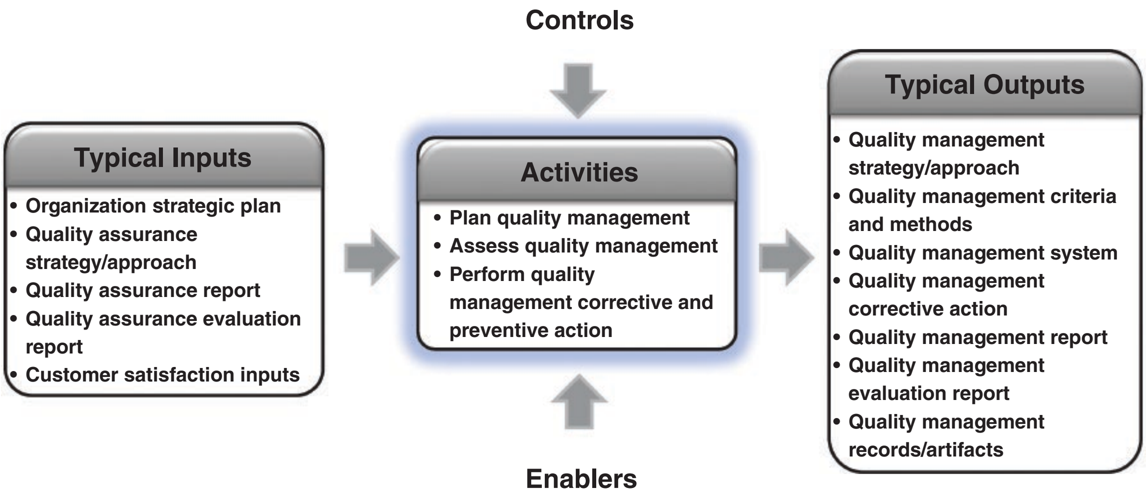

FIGURE 2.20 IPO diagram for the Quality Management process. INCOSE SEH original figure created by Shortell, Walden, and Yip. Usage per the INCOSE Notices page. All other rights reserved.

Process Activities The Quality Management process includes the following activities:

- *Plan quality management*.
  - Identify, assess, and prioritize quality guidelines consistent with the organization strategic plan. Establish QM guidelines-policies, standards, and procedures.
  - Establish organization and project QM goals and objectives, including QM Culture emphasis.
  - Establish organization and project QM responsibilities and authorities.
- *Assess quality management*.
  - Evaluate project assessments.
  - Assess customer satisfaction against compliance with requirements and objectives.
  - Continuously improve the QM guidelines.
- *Perform quality management corrective action and preventive action*.
  - Recommend appropriate action, when indicated.
  - Maintain open communications within the organization and with stakeholders.

Common Approaches and Tips

- Management's commitment to quality is reflected in the integration of QM principles in the strategic planning and budgeting of the organization, and the allocation of educational resources to achieve and sustain a reliable QM culture.
- A quality policy, mission, strategies, goals, and objectives provide essential inputs along with a description of an organization's fundamental values for supporting a growing QM culture.

Elaboration

QM Generally accepted theory and practice. The four generally accepted foundational values of quality are its definition, its system, the standard for quality, and the method for measuring quality. Philip Crosby called them the Four Absolutes of Quality (Crosby, 1979).

1. *The definition of quality is meeting the stakeholder's requirements, needs and expectations*. Organizations (and individuals) are both producers and users of systems. One organization or person (acting as an acquirer) can task another (acting as a supplier) for products or services. This transaction is achieved using agreements that promise to fulfil the stakeholder's requirements in exchange for something of value, usually money. Quality pioneer W. Edwards Deming stressed that meeting stakeholder needs represents the defining criterion for quality and that all members of an organization need to participate actively in "constant and continuous" quality improvement (Deming, 1986).
2. *The system of quality is prevention*. One of the two QM prevention methods is Quality Assurance (QA). QA can be described as "putting good things into our processes" so that they perform as designed and conform to our stakeholder's requirements. QA was born in the aerospace industry and was originally referred to as "reliability engineering." It is generally associated with activities such as failure testing and pre-inspecting batches of materials and system elements that are then certified for use, thus preventing errors and defects from occurring by building-in quality. The QA methodology also includes infusing processes with reliable human resources and appropriate policies, procedures, and training (SEH Section 2.3.4.8). Quality Control (QC) is the QM method for "taking bad things out of our processes after they occur" to prevent the defects that are discovered from reaching our stakeholders. QC includes checking, monitoring, and inspecting for defects and the removal, replacement, or rework of defective outcomes. One method of monitoring and statistically evaluating the stability and potential defect rates of processes is Statistical Process Control (SPC). Many manufacturing and high-volume service organizations use SPC to help achieve quality. Traditional SPC techniques include real-time, random sampling to test a fraction of the output for variances within critical tolerances (Juran, 1974).
3. *The standard for Quality is Zero Defects (ZD)*. It is important to make a distinction between the tracking of defects from feedback loops to improve our processes and progress toward a ZD count, and the more fundamental human term which is the Zero Defects Attitude (ZDA) (Kennedy, 2005). A ZDA is not about achieving perfection; it is a commitment to make each stakeholder's experience as close to what was promised as possible. No one can achieve perfection, nor attain and sustain ZD, so we cannot expect perfection from any of our staff or processes. Like the "pride of workmanship," people with a ZDA have a "heart attitude" that desires to prevent all defects and to reach the highest level of personal performance and customer satisfaction. People with a ZDA want to keep their promises to everyone and make things right when we fail. A ZDA, coupled with appropriate metrics and plans to progress toward ZD, will result in continuous and incremental improvement.
4. *The method for measuring quality is the price of non-conformance* (Crosby, 1979). It is a calculation of the expenses incurred by defects and their related rework, replacement, warranties, customer service, etc. The American Society for Quality calls it the "cost of poor quality." It is an essential factor in calculating the actual "cost of quality" which is determined by comparing the "price of non-conformance (or doing things wrong) that includes expense caused by re-work, defects, and warranties, with the "price of conformance (or doing things right)" which is a calculation of the expenses related to improving processes and applying preventive methods. The cost of quality includes a calculation of quantitative and qualitative parameters that are measured in both financial and human values. When the cost of doing things right is equal to or less than the price of non-conformance then, as Crosby said, "Quality is Free."

QM Culture. SE practitioners need to have sufficient process knowledge and a QM knowledge base to be able to evaluate prevention options and make continuous, incremental improvements. When engineering disciplines are supported by planning and budgeting skills that resonate with the organization, we can achieve Process Quality with effective, efficient, and profitable outcomes, low defect rates, and delighted stakeholders. Deming, in his "14 Points" emphasized the need to "create constancy of purpose for improving products and services" and that it should be supported